Housing and Inflation Inequality in the United States

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Abstract

This paper studies inflation inequality in the U.S. due to heterogeneity in housing consumption. I construct rent indices using the American Community Survey and show that they closely track the Consumer Price Index, both nationally and by metropolitan area. Using these indices, I estimate shelter inflation across sociodemographic groups. Shelter inflation from 2006 to 2019 was 5.5 percentage points higher for non-white householders, who sort into neighborhoods and housing varieties with faster-growing rents. In contrast, shelter inflation did not vary by householder education. High-education householders sort into cities with faster-growing rents, but into housing varieties with slower-growing rents. I develop a spatial equilibrium model with segmented housing markets and show that general non-homothetic preferences parsimoniously explain this pattern. Thus, the microgeography of housing supply and demand widens consumption inequality between education groups.

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1 Introduction

Housing represents the single largest expense for most U.S. households. It accounts for a third of total consumer expenditure, making housing costs central to understanding household finances and economic well-being (U.S. Bureau of Labor Statistics, 2026). Rental prices have also substantially outpaced the prices of other goods in recent decades, giving housing an outsized influence on aggregate inflation dynamics (Adams et al., 2024).[1] Notably, households live in markedly different neighborhoods and dwelling types, and this heterogeneity in housing consumption creates potential scope for inflation inequality.

A key challenge in studying differences in rental prices across space and time is that data on rents are far scarcer than data on house prices.[2] Ideally, one should construct segmented rent indices using a large, representative panel of rental units. With such a panel, repeat-rent estimators compare the rent of the same unit over time and therefore hold housing quality fixed by design (Ambrose, Coulson, and Yoshida, 2015).

In the U.S., no national rental panel is readily available at scale. Commercial datasets can be used to estimate repeat-rent indices but are typically non-representative since they only observe new contract rents for higher-quality units (Adams et al., 2024; Lyons et al., 2026). The American Housing Survey (AHS) follows a representative panel of rental units, but its sample is small and covers only a limited subset of metropolitan areas.[3] The Bureau of Labor Statistics (BLS) Housing Survey comes closest to the ideal: it follows a rotating panel of roughly 50,000 rental units to officially measure shelter inflation for the Consumer Price Index (CPI). However, the survey is not publicly available for research use.[4] As a result, researchers often turn to cross-sectional rents reported in the American Community Survey (ACS).[5] Comparisons of cross-sectional rents over time, though, potentially mix true price changes with changes in the quality of the rental stock.

This paper begins by validating changes in rental prices in the ACS. Using contract rents from the 2005–2024 ACS, I estimate hedonic regressions and use them to construct rent indices at the

[1] Rent-based components have a combined relative importance of roughly 34 percent in the Consumer Price Index as of December 2025. Rental housing accounts for 8 percent, and owner-occupied housing for a further 26 percent, measured using the owners' equivalent rent method (U.S. Bureau of Labor Statistics, 2026).

[2] In the U.S., housing transactions are administratively recorded by county assessor offices.

[3] While the AHS has been conducted since 1973 (annually until 1981 and biennially thereafter), the panel is not fully continuous because the Census Bureau redrew the national sample in 1985 and 2015. Since 2015, the AHS has included roughly 20,000 rental units and sampled 25 metropolitan areas in each survey year (U.S. Census Bureau, 2025). As of 2025, the AHS has been postponed and is being redesigned.

[4] To the best of my knowledge, the few papers that use the BLS Housing Survey (e.g., Adams et al., 2024) are authored by BLS staff. The BLS Housing Survey also lacks comprehensive coverage, because the BLS conducts the survey in only 75 metropolitan areas (U.S. Bureau of Labor Statistics, 2025).

[5] See Appendix B for a non-comprehensive literature review of publications that use rents from the ACS or the decennial Census.

level of a housing variety.[6] I show that aggregating these rent indices closely replicates the repeat-rent indices produced by the BLS, both nationally and across the 22 metropolitan areas for which the BLS publishes long-standing repeat-rent indices.[7] From 2006 to 2024, annual rent inflation in the ACS and the CPI had a correlation of 0.91, mean absolute deviation of 0.48 percentage points, and mean difference of 0.01 percentage points.[8] Two-year rent inflation across the 22 metropolitan areas had a correlation of 0.87, and the ACS and CPI produced similar rankings of metropolitan areas by average rent growth. In contrast, raw average rent changes in the ACS overstate rent inflation by 0.50 percentage points per year from 2006 to 2024, likely due to changes in the rental stock. Provided rents are hedonically adjusted, the ACS can therefore yield reasonable measures of rent inflation for households that official price indices do not cover.

I then measure housing-driven inflation inequality between householders of different sociodemographic groups and decompose the inequality into the contributions of location versus housing variety.[9] I aggregate these rent indices into sociodemographic-specific shelter indices—following the official methodology of the BLS. To study how housing costs have shaped consumption inequality, I embed these shelter indices in sociodemographic-specific aggregate price indices constructed using the Consumer Expenditure Survey (CEX).

I highlight three facts. First, rent inflation is heterogeneous across metropolitan areas. Among the 100 largest metropolitan areas, average annual rent inflation from 2006 to 2019 ranged from 1.1 percent in Bridgeport, CT, to 4.3 percent in San Jose, CA, and rent increases are strongly correlated with increases in household income. Second, shelter inflation is nearly identical for low- and high-education householders. From 2006 to 2019, the shelter indices of householders with at most a high school degree and of householders with at least a bachelor's degree rose 43.2 and 43.2 percent, respectively. Notably, high-education householders sort into metropolitan areas with faster-growing rents, but *within* an area they sort into housing varieties with slower-growing rents. Third, shelter inflation was higher for non-white householders than for white householders. The shelter index for non-white householders rose 46.8 percent, compared to 41.3 percent for white householders. Sorting across metropolitan areas explains little of the inequality; within an area, non-white householders sort into neighborhoods and housing varieties with faster-growing rents. I validate these facts (when possible) with repeat-rent indices estimated using the AHS and data

[6] I define a housing variety as the combination of Public Use Microdata Area, dwelling type, dwelling size, and dwelling age.

[7] The 22 metropolitan areas include Anchorage, Atlanta, Baltimore, Boston, Chicago, Dallas, Denver, Detroit, Honolulu, Houston, Los Angeles, Miami, Minneapolis, New York, Philadelphia, Phoenix, San Diego, San Francisco, Seattle, St. Louis, Tampa, and Washington, D.C. The BLS also publishes a repeat-rent index for Riverside beginning in 2017.

[8] The major exception is 2020, when rent inflation in the ACS was larger than in the CPI by 1.9 percentage points. This is likely due to the COVID-19 pandemic disrupting data collection for both surveys, and the BLS recording rent as $0 when landlords provided pandemic-related rent relief (Houck, 2023).

[9] More specifically, I define a household's demographic group by the characteristics of the head of household.

from Cotality.

Next, I examine why high-education householders sort into metropolitan areas with faster-growing rents but into housing varieties with slower-growing rents. I develop a spatial equilibrium model in the style of Rosen (1979) and Roback (1982) that rationalizes this sorting parsimoniously. The model has two key features that I empirically verify using the ACS: (i) households have general non-homothetic preferences over housing; and (ii) housing markets are segmented by housing variety, where construction favors new housing. Because housing is a necessity, higher incomes lead high-education householders to sort differentially into expensive housing. In particular, they choose desirable and productive metropolitan areas, where technological change differentially raises productivity and thus rents. Within an area, however, high-education householders choose newer (and thus nicer) housing varieties. New construction directly adds supply to these varieties and slows their rent growth. Non-homothetic preferences for housing quality therefore push shelter inflation for high-education householders in opposing directions.

Shocks calibrated to growth in income and households in the ACS from 2009 to 2019 generate differences in shelter inflation between high- and low-education householders of 1.4 log points across metropolitan areas and -1.4 within, close to the observed 1.5 across and -1.3 within. This highlights how modeling the microgeography of housing supply and demand matters for welfare evaluation. Low-education householders disproportionately live in varieties with an old dwelling age, whose supply is less elastic, so they bear more of the rent increases that come with demand shocks. When all housing varieties in a metropolitan area have identical supply elasticities, income growth from 2009 to 2019 raises welfare for low-education householders by 0.6 log points. When housing markets are segmented, their welfare instead decreases by 0.3 log points.[10]

Finally, I examine what the racial differences in shelter inflation imply for household welfare. When race-specific consumer price indices differ only in expenditure shares, prices for white and non-white households rose at nearly the same rate from 2006 to 2019: 1.9 and 1.9 percent per year, respectively. However, when I replace the official shelter index with each group's own shelter index, I find that prices for non-white households rose 0.1 percentage points per year faster than prices for white households. By the end of 2019, the price level for non-white households was 1.3 percentage points higher than for white households. Differences in shelter inflation therefore account for the entire racial inequality in overall inflation. Deflated by each group's own consumer price index, the wage gap between white and non-white workers fell 3.1 log points from 2005 to 2019, compared with 4.1 log points in nominal terms. Therefore, the racial wage gap closed about 24 percent less than nominal wages suggest.

This paper contributes to several strands of literature in urban and household economics. It re-

[10] An increase in welfare by one log point corresponds approximately to a one percent increase in non-housing consumption.

lates most directly to a growing literature measuring inflation inequality across households. Several papers use scanner data to document that low-income households experience higher inflation for consumer packaged goods, and that product innovation further lowers inflation for high-income households (Kaplan and Schulhofer-Wohl, 2017; Argente and Lee, 2021; Jaravel, 2019). Other papers study the full consumption basket, combining the BLS price series with consumer spending from the CEX (Hobijn and Lagakos, 2005; McGranahan and Paulson, 2006; Jaravel, 2019; Jaravel and Lashkari, 2024; Jaravel, 2026). However, these studies let households differ only in expenditure shares, so every household experiences the same shelter inflation.

Few studies examine inflation heterogeneity in shelter, with Moretti (2013) and Molloy (2024) as notable exceptions. Moretti (2013) finds that college graduates sorted into cities with higher rent inflation from 1980 to 2000. In contrast, Molloy (2024) uses the AHS and finds little difference in housing-driven inflation by income from 1985 to 2021. I reconcile these findings by showing that sorting across cities and sorting within cities offset each other. High-education householders sort into metropolitan areas with faster-growing rents, but into housing varieties with slower-growing rents. I also show that shelter inflation is higher for non-white householders, and that differences in shelter inflation, rather than in shelter expenditure shares, drive inflation inequality between races.

This paper additionally contributes to a substantial literature on spatial equilibrium models. Building on the foundational work of Rosen (1979) and Roback (1982), numerous papers use these models to empirically study the welfare implications of spatial sorting (e.g., Bayer, Ferreira, and McMillan 2007; Diamond 2016; Piyapromdee 2021; Su 2022; Couture et al. 2024; Fogli et al. 2026; see Diamond and Gaubert 2022 for a review). The set-up in this paper relates most directly to Ma (2025), who develops a spatial equilibrium model with segmented housing markets to study the distributional effects of zoning and new construction in Atlanta. I use a complementary set-up to examine heterogeneity in shelter inflation in the U.S.

Lastly, this paper contributes to a large literature on how to construct rent and house price indices. Repeat-sales estimators, introduced by Bailey, Muth, and Nourse (1963) and popularized by Case and Shiller (1987), control for quality by comparing the same housing unit over time and have been adapted to rents (Ambrose, Coulson, and Yoshida, 2015). Hedonic indices instead control for quality by pricing observed housing characteristics (Rosen, 1974). Because repeat-rent panels are rare, much of what is known about changes in rental prices comes from cross-sectional data (Appendix B). However, cross-sectional rent indices are rarely benchmarked against repeat-rent indices, especially below the national level. I find that hedonically adjusted indices built from the ACS closely track the BLS repeat-rent indices both nationally and in 22 metropolitan areas.

The paper proceeds as follows. Section 2 describes the data. Section 3 validates rent inflation in the ACS. Section 4 documents descriptive facts about rent inflation. Section 5 develops a spatial equilibrium model to explain the descriptive facts. Section 6 discusses welfare implications.

Section 7 concludes.

2 Data

2.1 Rent inflation

To measure rent and shelter inflation, I rely on three datasets: the American Community Survey, the American Housing Survey, and the Cotality Multiple Listing Service.

American Community Survey (ACS). The American Community Survey, conducted by the Census Bureau since 2000, randomly surveys 1 percent of U.S. households annually. I restrict my analysis to 2005 to 2024, since household Public Use Microdata Area (PUMA) of residence is available only for those years (Ruggles et al., 2025).[11] I assign each PUMA to the corresponding Core Based Statistical Area (CBSA) with the largest geographic overlap.

I classify households by three characteristics of the household head: (i) education (high school or less, some college, or a bachelor's degree or more); (ii) age (under 45, 45 to 64, or 65 or older); and (iii) race (white or non-white).[12] I measure household monthly rent using contract rent instead of gross rent, which additionally includes costs for utilities and fuel. I prefer contract rents for measuring inflation, since gross rents can move with the quantity of utilities consumed despite no change in prices.

The ACS collects three dwelling characteristics: dwelling type (i.e., number of units in structure), dwelling size (i.e., number of bedrooms), and dwelling age (i.e., the decade the structure was built). I define housing varieties as the combination of these dwelling characteristics along with PUMA.[13]

American Housing Survey (AHS). The American Housing Survey is a panel survey of U.S. housing units that has been conducted by the U.S. Census Bureau since 1973. The AHS has two key features: (i) it is nationally representative; and (ii) the unit of observation is the housing unit rather than the household. That is, the survey returns to the same physical address in each wave

[11] Prior to 2005, only household state of residence is available. PUMAs are non-overlapping statistical geographic areas defined by the U.S. Census Bureau to partition the U.S. into regions containing between 100,000 and 200,000 people.

[12] I define *white* as non-Hispanic white householders and *non-white* as all other householders. Hispanic origin is asked separately from race, and the race that Hispanic respondents report can shift with survey design. For example, the number of Hispanics reporting their race as white fell by half from the 2010 to the 2020 Census, largely because of changes to the race and Hispanic origin questions (Jones et al., 2021).

[13] I categorize the number of units in structure into single-family or multi-family. I categorize the number of bedrooms into studio or one-bedroom, two-bedroom, three-bedroom, and four-bedroom or larger.

regardless of who lives there, making it well suited for studying rent changes.[14]

However, the AHS has a small sample, covers only a limited set of metropolitan areas, and identifies only the Census region of a housing unit in the public-use survey data.[15] Therefore, unlike the ACS, the AHS cannot decompose inflation inequality into the contributions of location versus housing variety. I use the AHS only to validate facts from the ACS. In particular, I use the national survey in odd-numbered years from 1975 to 2023 and limit my sample to units rented for cash.[16] As in the ACS, I define household groups by the characteristics of the householder and measure monthly rent using contract rent.[17]

Cotality Multiple Listing Service (MLS). The Cotality Multiple Listing Service data compile the listings that real estate agents post to local databases across the U.S. from 2006 to 2024. As in the AHS, the unit of observation is the housing unit, so I can observe repeat rents. Unlike in the AHS, each listing also includes the address of the unit, so I can compare the rent changes of different housing varieties within a metropolitan area.

However, the MLS observes only units that are listed on the market, and thus records the asking rents of new tenants rather than the rents of all renters. Commercial listings data also tend to over-represent higher-quality units (Adams et al., 2024). Moreover, the MLS has no information on households, so unlike the ACS and AHS, it cannot measure inflation for groups of households directly. As with the AHS, I use the MLS only to validate facts from the ACS.

2.2 Consumption and wages

To translate differences in shelter inflation into differences in overall inflation and real wages, I use two additional datasets: the Consumer Expenditure Survey and the Current Population Survey.

Consumer Expenditure Survey (CEX). The Consumer Expenditure Survey is a quarterly interview of U.S. households conducted by the Bureau of Labor Statistics since 1980. Used to construct the Consumer Price Index, the CEX surveys a nationally representative sample of households on

[14]The panel is unfortunately not fully continuous, as the Census Bureau redrew the national sample in 1985 and 2015.

[15]Since 2015, the AHS has included roughly 20,000 rental units and sampled 25 metropolitan areas in each survey year (U.S. Census Bureau, 2025). Additionally, as of 2025, the AHS has been postponed and is being redesigned with a new sample.

[16]The national survey was conducted annually from 1973 to 1981 and biennially from 1983 onward. I only use odd-numbered years to keep the interval between survey waves constant.

[17]Contract rent is top-coded in the AHS, but the top code is only binding in 1985, when contract rent is top-coded at $751. I drop the 1985 top-coded observations when they serve as the base of a rent change. The Census Bureau additionally imputes contract rent for item nonresponse, and I include these imputed values in my analysis. Imputed values account for 8 percent of cash-rent observations overall, ranging from 3 percent in 1977 to 11 percent in 2013.

their expenditures. I use the harmonized data files constructed by Jaravel (2026). As in the ACS, I classify households as white or non-white by the race of the reference person.

Current Population Survey (CPS). The Current Population Survey, conducted by the Census Bureau since 1940, is a monthly U.S. labor force survey. I use the Annual Social and Economic Supplement (ASEC), which collects income and employment for the previous calendar year. I limit my sample to prime-age workers who worked full-time and full-year, and compute each worker's hourly wage from annual wage and salary income.[18] As in the ACS, I classify workers by race as either white or non-white.

2.3 Benchmarks

Consumer Price Index (CPI). To benchmark ACS estimates of rent and shelter inflation to official measures, I use the Bureau of Labor Statistics' Rent of Primary Residence in U.S. City Average CPI and Shelter in U.S. City Average CPI.[19] Both indices date back to at least 1952 and are constructed using a monthly survey of nationally representative housing units with a rotating panel design. I also use the Rent of Primary Residence CPI for the 22 metropolitan areas for which the BLS publishes long-standing indices.[20]

2.4 Summary statistics

Table 1 presents summary statistics for the 2019 ACS, which surveyed 1.2 million households representing 116 million households nationally. Housing consumption differs substantially across sociodemographic groups. For example, non-white householders are less likely to live in single-family homes than white householders (60 versus 79 percent, respectively). Similarly, high-education householders are more likely to live in larger and more recently built housing units. More specifically, 32 percent of householders with at least a bachelor's degree live in units with four or more bedrooms, versus 16 percent of householders with at most a high school degree, and 41 versus 25 percent live in units built in 1990 or later. This heterogeneity in housing consumption creates potential scope for inflation inequality.

[18] That is, I limit my sample to workers who are 25 to 54 years old, worked at least 35 usual hours per week and at least 40 weeks, and had positive wage and salary income. I compute the hourly wage as annual wage and salary income divided by the product of weeks worked and usual weekly hours.

[19] In particular, I use the series CUUR0000SEHA (Consumer Price Index for All Urban Consumers: Rent of Primary Residence in U.S. City Average) and CUUR0000SAH1 (Consumer Price Index for All Urban Consumers: Shelter in U.S. City Average).

[20] The 22 metropolitan areas include Anchorage, Atlanta, Baltimore, Boston, Chicago, Dallas, Denver, Detroit, Honolulu, Houston, Los Angeles, Miami, Minneapolis, New York, Philadelphia, Phoenix, San Diego, San Francisco, Seattle, St. Louis, Tampa, and Washington, D.C. The BLS also publishes a repeat-rent index for Riverside beginning in 2017.

3 Validating rents in the American Community Survey

I construct rent and shelter indices using the ACS and validate them against the official CPI. I find that hedonically adjusted rent and shelter indices constructed from the ACS closely track the CPI, both nationally and within metropolitan areas.

3.1 Index construction

Since the ACS only provides cross-sectional data, comparisons of rents over time must account for changes in the quality of the rental stock.[21] To do so, I first estimate the rent of each housing variety in each year using a hedonic regression.[22] I then aggregate these rents into price indices with expenditure weights, following the methodology of the BLS.

Hedonic rents. Let $v = (p, u, b, a)$ denote the housing variety with PUMA p, dwelling type u, dwelling size b, and dwelling age a. For each year t, I estimate the following hedonic regression:

$$\mathrm{E}[R_{it}] = \exp\left(\alpha_{put} + \alpha_{pbt} + \alpha_{pat}\right), \tag{1}$$

where R_{it} is the monthly contract rent of household i, whose dwelling is located in PUMA p and has type u, size b, and age a. The terms α_{put}, α_{pbt}, and α_{pat} are PUMA-specific fixed effects for dwelling type, size, and age, respectively. I estimate equation (1) via Poisson pseudo-maximum likelihood, weighting each household by its survey weight.[23] Intuitively, the additive specification allows the price of each dwelling characteristic to vary across PUMAs, while pooling information across varieties within the same PUMA.[24] The *hedonic rent* of variety v is then given by $\hat{R}_{vt} = \exp(\hat{\alpha}_{put} + \hat{\alpha}_{pbt} + \hat{\alpha}_{pat})$, which is also defined for varieties occupied by homeowners.[25]

[21] The Bureau of Economic Analysis (BEA) also uses ACS rents, but to construct cross-sectional indices of relative rents for its Regional Price Parities, which compare rent levels across states and metropolitan areas within a year (U.S. Bureau of Economic Analysis, 2025). In particular, the BEA averages the rents of tenant-occupied units in each PUMA, stratified by dwelling characteristics such as type of structure, number of bedrooms, and age. In contrast, I use the ACS to measure rent changes over time.

[22] I implicitly assume that the quality of the rental stock does not change *within* a housing variety over time. Therefore, I do not account for the depreciation or renovation of dwellings. Depreciation biases my rent indices downward, whereas renovation biases them upward. Nevertheless, my indices closely track the BLS repeat-rent indices, which adjust for both depreciation and renovation (Sections 3.2 and 3.3), suggesting the biases are likely small or offsetting.

[23] This specification ensures that I predict positive rental prices for each housing variety. Additionally, ordinary least squares on log rents is biased if the dispersion of rents differs across varieties (Santos Silva and Tenreyro, 2006).

[24] A fully saturated specification (i.e., with a fixed effect for each variety) fits the data only slightly better in sample. However, the typical variety contains few renters in a given year, so its hedonic rent would be noisy (Appendix C.1).

[25] To avoid prediction with disconnected fixed effects, I only keep varieties with hedonic rents in both $t-1$ and t (Abowd, Creecy, and Kramarz, 2002). Since the price change estimated for any given variety may be noisy, I implement empirical Bayes shrinkage as a robustness check. Results are quantitatively similar.

Price indices. Next, I aggregate hedonic rents into a chained price index using expenditure weights from the previous year (i.e., a chained Laspeyres index). For a group of households g, housing inflation in year t is given by:

$$\pi_t^g = \sum_v s_{v,t-1}^g \left(\log \hat{R}_{vt} - \log \hat{R}_{v,t-1}\right), \quad s_{v,t-1}^g = \frac{\omega_{v,t-1}^g \hat{R}_{v,t-1}}{\sum_{v'} \omega_{v',t-1}^g \hat{R}_{v',t-1}}, \tag{2}$$

where $s_{v,t-1}^g$ is the share of the group's housing expenditure spent on variety v in year $t-1$ and $\omega_{v,t-1}^g$ is the number of households in group g living in variety v in year $t-1$. I construct both a shelter index and a rent index. The *shelter index* sums over all households, where I value each owner-occupied dwelling at the hedonic rent of its variety.[26] The *rent index* sums over renter households only. To construct an index for a given metropolitan area, I restrict the sum to varieties located in the corresponding CBSA.[27] I construct confidence intervals using the 80 ACS replicate weights from IPUMS.

Sample splitting. Measuring expenditure weights and price changes using the same households can mechanically bias index estimation, which is commonly referred to as formula bias (Reinsdorf, 1998).[28] To avoid this bias, I implement a standard sample-splitting procedure. Specifically, I randomly assign each household to one of five folds. For households in fold k, I compute expenditure weights using hedonic rents estimated on fold k alone, and rent changes using hedonic rents estimated on the other four folds. The sum in equation (2) then averages over all variety–fold combinations.

3.2 National benchmark

I benchmark my national rent and shelter indices to corresponding indices produced by the BLS: the Rent of Primary Residence in U.S. City Average CPI and Shelter in U.S. City Average CPI. These indices are constructed using a repeat-rent methodology that adjusts for depreciation and renovation (Houck, 2023). Panel A of Figure 1 presents annual rent inflation in the ACS versus the Rent CPI. The ACS rent index closely tracks the Rent CPI. From 2006 to 2019, annual rent inflation in the two series has a correlation of 0.96, mean absolute deviation of 0.30 percentage

[26] This approach mirrors how the BLS imputes owners' equivalent rent using the rents of comparable rental units.

[27] Because PUMA boundaries changed in 2012 and 2022, I use the consistent PUMAs constructed by IPUMS: the 2000–2010 consistent PUMAs through 2011 and the 2010–2020 consistent PUMAs thereafter. I measure the 2011–2012 rent change using the 2000–2010 consistent PUMAs.

[28] Sampling error that lowers the hedonic rent of a variety in year $t-1$ both lowers its expenditure weight and raises its measured rent change. Expenditure weights and rent changes are therefore mechanically negatively correlated, particularly for thinly populated varieties.

points, and mean difference of −0.19 percentage points.[29] Panel B of Figure 1 presents annual shelter inflation in the ACS versus the Shelter CPI. The ACS shelter index also closely tracks the Shelter CPI. From 2006 to 2019, annual shelter inflation in the two series has a correlation of 0.88, mean absolute deviation of 0.42 percentage points, and mean difference of 0.07 percentage points.

However, rent and shelter inflation in the ACS diverge from the CPI in 2020. Rent inflation in 2020 is 5.0 percent in the ACS versus 3.1 percent in the CPI, and shelter inflation in 2020 is 5.2 percent in the ACS versus 2.4 percent in the CPI. This divergence is likely caused by the COVID-19 pandemic, which disrupted data collection for both surveys.[30] Furthermore, the BLS recorded zero rent for units that received pandemic-related rent relief, which mechanically lowered CPI rent inflation (Houck, 2023). After 2021, rent and shelter inflation in the ACS closely track the CPI again.

The hedonic adjustment described in Section 3.1 is necessary to ensure that rent inflation in the ACS closely tracks the CPI. To show this, Appendix Figure A1 replaces the ACS rent index with annual changes in average contract rent in the ACS. These raw rent changes track the CPI less closely and overstate rent inflation. From 2006 to 2024, the correlation with the CPI is 0.86 for raw rent changes versus 0.91 for the ACS rent index, and the mean absolute deviation is 0.65 versus 0.48 percentage points. Raw rent changes are on average 0.50 percentage points higher than rent inflation in the CPI, whereas the ACS rent index has a mean difference of only 0.01 percentage points. This overstatement likely reflects improvements in the quality of the rental stock over time (e.g., through new construction).

3.3 Metropolitan area benchmark

Next, I benchmark my metropolitan area rent indices to the CPI. Figure 2 presents rent inflation in the ACS and in the Rent of Primary Residence CPI for each of the 22 metropolitan areas for which the BLS publishes a rent index going back to 2005. Since samples within a metropolitan area can be small in both the ACS and the BLS Housing Survey, I average annual inflation over two-year periods. From 2006 to 2019, two-year rent inflation in the two sources has a correlation of 0.86 when pooled across metropolitan areas and periods, a mean absolute deviation of 0.73 percentage points, and a mean difference of −0.17 percentage points.[31] Within metropolitan areas, the correlation exceeds 0.7 in 18 of the 22 areas.

Furthermore, the ACS ranks metropolitan areas by average rent growth from 2006 to 2019 sim-

[29] Cumulative rent growth from 2006 to 2019 is lower in the ACS than in the CPI: 48.4 compared to 52.3 percent. This difference potentially reflects the BLS age-bias adjustments for depreciation, which add 0.0 to 0.3 percentage points to each unit's six-month rent change (Houck, 2023).

[30] In particular, the 2020 ACS collected only about two-thirds of its usual responses, and respondents had higher incomes and more education than nonrespondents (U.S. Census Bureau, 2021).

[31] As with the national benchmark, rents in the ACS temporarily diverge from the CPI in 2020–2021.

ilarly to the CPI. For example, both sources show the fastest rent growth in Seattle, San Francisco, and Denver, and the slowest rent growth in St. Louis and Detroit. The ACS can therefore provide reasonable measures of rent inflation for market segments that official price indices do not cover, as long as rents are hedonically adjusted.

4 Shelter inflation across households

Having validated the ACS rent and shelter indices estimated in Section 3, I document three stylized facts about rent and shelter inflation. These facts provide motivation for the spatial equilibrium model in Section 5.

4.1 Fact 1: Rent inflation varies widely across metropolitan areas

First, rent inflation is heterogeneous across metropolitan areas. Appendix Figure A2 presents average annual rent inflation in the ACS from 2006 to 2019 for the 100 largest CBSAs, plotted against the average growth of nominal household income. Rent inflation ranged from a minimum of 1.1 percent per year in Bridgeport, CT, to a maximum of 4.3 percent in San Jose, CA.[32] The 25th and 75th percentiles of average rent inflation are 2.3 and 3.0 percent per year, cumulating to a difference of 14 percentage points from 2006 to 2019.

Furthermore, rent inflation is strongly correlated with household income growth. The correlation across CBSAs is 0.49, and a one percentage point increase in household income growth is associated with 0.41 percentage points higher rent inflation. The BLS does not publish an official rent index for the vast majority of metropolitan areas. Deflating local incomes by the national CPI therefore overstates real income growth in the metropolitan areas where incomes grew fastest, since rents also grew fastest in those areas.[33]

4.2 Fact 2: Shelter inflation is similar across education and age groups

Second, shelter inflation is similar across education and age groups.[34] Figure 3 presents the ACS shelter index from equation (2) by household group, normalized to 100 in 2005. Panel A shows that householders with at most a high school degree and householders with at least a bachelor's degree experienced the same shelter inflation from 2006 to 2019. Their shelter indices rose 43.2 and 43.2 percent, respectively. Similarly, Panel B shows a small difference in shelter inflation by

[32] Rents grew fastest in San Jose, Seattle, and Portland and slowest in Bridgeport, Las Vegas, and Tucson.

[33] This finding is not new (e.g., Moretti, 2013), though I replicate it for more recent years using my rent indices.

[34] I segment by householder education instead of income since income is endogenous to where a household chooses to live (e.g., Card, Rothstein, and Yi, 2025).

age from 2006 to 2019. The shelter index of householders under 45 rose 44.3 percent, compared to 42.8 percent for householders 65 or older.[35]

Decomposition of shelter inflation by education. Notably, the similar shelter inflation of high- and low-education householders does not reflect similar housing consumption, but rather two margins of sorting that are offsetting. To show this, I conduct the following decomposition exercise. I recompute equation (2) by education while coarsening the level of aggregation at which rental prices may differ. This distinguishes two reasons why a householder may experience higher shelter inflation: (i) they live in a metropolitan area or neighborhood where rents grew faster; or (ii) they occupy housing varieties whose rents grew faster within the same neighborhood. I use four levels of aggregation: (1) CBSA; (2) PUMA; (3) CBSA interacted with dwelling type, size, and age; and (4) PUMA interacted with dwelling type, size, and age. The last level reproduces the index in Section 3.1.[36]

Figure 4 presents the decomposition of shelter inflation by education. When every householder in a CBSA experiences the same rent change, cumulative shelter inflation from 2006 to 2019 is 1.9 percentage points higher for high-education householders. That is, high-education householders sort into metropolitan areas where rents grew faster. Sorting across neighborhoods does not narrow this gap, which remains 1.7 percentage points with PUMA as the level of aggregation. In contrast, allowing rent changes to differ across dwelling types, sizes, and ages within a CBSA closes the gap entirely. Therefore, within metropolitan areas, high-education householders sort into higher-quality housing varieties whose rents grew more slowly.[37] I further examine this sorting across housing varieties in Section 5.

4.3 Fact 3: Shelter inflation is higher for non-white households

Third, shelter inflation is higher for non-white households. Figure 3 presents the ACS shelter index from equation (2) by household group, normalized to 100 in 2005. Panel C of Figure 3 shows that from 2006 to 2019, the shelter index of non-white householders rose 46.8 percent, compared to 41.3 percent for white householders, a difference of 5.5 percentage points. Non-white householders experienced higher shelter inflation in 10 of the 14 years, and the difference

[35] Appendix Figure A6 presents the annual shelter inflation underlying Figure 3.

[36] Appendix Figure A3 presents the same decomposition by age. From 2006 to 2019, the cumulative gap between householders 65 or older and householders under 45 is -0.4, -1.0, -1.0, and -1.4 percentage points at the four levels of aggregation, respectively. Appendix Figure C3 presents the dwelling characteristics of older relative to younger householders.

[37] Appendix Figure C4 presents the dwelling characteristics of high- relative to low-education householders. For instance, from 2005 to 2019, high-education householders live in dwellings built 6.5 years later on average than low-education householders within the same PUMA.

averaged 0.27 percentage points per year—approximately 11 percent of the 2.5 percent average annual shelter inflation of white householders.

Appendix Figure A4 repeats the decomposition of shelter inflation in Section 4.2, but for race. I find that sorting across metropolitan areas explains little of the racial difference in shelter inflation. When every householder in a CBSA experiences the same rent change, cumulative shelter inflation from 2006 to 2019 is only 1.9 percentage points higher for non-white householders. Allowing rent changes to differ across neighborhoods within a CBSA increases the gap to 4.5 percentage points, and allowing rent changes to differ across dwelling types, sizes, and ages within a CBSA increases the gap to 3.9 percentage points. Therefore, non-white householders experience faster-growing rents both because of where they live within a metropolitan area and because of the housing varieties they prefer.[38]

4.4 Comparison with repeat-rent indices

American Housing Survey. As a validation exercise, I compare the ACS rent indices for household groups constructed in Section 3.1 to repeat-rent indices for the same groups estimated using the AHS.[39] Unlike the ACS indices, which adjust for housing quality hedonically, the AHS indices compare the rent of the same unit over time and therefore hold dwelling characteristics fixed. The drawback to the AHS is that its sample size is small, so indices are imprecisely estimated. Furthermore, the AHS only identifies Census region in the public-use survey data, meaning I cannot decompose inflation inequality into the contributions of location versus housing variety.[40]

I estimate the AHS indices with the repeat-sales estimator of Case and Shiller (1987), using a Huber robust regression to limit the influence of extreme outliers. I assign each rent change to the group of the householder living in the unit at the start of the interval. Since the AHS is conducted every two years, Figure 5 compares these indices to two-year averages of the ACS rent indices over the years covered by both surveys. The AHS indices show qualitatively similar patterns to the ACS indices, and the differences between them are not statistically significant. Appendix D provides further detail on the estimation of the repeat-rent indices as well as additional results.

Cotality Multiple Listing Service. Since the AHS does not identify location within a Census region, I additionally estimate repeat-rent indices within metropolitan areas using Cotality MLS

[38] Appendix Figure C5 presents the dwelling characteristics of non-white relative to white householders. For instance, from 2005 to 2019, non-white householders live in dwellings with 0.24 fewer bedrooms on average than white householders, and are 19 percentage points less likely to live in a single-family home.

[39] I do not construct a shelter index from the AHS. The public-use data identify only the Census region of each unit, so imputing rents for owner-occupied units is challenging.

[40] In addition, the AHS panel is not fully continuous, because the Census Bureau redrew the national sample in 1985 and 2015. As of 2025, the AHS has been postponed and is being redesigned.

rental listings. Specifically, I test whether rents grew faster for lower-quality housing units than for higher-quality housing units in the same metropolitan area. To do so, I first regress log rent on unit fixed effects and CBSA-by-year fixed effects, and classify a housing unit as below-median quality if its unit fixed effect is below the median among units in its CBSA. I then regress log rent on unit fixed effects, CBSA-by-year fixed effects, and year fixed effects interacted with an indicator for below-median quality, normalized to zero in 2010. Since the CBSA-by-year fixed effects absorb differences in rent growth across metropolitan areas, this index compares the rents of below- and above-median quality units in the same CBSA and year. Appendix E provides further detail on the estimation of the repeat-rent indices.

Appendix Figure E1 presents the repeat-rent index of below-median relative to above-median quality housing units. I find that rents grew faster for lower-quality units within a metropolitan area. Between 2010 and 2024, rents of below-median quality units rose by 18.9 percent relative to rents of above-median quality units in the same CBSA. These results are consistent with the ACS, in which high-education householders sort into higher-quality housing varieties with slower-growing rents within a metropolitan area.

5 Model

Section 4 shows that high-education householders sort into metropolitan areas with faster-growing rents, but into housing varieties with slower-growing rents. To examine why, I develop a spatial equilibrium model in which households first choose a metropolitan area in which to live and then a housing variety within it. The set-up is a variant of the classic Rosen (1979) and Roback (1982) framework, with two key features: (i) households have general non-homothetic preferences over housing; and (ii) housing markets are segmented by housing variety, where construction primarily builds new housing. I empirically verify both features and derive comparative statics. I then calibrate the model to the 2009 ACS, and simulate changes in household income by metropolitan area and education, and in the number of households by education, through 2019.

5.1 Set-up

Households. A unit measure of households chooses a metropolitan area $c \in \mathscr{C}$ to live in. Each metropolitan area offers housing varieties $v \in \mathscr{V}_c$, where a variety is a combination of PUMA and dwelling age (i.e., decade built).[41] Households differ by education e, and N_e denotes the number of households with education e.

[41] Results are quantitatively similar when I define varieties instead as the combination of PUMA, dwelling type, dwelling size, and dwelling age specified in Section 3.1.

A household i with education e that lives in variety v of metropolitan area c derives utility:

$$U_{iv} \;=\; \log\left(X_{ec} - R_v\right) \;+\; q_{ev} \;+\; \sigma\varepsilon_{iv},$$

where X_{ec} is the household budget, R_v is the rent of variety v, q_{ev} is the quality of the variety, and ε_{iv} is an idiosyncratic taste shock with scale σ. Here, quality captures the characteristics of the dwelling as well as the amenities of its neighborhood and metropolitan area.

Since income enters utility only through the budget left after paying rent, and utility is concave in this budget, preferences are non-homothetic. A dollar of rent "costs" a low-income household more utility than a high-income household, since the marginal utility of the remaining budget falls as the budget rises. As a result, within a metropolitan area, higher-income households are over-represented in more expensive varieties, which are of higher quality. Similarly, higher-income households are over-represented in expensive metropolitan areas, which are more desirable and productive. Since high-education householders have higher incomes, non-homotheticity alone sorts them into expensive metropolitan areas, and into expensive varieties within them. I formalize this intuition in Section 5.2.

I assume that taste shocks follow the generalized extreme value distribution of a nested logit, with metropolitan areas as nests. The nesting parameter $\lambda \in (0,1]$ measures the independence of taste shocks across varieties in the same metropolitan area.[42] Intuitively, the parameter σ governs how much households move across metropolitan areas in response to income and rents, whereas the product $\sigma\lambda$ governs how much they substitute across varieties within a metropolitan area.

A household with education e then chooses variety v in metropolitan area c with probability:

$$P_v(e) \;=\; \underbrace{\frac{\exp\left(\lambda I_{ec}\right)}{\sum_{c'}\exp\left(\lambda I_{ec'}\right)}}_{P_c(e)} \;\times\; \underbrace{\frac{\exp\left(V_{ev}/\lambda\right)}{\sum_{v'\in\mathscr{V}_c}\exp\left(V_{ev'}/\lambda\right)}}_{P_{v|c}(e)}, \tag{3}$$

where

$$V_{ev} \;=\; \frac{\log\left(X_{ec} - R_v\right) + q_{ev}}{\sigma} \qquad \text{and} \qquad I_{ec} \;=\; \log \sum_{v\in\mathscr{V}_c} \exp\left(V_{ev}/\lambda\right)$$

are the systematic utility of variety v and the inclusive value of metropolitan area c, respectively.

Labor demand. A representative firm in each metropolitan area produces a tradable good sold at a common national price. The firm has constant returns to scale in production and exogenous productivity, and takes wages as given. The income of each education group in each metropolitan area is therefore pinned down by the group's marginal product of labor. For tractability, I abstract

[42] The term $1-\lambda^2$ can be interpreted as the correlation of taste shocks within a metropolitan area. When $\lambda = 1$, the model reduces to a standard multinomial logit over all varieties.

from income heterogeneity within an education group. That is, every household in education group e and metropolitan area c earns income w_{ec} and thus has budget $X_{ec} = f(w_{ec})$, where f maps household income into total expenditure.[43]

Housing supply. Housing markets are segmented by variety. The supply of variety v is given by:

$$S_v(R_v) \;=\; H_v \left(R_v / R_v^0\right)^{\gamma_v}, \tag{4}$$

where H_v is a supply shifter, R_v^0 is the rent of the variety in 2009, and γ_v is the supply elasticity. Let γ_c denote the supply elasticity of the housing stock of metropolitan area c, and γ_c^{new} the part of γ_c that comes from new construction. Since varieties with an old dwelling age cannot, by definition, be newly built, I assign the supply response from new construction, γ_c^{new}, to varieties with a new dwelling age and the remainder, $\gamma_c - \gamma_c^{\text{new}}$, to every other variety:

$$\gamma_v \;=\; \begin{cases} \gamma_c^{\text{new}}\, S_c^0 / S_c^{0,\text{new}} & \text{if } v \text{ has a new dwelling age,} \\ (\gamma_c - \gamma_c^{\text{new}})\, S_c^0 / \left(S_c^0 - S_c^{0,\text{new}}\right) & \text{otherwise,} \end{cases} \tag{5}$$

where S_c^0 and $S_c^{0,\text{new}}$ are the total housing stock of metropolitan area c and the stock of its varieties with a new dwelling age in 2009. The stock-weighted mean of γ_v across the varieties of metropolitan area c is therefore γ_c, so the housing stock of the area responds to a uniform increase in its rents with elasticity γ_c. As a result, varieties with a new dwelling age, which high-education householders disproportionately occupy, have a much more elastic supply than other varieties.

Equilibrium. An equilibrium is a vector of rents such that household demand equals housing supply for every variety:

$$\sum_e N_e P_v(e) \;=\; S_v(R_v). \tag{6}$$

5.2 Comparative statics

To clarify the mechanisms behind the model, I consider a restricted version with two education groups, low and high, $e \in \{L, H\}$; two metropolitan areas, $c \in \{A, B\}$; and, in each area, two housing varieties that differ in dwelling age, old and new, $v \in \{o, n\}$. Quality is common across education groups, $q_{ev} = q_{cv}$, and $\lambda = 1$, so choices follow a multinomial logit.

For simplicity, assume education-specific income is the same across metropolitan areas, $X_{ec} =$

[43] The function f is the regression of total expenditure on a natural cubic spline of household income in the CEX (Appendix Figure C1). Using total expenditure rather than income accounts for taxes and transfers.

X_e, with high-education households having higher income, $X_H > X_L$.[44] Assume new housing has higher quality and rent than old housing, $q_{cn} > q_{co}$ and $R_{cn} > R_{co}$. I hold quality and supply shifters fixed.

Decomposing shelter inflation. Denote local rent inflation by $\rho_{cv} = d \log R_{cv}$. Let s_{ec} be the share of group e's initial housing expenditure spent in metropolitan area c, and θ_{ec} the share of its expenditure within that area spent on new housing. Evaluated at these initial shares, local shelter inflation is $\pi_{ec} = (1-\theta_{ec})\rho_{co} + \theta_{ec}\rho_{cn}$. Let $\bar{\rho}_c$ be local rent inflation averaged with the initial housing expenditure of all households. As in Section 4.2, I split shelter inflation into components between and within metropolitan areas:

$$\pi_e = \underbrace{\sum_c s_{ec}\bar{\rho}_c}_{\pi_e^B} + \underbrace{\sum_c s_{ec}(\pi_{ec} - \bar{\rho}_c)}_{\pi_e^W}. \tag{7}$$

The between-area gap, $\pi_H^B - \pi_L^B$, depends on whether high-education households devote more of their housing expenditure than low-education households to the metropolitan area with faster rent growth. The within-area gap depends on the local gap in each metropolitan area:

$$\pi_{Hc} - \pi_{Lc} = (\theta_{Hc} - \theta_{Lc})(\rho_{cn} - \rho_{co}). \tag{8}$$

Each gap is the product of a difference in sorting and a difference in rent growth (Appendix G.1). I first sign the sorting terms and then the rent growth terms.

Proposition 1 (Sorting)**.** *In each metropolitan area, $\theta_{Hc} > \theta_{Lc}$. If every rent in metropolitan area A exceeds every rent in metropolitan area B, then $s_{HA} > s_{LA}$.*

Relative to low-education households, high-education households are more likely to choose expensive housing. For any two housing alternatives j and k with $R_j > R_k$, a higher budget raises the utility of the more expensive alternative relative to the cheaper one:

$$\frac{\partial}{\partial X}\left[\log(X - R_j) - \log(X - R_k)\right] = \frac{R_j - R_k}{(X - R_j)(X - R_k)} > 0. \tag{9}$$

A rent premium therefore costs a high-education household less utility than a low-education household, even when both value quality equally.

Within each metropolitan area, high-education households devote a larger share of their housing expenditure to new housing. If every rent in metropolitan area A exceeds every rent in metro-

[44] With slight abuse of terminology, I call X income rather than budget, since budget is a monotonic function of income.

politan area B, they also devote a larger share of their housing expenditure to metropolitan area A. See Appendix G.2 for the proof.

Proposition 2 (Rent inflation across metropolitan areas)**.** *Consider one housing variety per metropolitan area, and hold the national number of households in each education group fixed. If $\gamma_A, \gamma_B > 0$ and income grows in metropolitan area A alone, with $dX_{eA} \geq 0$ for both groups and a strict increase for at least one, then $\rho_A > 0 > \rho_B$.*

Income growth in metropolitan area A causes households to migrate toward A, which raises housing demand there and lowers it in B. If A is also the expensive metropolitan area, its rent inflation is therefore higher. Since high-education households initially devote more of their housing expenditure to the expensive metropolitan area (Proposition 1), they then experience higher inflation between metropolitan areas. Here, faster income growth in the expensive metropolitan area is an assumption about income shocks rather than a consequence of non-homothetic preferences. See Appendix G.3 for the proof.

Proposition 3 (Rent inflation within a metropolitan area)**.** *Consider the two housing varieties in one metropolitan area and suppress c. Suppose the local number of households in each education group, N_e, grows at a common rate n, and each group's income grows at a common rate g. Let ζ be the initial share of new housing in the local stock and $\bar{\gamma} = \zeta\gamma_n + (1-\zeta)\gamma_o$ its stock-weighted mean supply elasticity, with at least one supply elasticity positive. If household growth and income growth satisfy $n > \bar{\gamma}g \geq 0$, and new housing has at least as elastic a supply as old housing, $\gamma_n \geq \gamma_o$, then rents grow faster for old housing than for new housing: $\rho_o > \rho_n$.*

Household growth in a metropolitan area raises demand for both old and new varieties, whereas income growth shifts demand toward the new variety. Proposition 3 gives a sufficient condition for the force of household growth to outweigh the force of income growth, such that rent inflation is higher for the old variety. Household growth—or equivalently, depreciation of the housing stock—is essential.[45] With no household growth and positive income growth, rents grow faster for the new variety than for the old variety for any admissible supply elasticities. Supply differences alone therefore do not generate higher rent inflation for the old variety. See Appendix G.4 for the proof.

Combined with the sorting in Proposition 1, Propositions (2) and (3) have two implications. First, high-education households experience higher shelter inflation between metropolitan areas if income grows in the more expensive metropolitan area alone. Second, they experience lower shelter inflation within metropolitan areas if new housing is at least as elastically supplied as old

[45] Common household growth is isomorphic to proportional depreciation of the housing stock, as shown in Appendix F.

housing and household growth is sufficiently high relative to income growth. Sorting due to non-homothetic preferences therefore pushes the two components of shelter inflation in opposite directions. The components do not need to offset exactly, and whether their magnitudes are similar is an empirical question.

5.3 Empirical validation

The model in Section 5.1 has two features: (i) non-homothetic preferences sort high-education households into more expensive metropolitan areas and housing varieties; and (ii) construction primarily adds to varieties with a new dwelling age. I empirically verify these features using the ACS.

Sorting by education. High-education householders sort into metropolitan areas with higher rents and faster income growth (Appendix Figure F1). Across metropolitan areas, a 10 percentage point higher share of householders with at least a bachelor's degree in 2009 is associated with 27 percent higher rents in 2009 and 1.6 percentage points faster income growth from 2009 to 2019. Within a metropolitan area, high-education householders sort into newer construction (Appendix Figure C4). For example, from 2009 to 2019, householders with at least a bachelor's degree live in dwellings built 7.4 years later on average than householders with at most a high school degree within the same CBSA.

The model additionally allows quality to differ by education, so that education groups may differ in their tastes for metropolitan areas and housing varieties beyond what their incomes imply. Differences in quality across groups then absorb any sorting that differences in income do not explain. To measure how much sorting income alone explains, I compare the quality of each group with a common quality that rationalizes the total number of households in each variety when all groups share the same quality.[46] If differences in income fully explained sorting, the quality of each group would equal the common quality up to a constant, so the two would move one-for-one across varieties.[47] In practice, the quality of each group moves nearly one-for-one with the common quality (Appendix Figure F4). Consistent with this, the common quality predicts the educational composition of each variety well, even though it targets only the variety's total number of households (Appendix Figure F5).[48] Together, these results suggest that non-homothetic preferences explain the majority of the sorting across education groups.

[46] See Appendix F for more detail on how I compute a common quality.

[47] Quality is only identified up to a constant within each group, since adding the same constant to the quality of every variety does not change the group's choices.

[48] The slope of the actual on the predicted share of a variety's households is 0.60 for householders with at most a high school degree and 0.90 for householders with at least a bachelor's degree (Appendix F).

Supply of new varieties. Equation (5) assumes that new construction adds only to varieties with a new dwelling age. From 2009 to 2019, varieties with a new dwelling age (i.e., built in 2000–2009 or 2010–2019) accounted for all of the growth in the national housing stock, as the stock of varieties with an old dwelling age shrank slightly. The same holds across metropolitan areas. Appendix Figure F2 plots the contribution of each dwelling age to the growth in the housing stock of each CBSA against its total growth. Among CBSAs whose stock grew, the contribution of varieties with a new dwelling age rises one-for-one with total growth, whereas varieties with an old dwelling age contribute almost nothing. Among the CBSAs whose stock shrank, the decline comes entirely from varieties with an old dwelling age. Since high-education householders disproportionately occupy new housing, new construction should disproportionately slow the rent growth they face within a metropolitan area.

5.4 Calibration

I calibrate the model to householders in the 2009 ACS, which covers 431 CBSAs and 8,400 housing varieties.[49] Table 2 lists the parameters of the model and how I calibrate each.

Rents and budgets. I set the rent R_v^0 of each variety to its annualized mean contract rent in the 2009 ACS. To measure budgets, I compute the mean household income w_{ec} of householders in each education group and CBSA, and convert it into total expenditure using the CEX (Appendix Figure C1).

Preference parameters. I set the migration elasticity to $1/\sigma = 4$ to match estimates for the U.S. in Diamond (2016). I set the nesting parameter to $\lambda = 0.4$ to match estimates for the U.S. in Huang (2025). Since I do not estimate migration elasticities, I examine the sensitivity of my results to the parameters in Section 5.5.

Housing supply. I calculate metropolitan-area-level housing supply elasticities γ_c using Census tract-level elasticities from Baum-Snow and Han (2024). Specifically, I average over the Census tracts of each CBSA with weights equal to occupied housing units in 2000.[50] I calculate γ_c^{new} in the same way from their tract-level elasticities of new construction.[51] Housing varieties with a new

[49] I define varieties using the 2000–2010 consistent PUMAs. Dwelling age has eight categories in 2009: built before 1940, each decade from the 1940s to the 1990s, and 2000 to 2009. I exclude mobile homes, as well as the small number of households who live in varieties with hedonic rents greater than their predicted budget. I use the three education groups of Section 2.

[50] CBSAs without any estimate are assigned the average elasticities of the other CBSAs, weighted by occupied housing units in 2000.

[51] I set γ_c^{new} to zero in CBSAs where the average is negative.

dwelling age have a mean supply elasticity of 1.13, compared to 0.24 for other varieties and 0.36 for metropolitan areas as a whole.

Quality. Given these parameters, budgets, and 2009 rents, I choose the quality q_{ev} of each variety for each education group so that the model reproduces the observed share of the group that chooses each variety.[52] I then set the supply shifter H_v equal to the observed stock S_v^0, so that every variety clears at rents in 2009.[53]

Counterfactual. I simulate the change from 2009 to 2019 by changing two primitives, holding all others at their values in 2009. First, productivity changes so that budgets move to $X'_{ec} = f(w'_{ec}/P')$, where w'_{ec} is the mean household income of education group e in CBSA c in 2019, and P' is the change in the CPI for all items less shelter.[54] Second, the number of households N_e in each education group changes as in the ACS. I then solve equation (6) for counterfactual rents.

Shelter inflation. To compare the model with Fact 2, I compute shelter inflation for each education group as in equation (2), weighting the rent change of each variety by the group's housing expenditure in the base year. To separate sorting across and within metropolitan areas, I then replace the rent change of each variety with the expenditure-weighted mean rent change of its CBSA, $\bar{\rho}_c$, as in Section 5.2:

$$\pi_e^B = \sum_v s_{ev} \bar{\rho}_{c(v)}, \qquad \pi_e^W = \pi_e - \pi_e^B, \tag{10}$$

where s_{ev} is the share of the 2009 housing expenditure of group e spent on variety v, and $c(v)$ is the CBSA of variety v. This mirrors the decomposition with CBSA and PUMA-by-variety cells in Section 4.2, except that the model compares two equilibria rather than chaining annual rent changes.

5.5 Results

Figure 8 presents shelter inflation in the model by education group, decomposed into sorting across and within metropolitan areas following equation (10). The model reproduces the offsetting forces of Fact 2. When each household is assigned the rent change of its CBSA, shelter inflation is 27.2 log points for householders with at least a bachelor's degree, compared to 25.7 log points for householders with at most a high school degree, a gap of 1.4 log points. Within metropolitan areas, however, the difference is -1.4 log points. As a result, total shelter inflation is nearly identical for

[52] The few varieties in which no household of a group lives in 2009 get a quality of $-\infty$ for that group.

[53] Appendix F describes the inversion in detail.

[54] I exclude shelter from the deflator, since the marginal product of labor scales with the price of the tradable national good.

the two groups: 26.4 versus 26.5 log points.[55] Over the same period, the corresponding differences in the ACS are 1.5, -1.3, and 0.2 log points, respectively.[56]

Intuitively, since housing is a necessity, higher incomes lead high-education householders to sort differentially into expensive housing. In particular, they choose desirable and productive metropolitan areas, where technological change differentially raises productivity and thus rents. Within an area, however, high-education householders choose newer, nicer housing varieties. New construction directly adds supply to these varieties and slows their rent growth. Preference for housing quality therefore pushes shelter inflation for high-education householders in opposite directions.

Since I do not estimate the migration elasticity or the nesting parameter, Figure 9 presents the education gaps for $1/\sigma$ from 2 to 6 and λ from 0.2 to 0.6. I find qualitatively similar results across parameter specifications, where the across- versus within-city sorting effects on shelter inflation are offsetting for high-education households.

5.6 The microgeography of housing supply and demand

Modeling the microgeography of housing supply and demand matters for distributional welfare evaluation. Low-education householders disproportionately live in varieties with an old dwelling age, whose supply is less elastic, so they bear more of the rent increases that come with positive demand shocks. To quantify how much the microgeography of housing supply and demand can change welfare evaluation, I compute welfare in the model and compare it with a model in which all housing varieties in a metropolitan area have the same supply elasticity.

To evaluate welfare, holding quality at its 2009 level, I compute the expected utility of each education group, $W_e = \sigma \log \sum_c \exp(\lambda I_{ec})$. Since the budget enters utility through the log of non-housing consumption $X_{ec} - R_v$, a change in W_e equals the log change in non-housing consumption in every variety that would leave the group equally well off. Figure 10 presents the change in welfare from 2009 to 2019 by education group, split into the change from higher budgets at 2009 rents and the remainder from changes in rents. Welfare changes by 1.9 log points for householders with at least a bachelor's degree and by -0.3 log points for householders with at most a high school degree, a gap of 2.2 log points. In terms of income, these changes equal a uniform change in 2009 household income of 2.4 and -0.3 percent, respectively, a gap of 2.7 percentage points.[57] Rising

[55] Results are quantitatively similar when I instead define varieties as in Section 3.1, by PUMA, dwelling type, dwelling size, and dwelling age with four categories. With these varieties, the gap between high- and low-education householders is 1.2 log points across metropolitan areas and -2.0 log points within them, for a total of -0.8 log points.

[56] I compute the gaps in the ACS from the indices in Figure 4, cumulating annual shelter inflation from 2009 to 2019. The between gap uses CBSA cells, and the total gap uses PUMA-by-variety cells.

[57] For each education group, I find the percent change in the 2009 income w_{ec} of every metropolitan area that, through the budget $X_{ec} = f(w_{ec})$ and at 2009 rents, changes W_e by as much as the counterfactual. I allow households to sort in response to this change in income.

rents offset much of the gain from rising incomes, especially for low-education householders.

To isolate the role of the microgeography of housing supply, I solve the model again with a single supply elasticity for all varieties in a metropolitan area, $\gamma_v = \gamma_c$. Importantly, the total supply elasticity of each metropolitan area is the same in both cases. Under equation (5), the elasticities of the varieties in metropolitan area c, weighted by their 2009 stock, average to exactly γ_c, so the housing stock of the area responds to a uniform increase in its rents with elasticity γ_c either way.[58] The single elasticity therefore removes only the difference between varieties with a new and an old dwelling age, holding fixed how elastic the housing supply of each metropolitan area is.

With a single elasticity, rising rents lower welfare by less for every education group, and especially for low-education householders, who disproportionately live in varieties with an old dwelling age. From 2009 to 2019, the welfare of low-education householders instead rises by 0.6 log points, rather than falling by 0.3 log points with segmented housing markets. The welfare gap between the two groups therefore narrows from 2.2 to 1.6 log points, and the equivalent gap in income narrows from 2.7 to 2.0 percentage points. A single supply elasticity for all varieties thus understates the welfare gap by education by about 24 percent.

6 Racial inflation inequality and real wages

Lastly, I examine what the differences in shelter inflation documented in Section 4 imply for household welfare. Shelter is the largest item in the consumption basket, with a weight of roughly one-third in the CPI. Therefore, small differences in shelter inflation can cumulate to large differences in the overall price level experienced by a household. To quantify this, I construct race-specific consumer price indices that combine differences in expenditure shares across the full consumption basket with differences in shelter inflation, and use these indices to deflate wages.

6.1 Race-specific price indices

Methodology. To construct race-specific price indices, I follow Jaravel (2026), who builds distributional CPIs by replicating the methodology of the official CPI for groups of households. Let j index CPI items, p_{jm} denote the official BLS price index for item j in month m, and P_m^g denote the price level of group g. Monthly inflation for group g is given by:

$$\frac{P_m^g}{P_{m-1}^g} = \sum_j \phi_{j,m-1}^g \frac{p_{jm}}{p_{j,m-1}}, \qquad \phi_{j,m-1}^g \propto \kappa_{j,m-1} \frac{\eta_{j,m-1}^g}{\eta_{j,m-1}}, \tag{11}$$

[58] The two cases have the same elasticity at 2009 rents. For large changes in rents, the response of the housing stock of a metropolitan area also depends on how its elasticity is distributed across varieties.

where $\kappa_{j,m-1}$ is the BLS relative importance of item j, and $\eta^g_{j,m-1}$ and $\eta_{j,m-1}$ are the shares of item j in the CEX expenditure of group g and of all households, respectively. The group-specific weights scale the official CPI weights by how much more or less group g spends on each item relative to the average household.[59] In equation (11), groups only differ in the composition of their consumption baskets, and every group experiences the same item price index p_{jm}.

To incorporate differences in shelter inflation, I replace the national price indices for rent of primary residence and for owners' equivalent rent with group-specific indices. For the two shelter items and every month m from July of year $\tau-1$ through June of year τ, I set:

$$\frac{p^g_{jm}}{p^g_{j,m-1}} = \frac{p_{jm}}{p_{j,m-1}} \exp\left(\frac{\pi^g_\tau - \pi_\tau}{12}\right), \tag{12}$$

where π^g_τ is the ACS shelter inflation of group g in year τ from equation (2), and π_τ is the same quantity for all households. That is, equation (12) scales the official shelter index by the group's shelter inflation relative to all households in the ACS. Since only the differential enters, the level of each group's shelter index remains anchored to the CPI. I apply the same differential to owners' equivalent rent, since the BLS imputes owners' equivalent rent using the rents of comparable rental units.[60]

Results. Figure 6 presents the price indices for white and non-white households, normalized to 100 in January 2006, with and without the shelter adjustment. With the official shelter index, the price levels of the two groups rose almost identically between January 2006 and December 2019: by 29.6 percent for white households and 29.4 percent for non-white households, or 1.9 and 1.9 percent per year, respectively. Thus, differences in expenditure shares alone leave cumulative inflation of non-white households 0.1 percentage points below that of white households by the end of 2019.

Using group-specific shelter indices reverses this result. From January 2006 to December 2019, the price level of non-white households rose by 30.4 percent, compared to 29.1 percent for white households. This corresponds to 1.9 and 1.8 percent per year, respectively, or a difference of 0.1 percentage points per year.

[59] I map each CEX expenditure category to a CPI item using the BLS crosswalk. As in the CPI, owner-occupied housing enters through owners' equivalent rent.

[60] The ACS compares annual average rents, so its inflation for year τ is centered on the twelve months from July of $\tau-1$ to June of τ. I also leave the weights in equation (11) unchanged. Between weight updates, the BLS carries expenditure shares forward using official item prices, so a group's shelter weight would in principle drift from the official one. However, this drift is negligble since the shares are rebuilt from CEX spending every one to two years.

6.2 Real wage differences

To measure the racial wage gap, I use the CPS Annual Social and Economic Supplement. I limit my sample to prime-age workers who worked full-time and full-year.[61] I then define the nominal log wage gap in year τ as the difference in the survey-weighted mean of log hourly wages between white and non-white workers.

To construct the real wage gap, I deflate each group's wage by its own price index, including the shelter adjustment. Let P_τ^{w} and P_τ^{nw} denote the price levels of white and non-white households in January of year $\tau+1$. The real log wage gap is then the nominal gap plus $\log(P_\tau^{\text{nw}}/P_\tau^{\text{w}})$.[62] Figure 7 presents the nominal and real racial wage gaps. I find that the nominal gap fell from 0.26 in 2005 to 0.22 in 2019, a decline of 4.1 log points. However, since prices rose faster for non-white households, the real gap fell less, to 0.23, a decline of 3.1 log points. Price differences therefore offset 1.0 log points, or approximately 24 percent, of the convergence in nominal wages. Nearly all of this offset is driven by shelter: with the official shelter index, cumulative inflation of the two groups differs by only 0.1 percentage points in 2019.

7 Conclusion

Housing is the single largest expense for most U.S. households, yet representative panel data on rents are scarce. Researchers therefore often rely on cross-sectional rents, which potentially mix true price changes with changes in the quality of the rental stock. This paper validates rent indices constructed from the American Community Survey and uses them to study inflation inequality due to heterogeneous housing consumption. I find that hedonic rent indices built from the ACS closely track the Bureau of Labor Statistics' repeat-rent indices nationally and in the 22 metropolitan areas covered by the BLS. In contrast, raw average rents track the official indices less closely and overstate rent inflation. With hedonic adjustment, the ACS can therefore yield reasonable measures of rent inflation for households that official price indices do not cover.

Using these indices, I document three stylized facts about shelter inflation. First, rent inflation is heterogeneous across metropolitan areas, with average rent inflation from 2006 to 2019 ranging from 1.1 to 4.3 percent per year across large metropolitan areas. Rent inflation is strongly correlated with household income growth: a one percentage point increase in household income growth is associated with 0.41 percentage points higher rent inflation. Deflating local incomes by the national CPI therefore overstates real income growth in the metropolitan areas where incomes grew

[61] That is, workers who are 25 to 54 years old, worked at least 35 usual hours per week and at least 40 weeks, and had positive wage and salary income. I compute the hourly wage as annual wage and salary income divided by the product of weeks worked and usual weekly hours.

[62] Since both price indices equal 100 in January 2006, the real and nominal gaps coincide in 2005 and only diverge as the price levels of the two groups drift apart.

fastest, since rents also grew fastest in those areas.

Second, shelter inflation is nearly identical across education groups, but only because two forces offset. High-education households live in metropolitan areas with faster-growing rents but, within those areas, in housing varieties whose rents grow more slowly. A spatial equilibrium model with non-homothetic preferences and segmented housing markets reproduces this pattern parsimoniously. This highlights how modeling the microgeography of housing supply and demand matters for distributional welfare evaluation. Low-education householders disproportionately live in varieties with an old dwelling age, whose supply is less elastic, so they bear more of the rent increases that come with positive demand shocks.

Third, non-white households experienced shelter inflation that was 0.27 percentage points per year higher than that of white households. Most of this difference is explained by the neighborhoods and housing varieties that non-white households occupy within metropolitan areas, rather than by the metropolitan areas themselves. These differences in shelter inflation have implications for household welfare. Since shelter accounts for a third of the consumption basket, the difference in shelter inflation accounts for the entire racial gap in overall inflation. As a consequence, the convergence of white and non-white wages since 2005 is approximately 24 percent smaller in real terms than in nominal terms. These findings suggest that an official shelter index that assigns every household the same rent inflation can miss important differences in the cost of living across households.

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Table 1: Summary statistics, 2019 ACS

		Race		Education	
	All	White	Non-white	High school or less	Bachelor's or more
Panel A. Households					
Households (millions)	116.0	77.0	39.0	37.3	43.4
Sample households (thousands)	1,203	884	319	372	468
Renters (%)	36	28	53	44	28
Monthly contract rent, renters ($)	1,096	1,128	1,062	880	1,442
Panel B. Dwellings (%)					
Single-family	72	79	60	69	74
Studio or one bedroom	13	11	17	15	12
Two bedrooms	24	22	27	28	20
Three bedrooms	39	41	35	41	36
Four or more bedrooms	24	25	20	16	32
Built before 1950	18	18	17	21	16
Built 1950–1969	22	21	23	25	18
Built 1970–1989	27	27	28	29	25
Built 1990–2009	26	27	25	21	31
Built 2010 or later	7	7	8	4	10

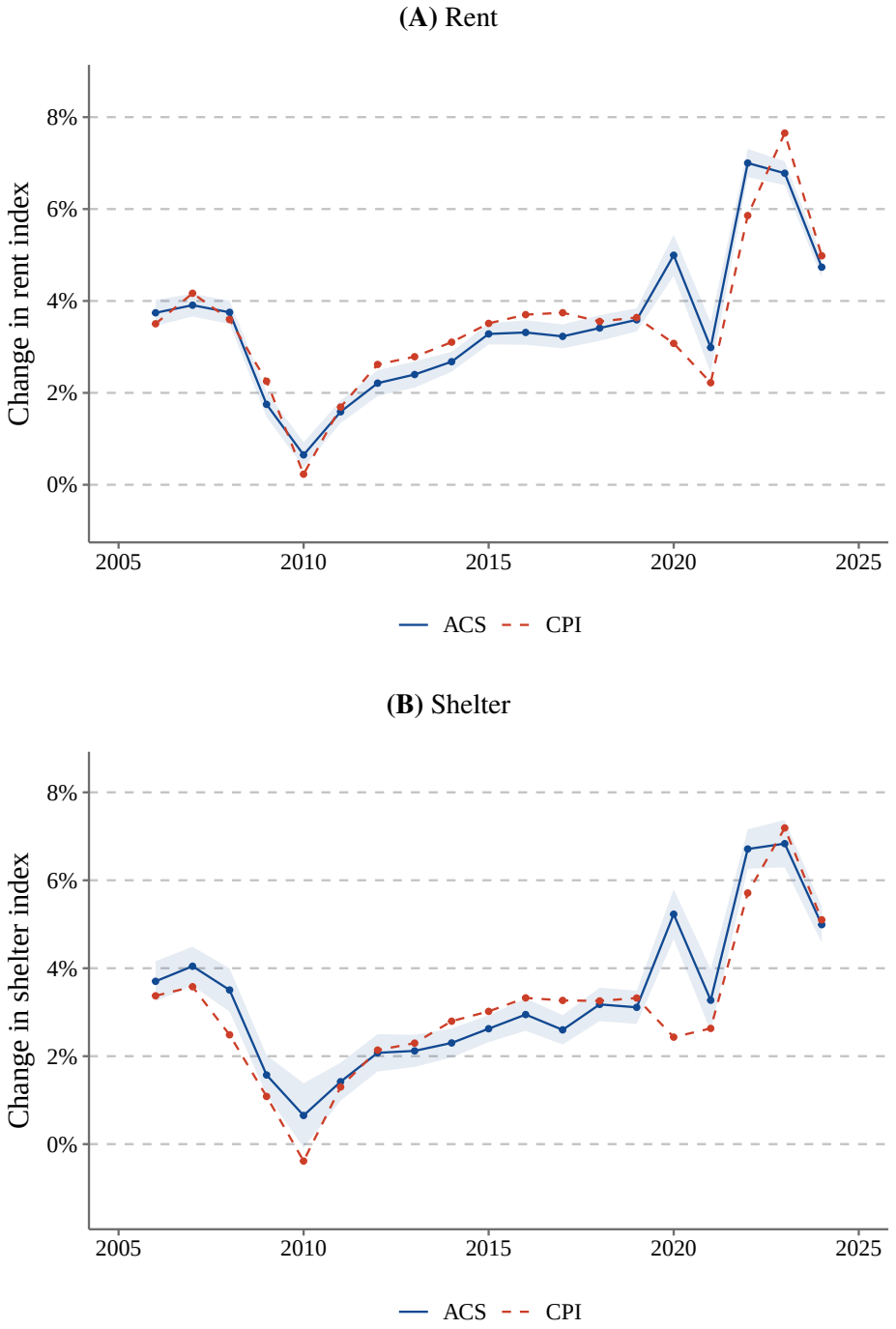
Notes: This table presents summary statistics for householders in the 2019 ACS by the race and education of the householder. The sample excludes group quarters, mobile homes, boats, RVs, and vans. Contract rent is the average monthly contract rent of renter households. Dwelling shares are calculated over households with a non-missing characteristic. The education columns omit householders with some college.

Table 2: Model calibration

Parameter	Description	Value or target	Source
Panel A: Households			
$1/\sigma$	Migration elasticity	4	Diamond (2016)
λ	Nesting parameter	0.4	Huang (2025)
X_{ec}	Budget	Expenditure at mean income by group and CBSA	2009 ACS; CEX
q_{ev}	Quality	Matches each group's choice shares	2009 ACS
N_e	Number of households	Householders in each group	2009 ACS
Panel B: Housing supply			
R_v^0	Rent in 2009	Annualized mean contract rent	2009 ACS
γ_c	Supply elasticity of the housing stock	Housing-weighted mean over tracts	Baum-Snow and Han (2024)
γ_c^{new}	Part of γ_c from new construction	Housing-weighted mean over tracts	Baum-Snow and Han (2024)
H_v	Supply shifter	Observed housing stock S_v^0	2009 ACS

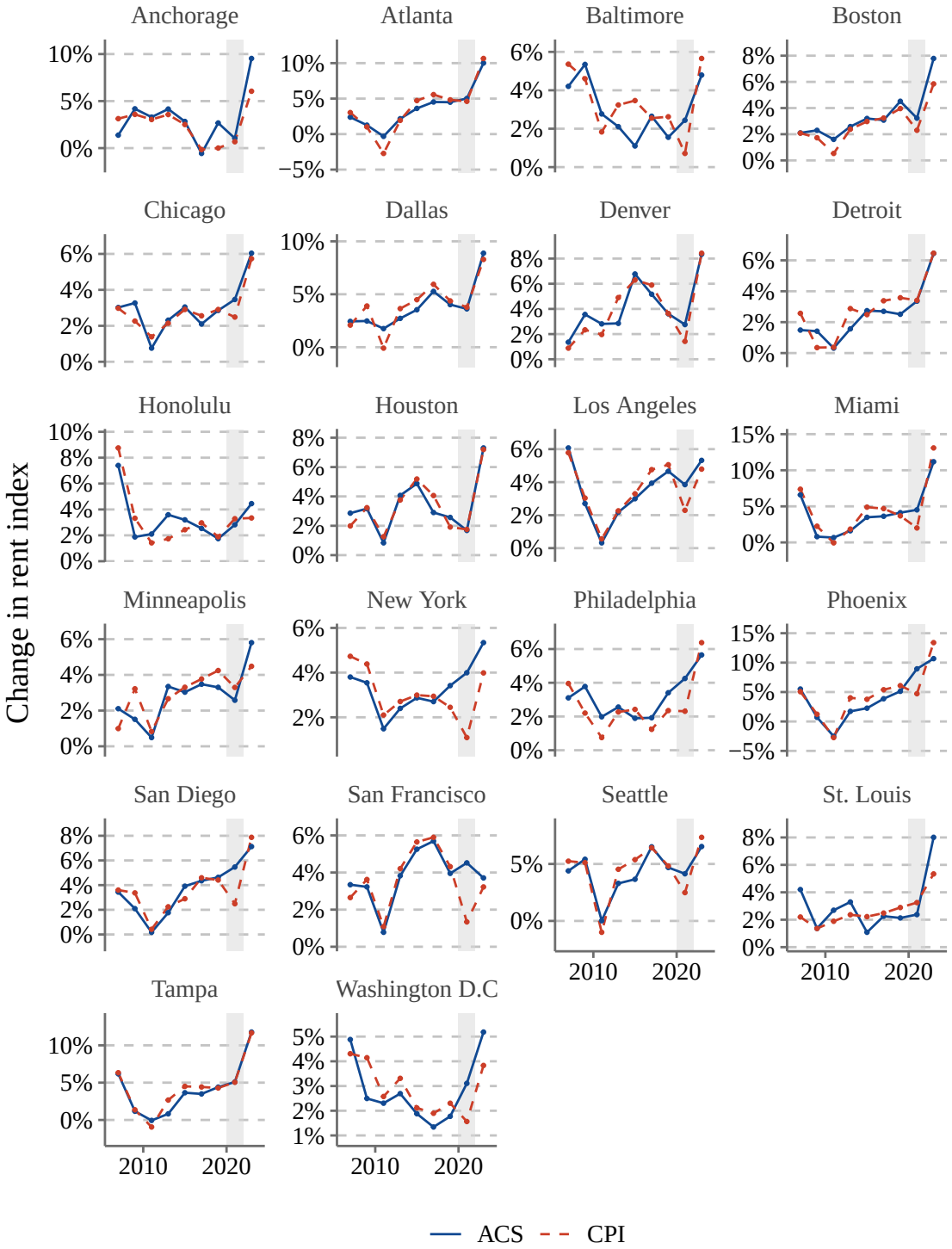
Notes: This table presents the parameters of the model in Section 5.1 and their calibration sources.

Figure 1: Rent and shelter inflation in the ACS and the CPI, 2006–2024



Notes: Panel A of this figure presents annual rent inflation in the ACS from equation (2) and the annual log change in the Rent of Primary Residence CPI. Panel B of this figure presents annual shelter inflation in the ACS and the annual log change in the Shelter CPI. Shaded areas present 95 percent confidence intervals for the ACS, constructed using ACS replicate weights.

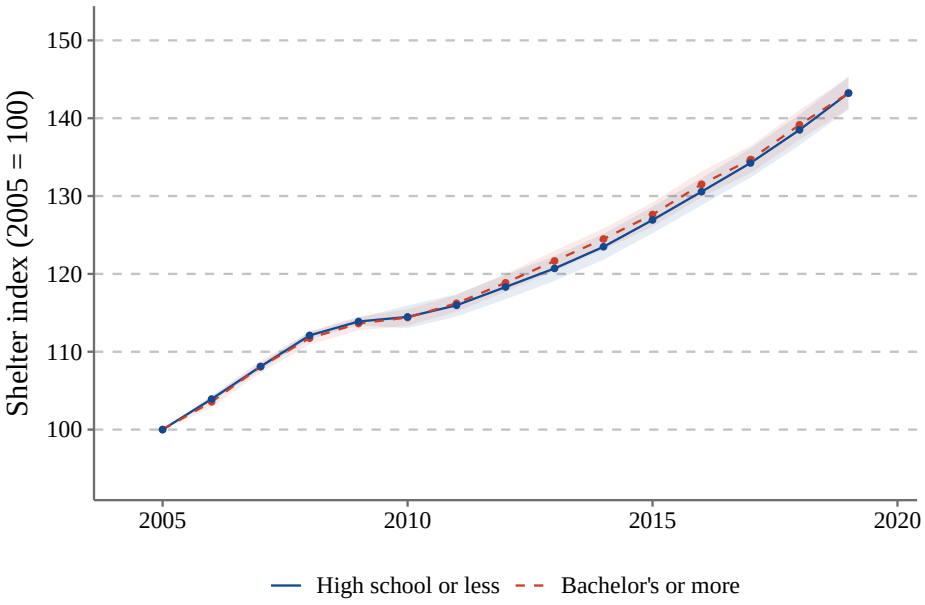
Figure 2: Rent inflation in the ACS and the CPI by metropolitan area, 2006–2023



Notes: This figure presents rent inflation in the ACS from equation (2), computed in each BLS metropolitan area, and the annual log change in the Rent of Primary Residence CPI for the same area. Both series are averaged over two-year periods and plotted at the second year of each period. The shaded area marks the period when data collection was affected by the COVID-19 pandemic.

Figure 3: Shelter index by household group in the ACS, 2005–2019

(A) Education



(B) Age

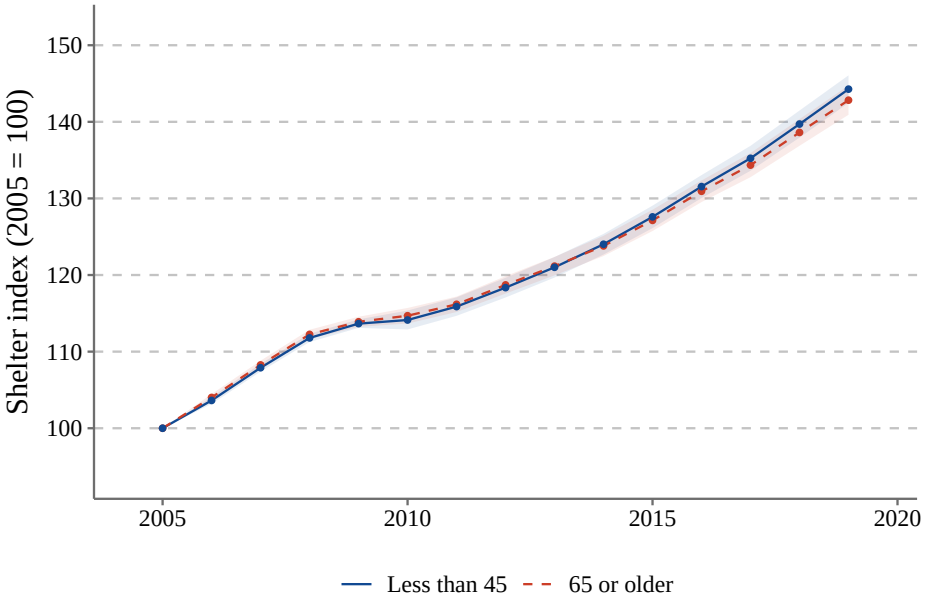
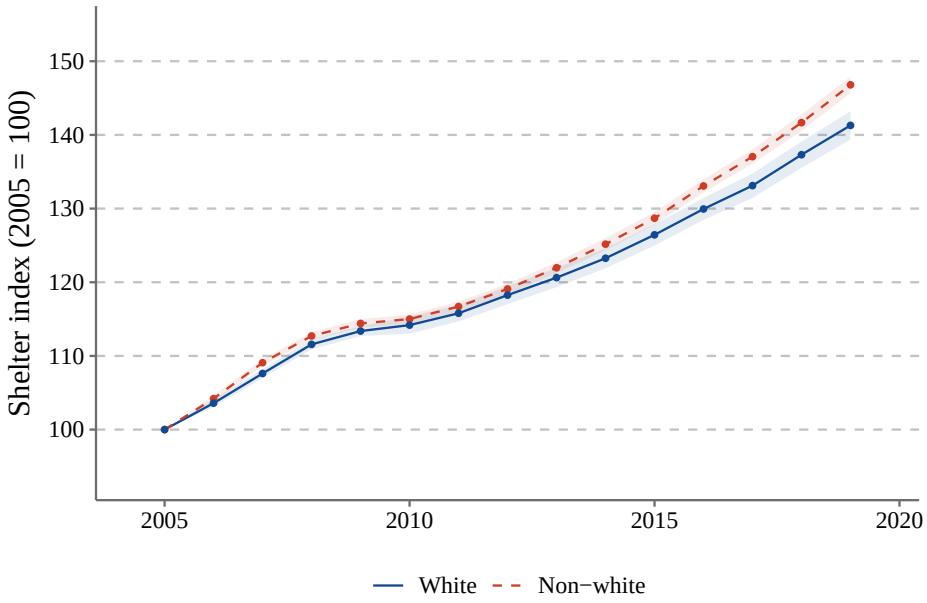


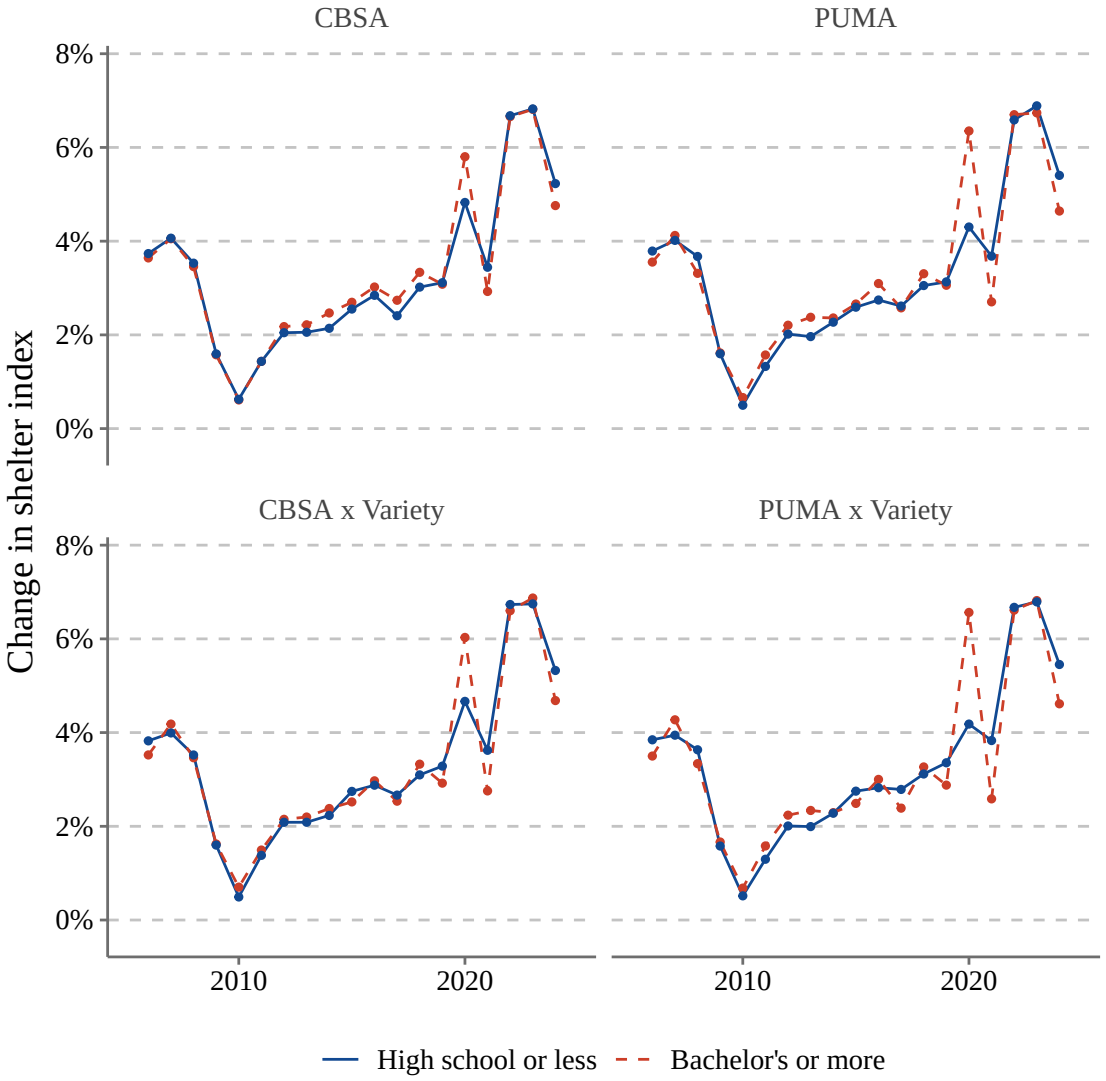
Figure 3: Shelter index by household group in the ACS, 2005–2019 (continued)

(C) Race



Notes: This figure presents the ACS shelter index by the education, age, and race of the householder, chained from the annual inflation rates in equation (2) and normalized to 100 in 2005. Shaded areas present 95 percent confidence intervals constructed using ACS replicate weights. Panels A and B omit the middle education group (some college) and the middle age group (45 to 64).

Figure 4: Shelter inflation by education with coarser aggregation, 2006–2024



Notes: This figure presents annual shelter inflation in the ACS from equation (2) for householders with at most a high school degree and householders with at least a bachelor's degree, varying the level of aggregation at which rental prices may differ: (1) CBSA; (2) PUMA; (3) CBSA interacted with dwelling type, size, and age (CBSA × Variety); and (4) PUMA interacted with dwelling type, size, and age (PUMA × Variety). Each household is assigned the expenditure-weighted average rent change of all households in the level of aggregation.

Figure 5: Gap in rent inflation between household groups in the ACS and the AHS, 2005–2023

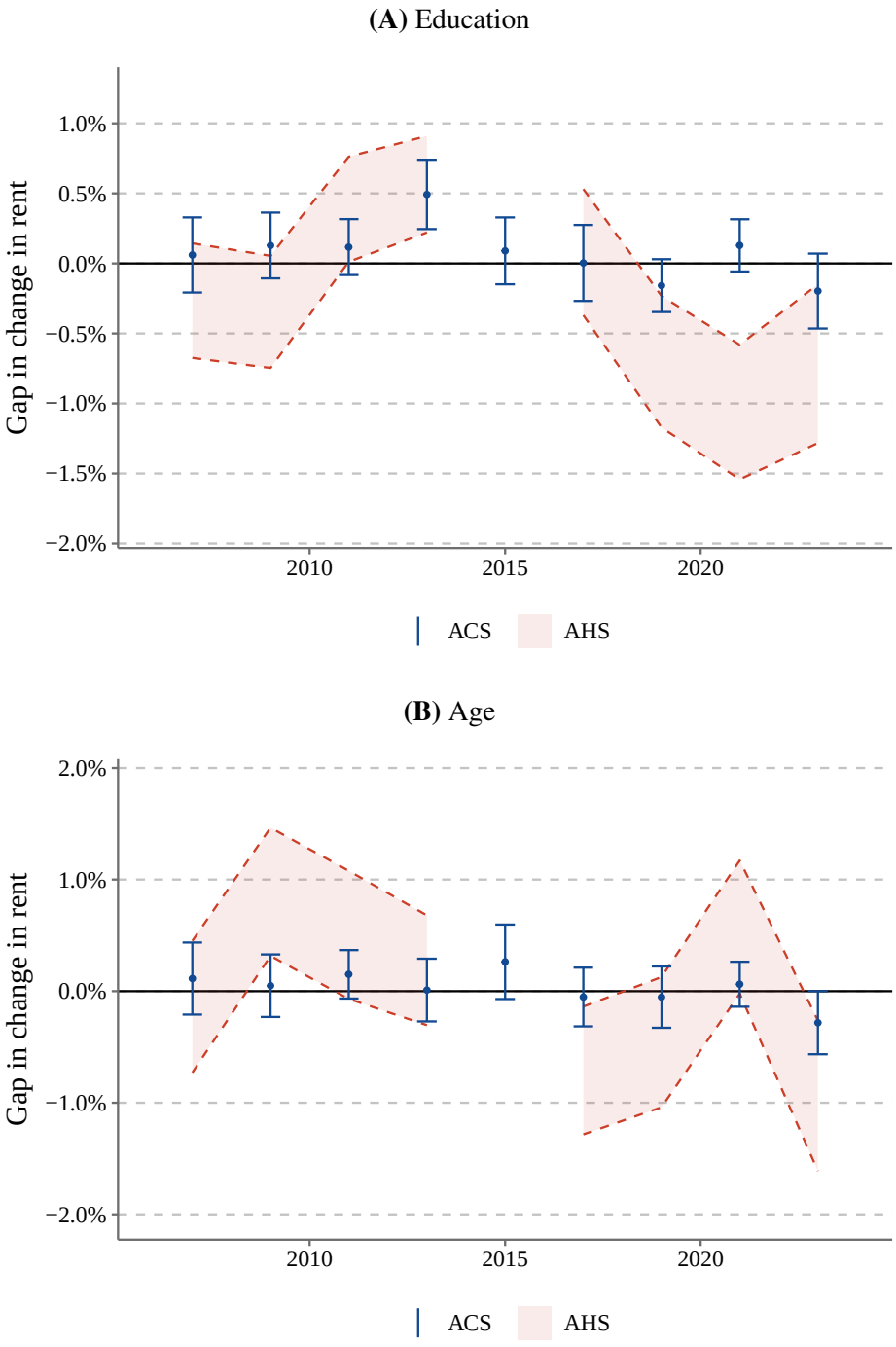
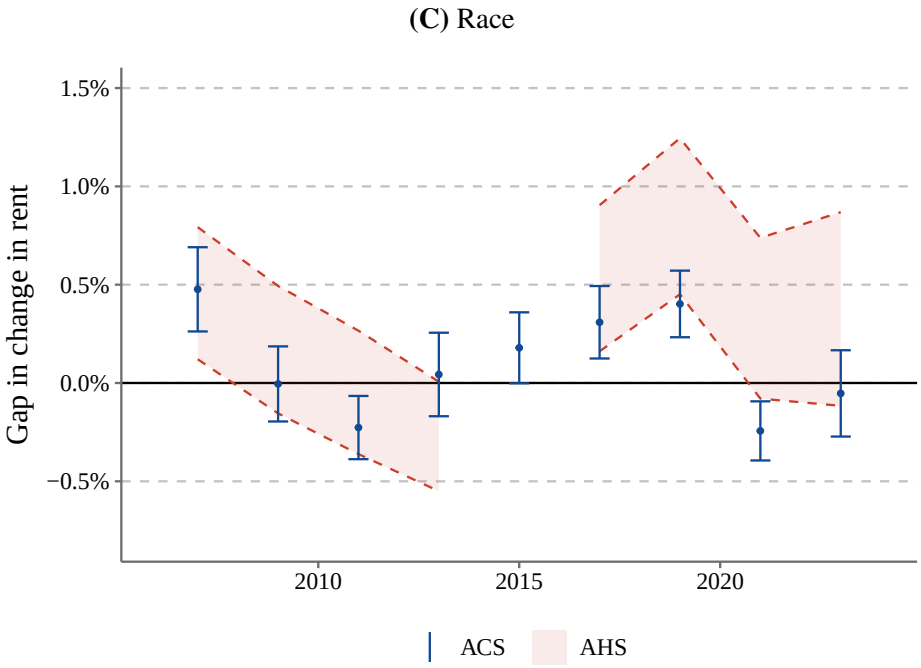
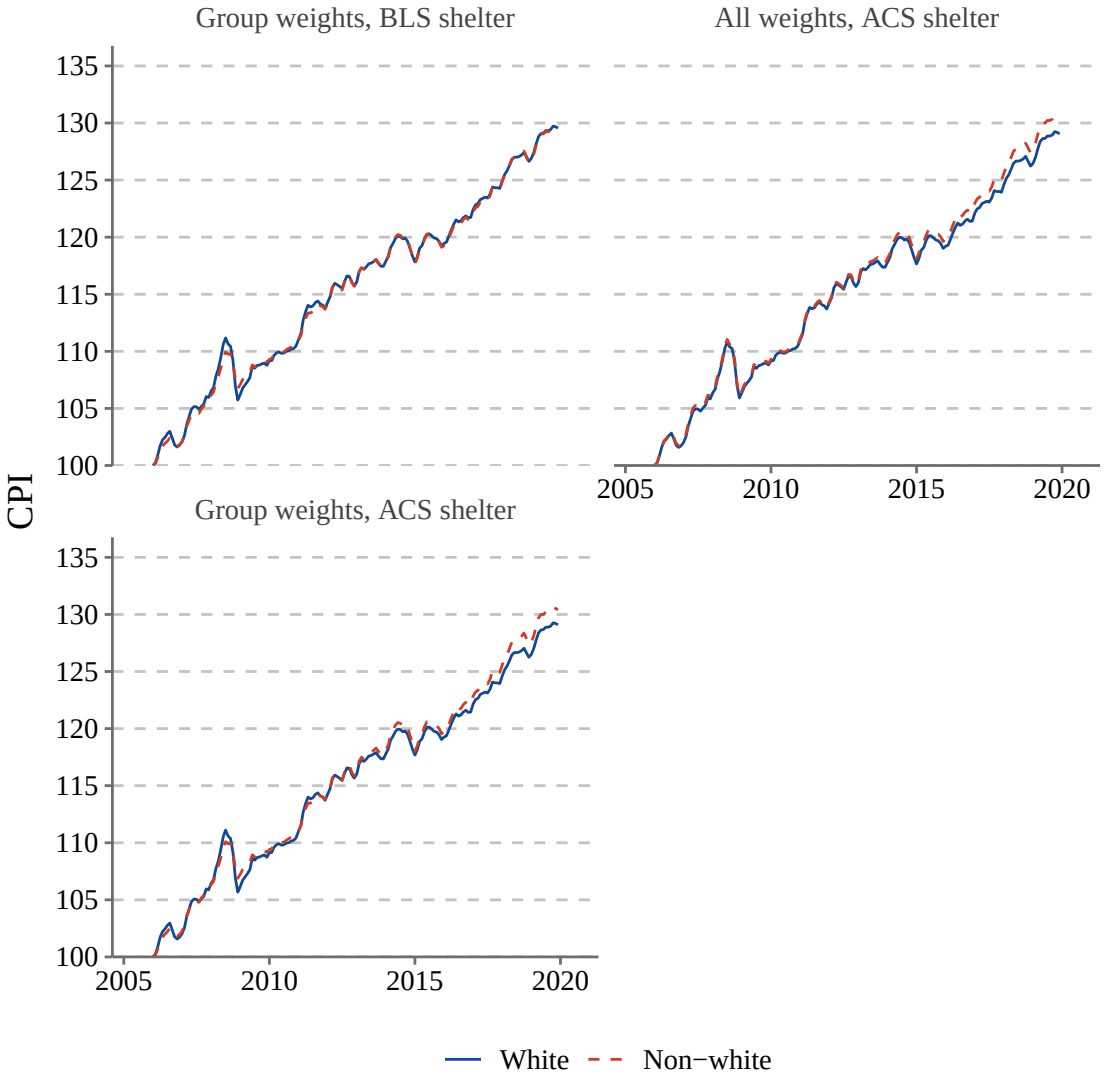


Figure 5: Gap in rent inflation between household groups in the ACS and the AHS, 2005–2023 (continued)



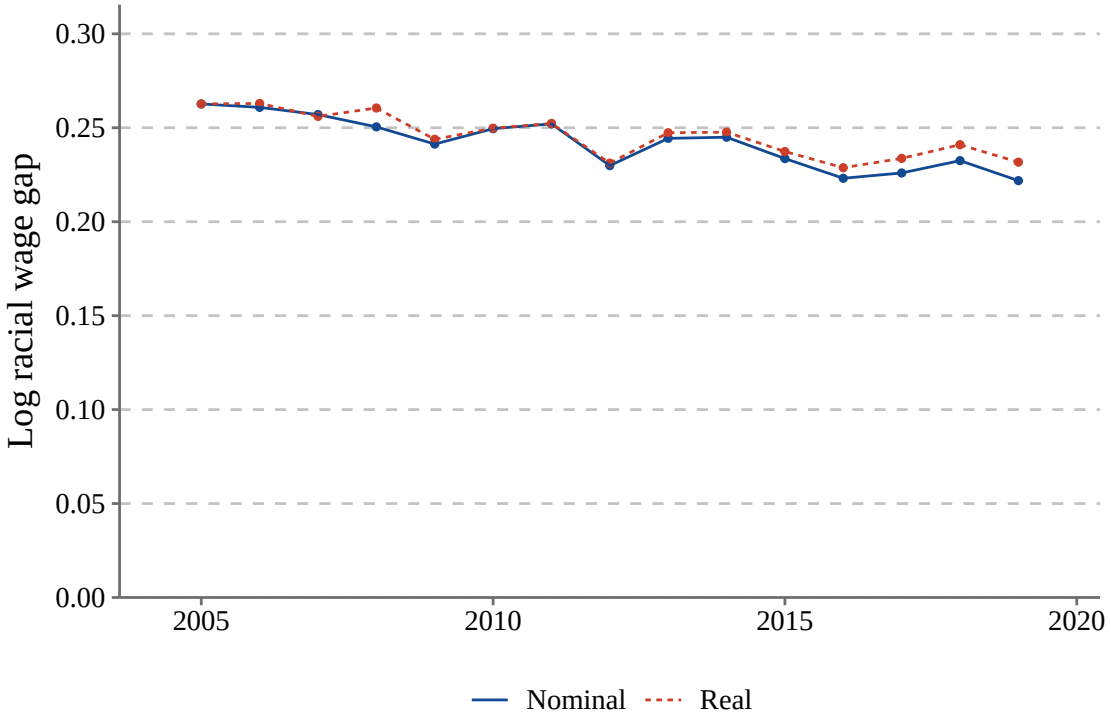
Notes: This figure presents the gap in rent inflation between household groups in the ACS and the AHS. Panel A presents the difference between householders with at least a bachelor's degree and householders with at most a high school degree, Panel B between householders 65 or older and householders under 45, and Panel C between non-white and white householders. Points present the gap in rent inflation in the ACS from equation (2), averaged over two-year periods, and bars present 95 percent confidence intervals constructed using ACS replicate weights. Shaded bands present 95 percent confidence intervals of the same gap in the AHS repeat-rent indices (Appendix D). Since the AHS redrew its sample in 2015, there is no AHS estimate for 2013–2015.

Figure 6: Consumer price indices by race, 2006–2019



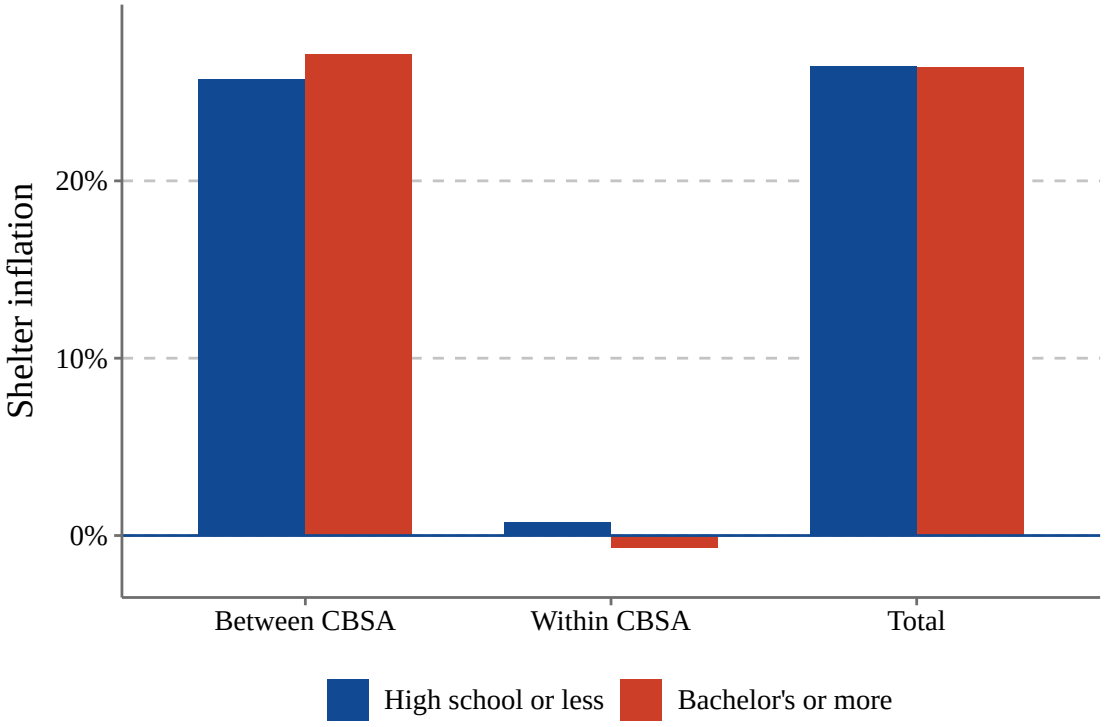
Notes: This figure presents the price index from equation (11) for white and non-white households, normalized to 100 in January 2006. Panels differ in the expenditure weights and the shelter index. "Group weights" are the CEX expenditure shares of each group, and "all weights" are those of all households. "BLS shelter" uses the official price indices, and "ACS shelter" replaces them with the group-specific indices from equation (12).

Figure 7: Racial wage gap in the CPS, nominal and real, 2005–2019



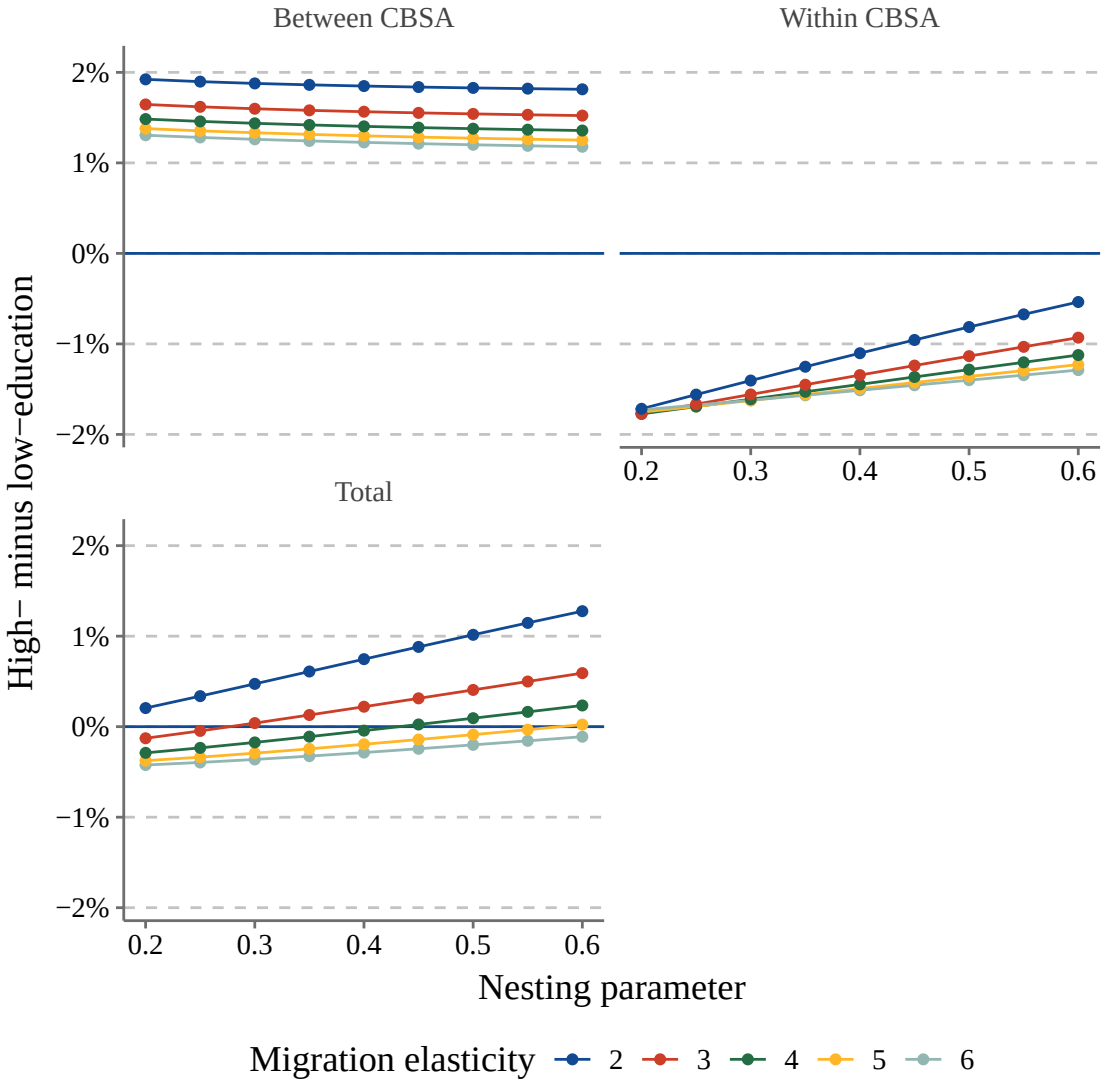
Notes: This figure presents the difference in mean log hourly wages between white and non-white workers in the CPS. The sample includes workers aged 25 to 54 who worked at least 35 usual hours per week and 40 weeks. The real gap adds the log ratio of the non-white to the white price level in January of the following year, using the price indices with the shelter adjustment of equation (12).

Figure 8: Shelter inflation by education in the model, 2009–2019



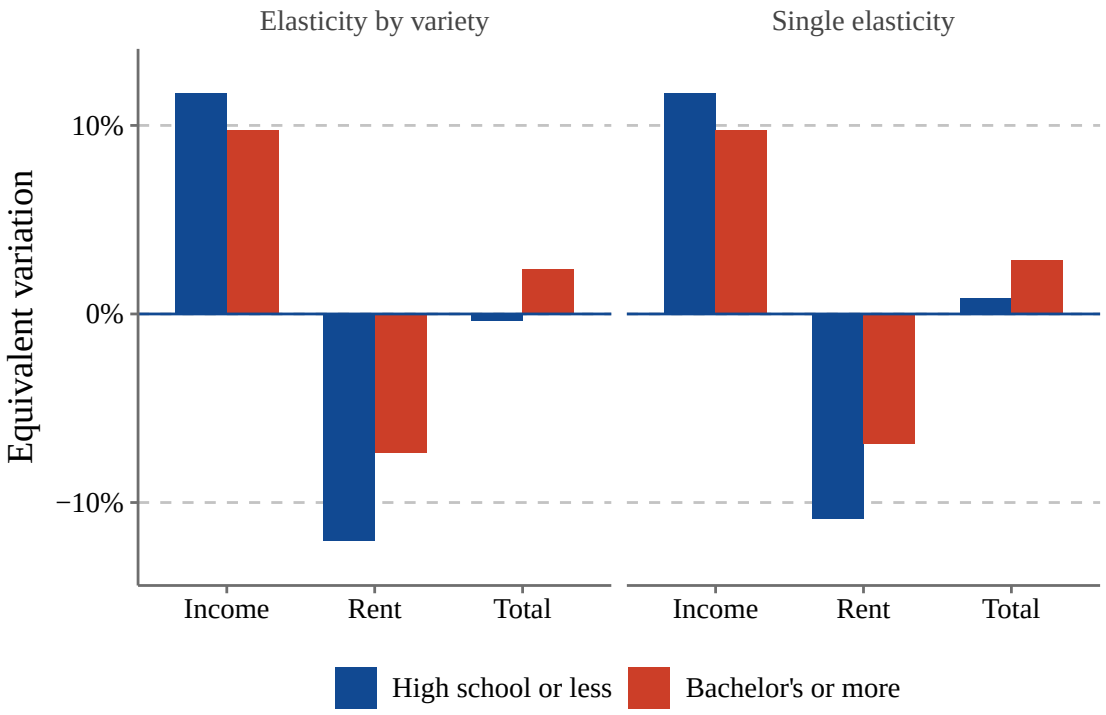
Notes: This figure presents shelter inflation in the model from 2009 to 2019 for householders with at most a high school degree and householders with at least a bachelor's degree. The between term assigns each household the expenditure-weighted mean rent change of its CBSA, and the within term is the remainder (equation (10)).

Figure 9: Education gap in shelter inflation in the model by parameter, 2009–2019



Notes: This figure presents the difference in shelter inflation in the model from 2009 to 2019 between householders with at least a bachelor's degree and householders with at most a high school degree, by migration elasticity $1/\sigma$ and nesting parameter λ. For each combination of parameters, qualities are re-inverted. The panels present the between term, the within term, and the total of equation (10). The main results calibrate $1/\sigma = 4$ and $\lambda = 0.4$.

Figure 10: Welfare by education in the model, 2009–2019

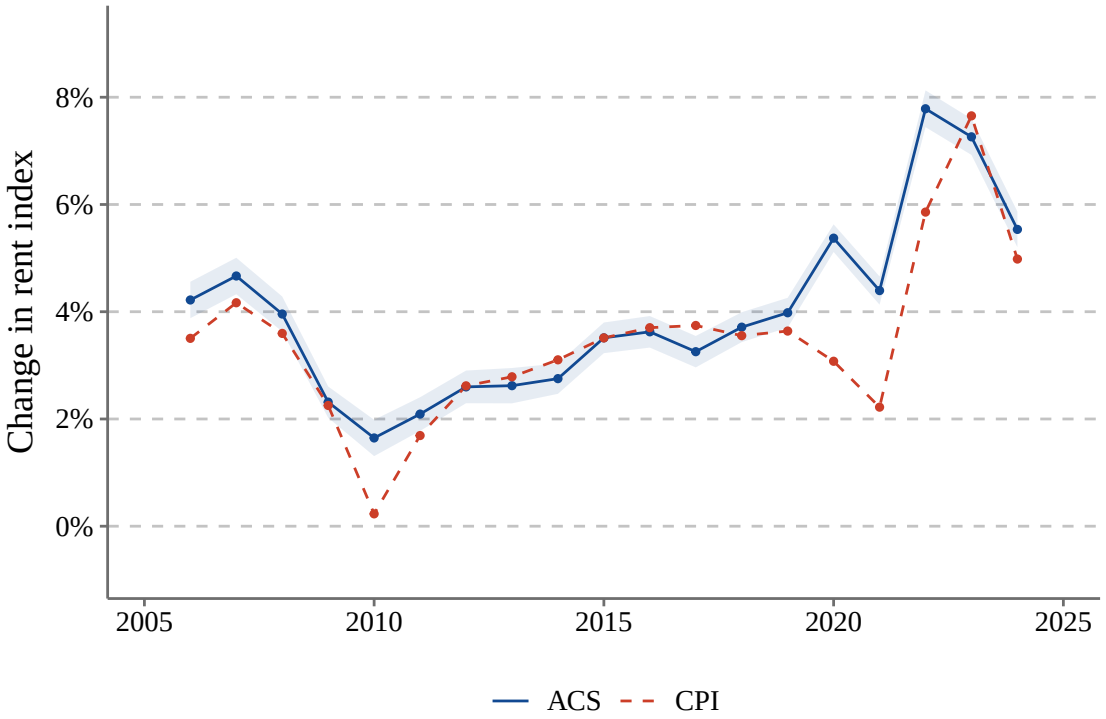


Notes: This figure presents the change in expected utility W_e in the model from 2009 to 2019 for householders with at most a high school degree and householders with at least a bachelor's degree, in terms of equivalent variation. Income is the change in welfare from budgets in 2019 at rents in 2009, and rent is the remainder from counterfactual rents. The left panel sets the supply elasticity of each variety as in equation (5). The right panel instead sets the supply elasticity of every variety to that of its metropolitan area, $\gamma_v = \gamma_c$.

Appendix

A Additional figures

Figure A1: Raw average rents in the ACS and the CPI, 2006–2024



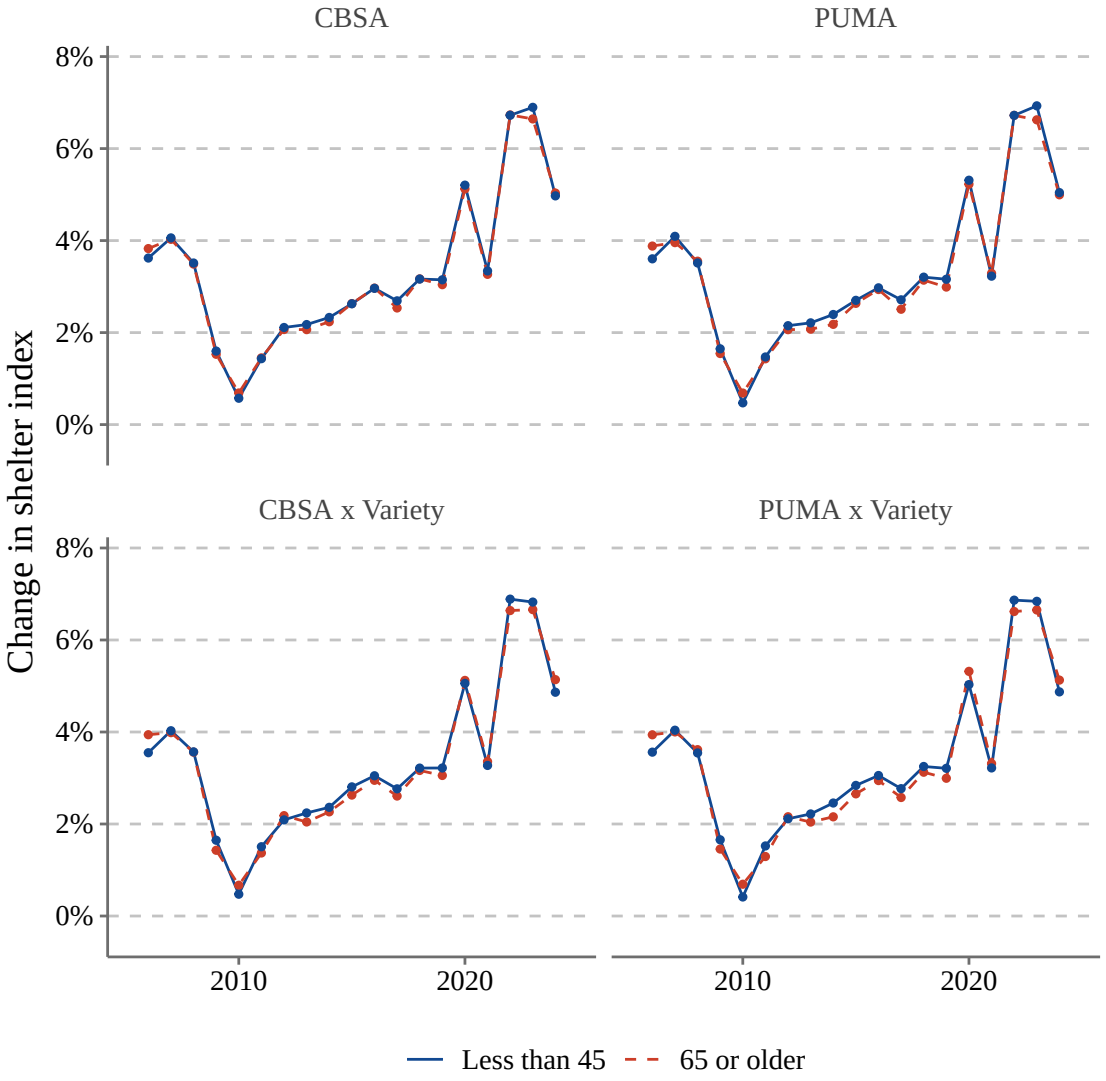
Notes: This figure presents the annual log change in the mean contract rent of renter households in the ACS and the annual log change in the Rent of Primary Residence CPI. Shaded areas present 95 percent confidence intervals constructed using ACS replicate weights.

Figure A2: Rent inflation and income growth across metropolitan areas, 2006–2019



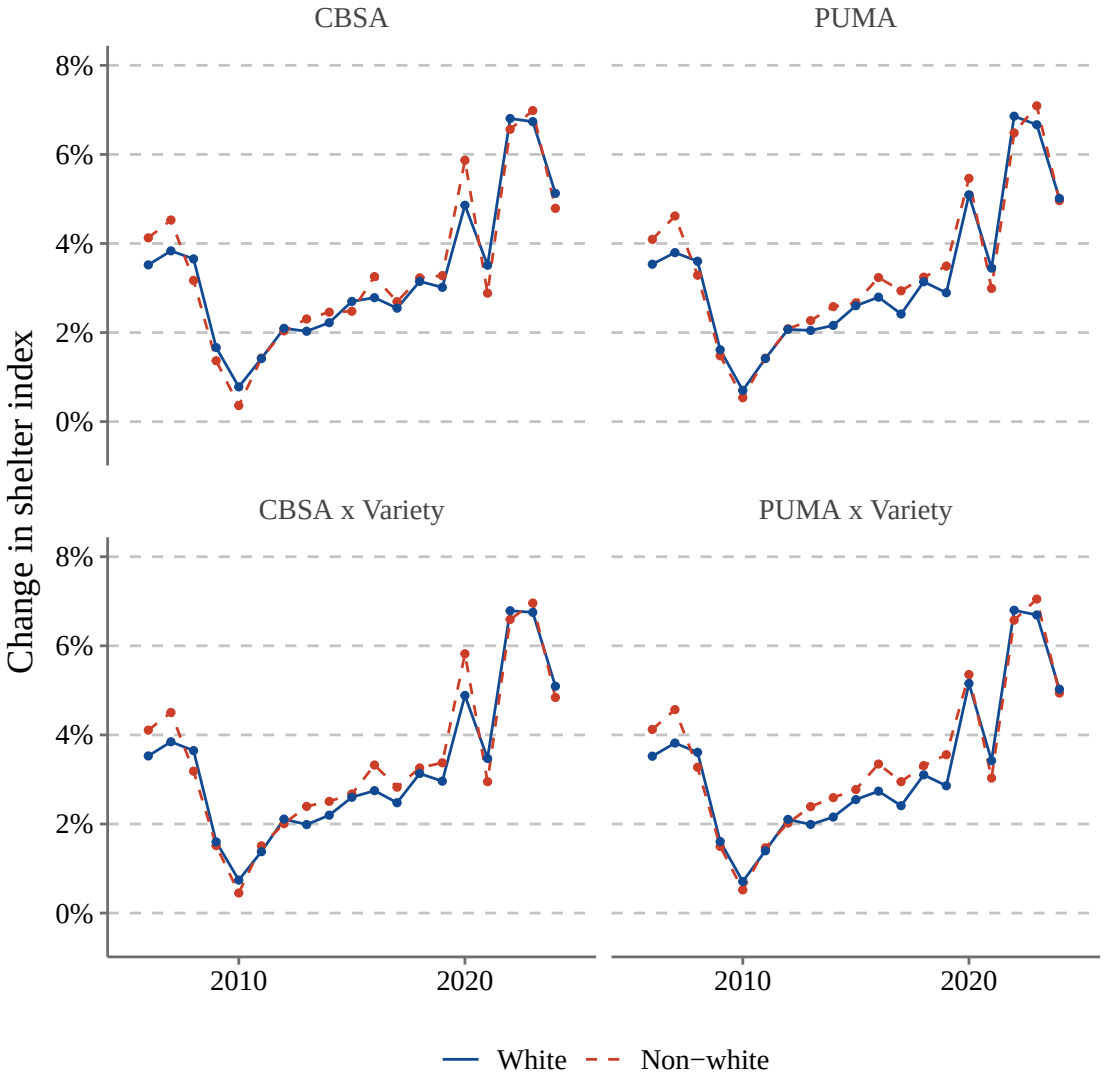
Notes: This figure presents average annual rent inflation by average annual household income growth from 2006 to 2019 for the 100 largest CBSAs by number of households in 2005. Rent inflation is measured in the ACS using equation (2). Income growth is the average annual log change in the mean household income for the CBSA. The dashed line is the 45-degree line. Labels mark the five CBSAs with the fastest and the slowest rent inflation.

Figure A3: Shelter inflation by age with coarser aggregation, 2006–2024



Notes: This figure presents annual shelter inflation in the ACS from equation (2) for householders under 45 and householders 65 or older, varying the level of aggregation at which rental prices may differ: (1) CBSA; (2) PUMA; (3) CBSA interacted with dwelling type, size, and age (CBSA × Variety); and (4) PUMA interacted with dwelling type, size, and age (PUMA × Variety). Each household is assigned the expenditure-weighted average rent change of all households in the level of aggregation.

Figure A4: Shelter inflation by race with coarser aggregation, 2006–2024



Notes: This figure presents annual shelter inflation in the ACS from equation (2) for white and non-white householders, varying the level of aggregation at which rental prices may differ: (1) CBSA; (2) PUMA; (3) CBSA interacted with dwelling type, size, and age (CBSA × Variety); and (4) PUMA interacted with dwelling type, size, and age (PUMA × Variety). Each household is assigned the expenditure-weighted average rent change of all households in the level of aggregation.

Figure A5: Annual rent inflation by household group in the ACS, 2006–2024

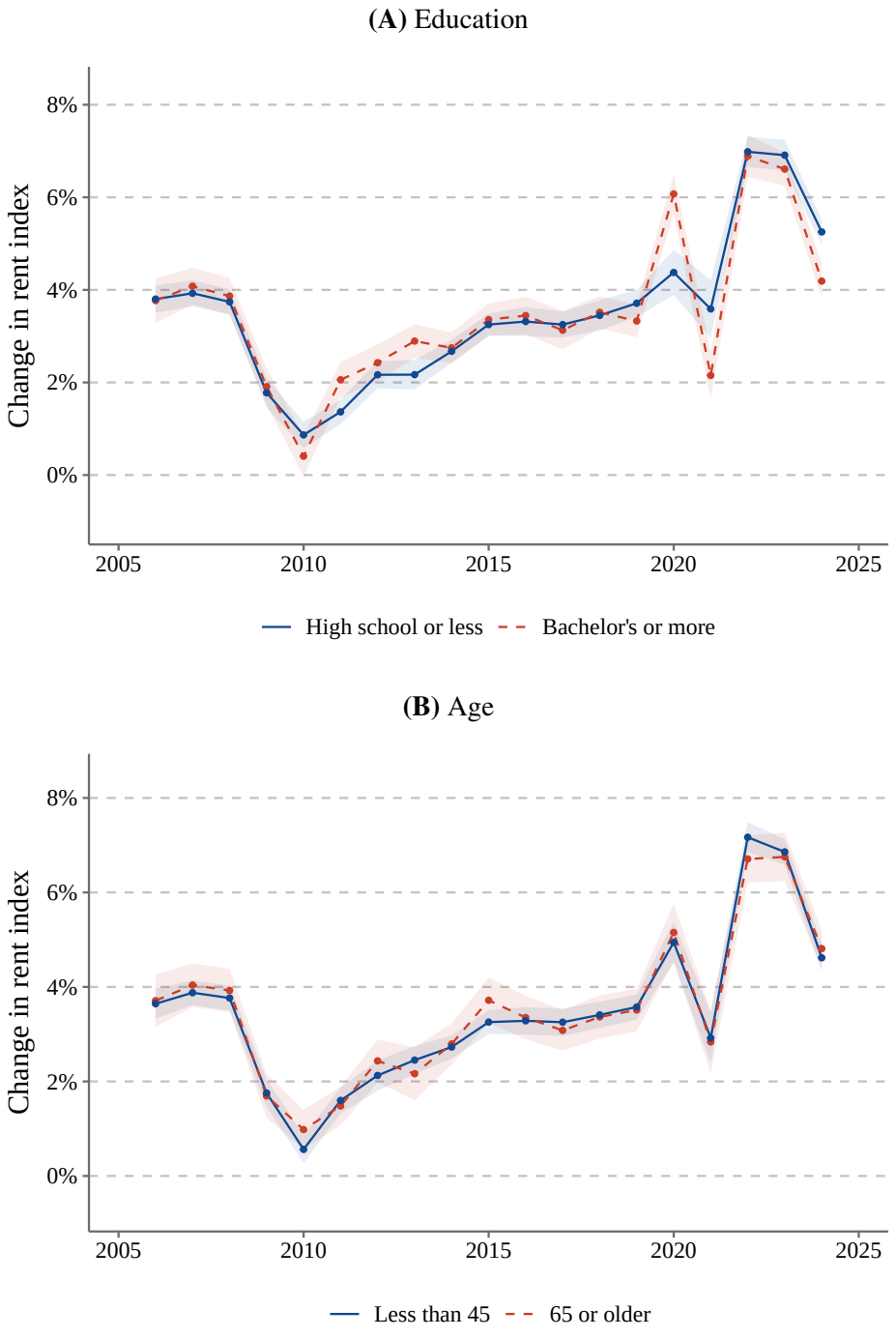
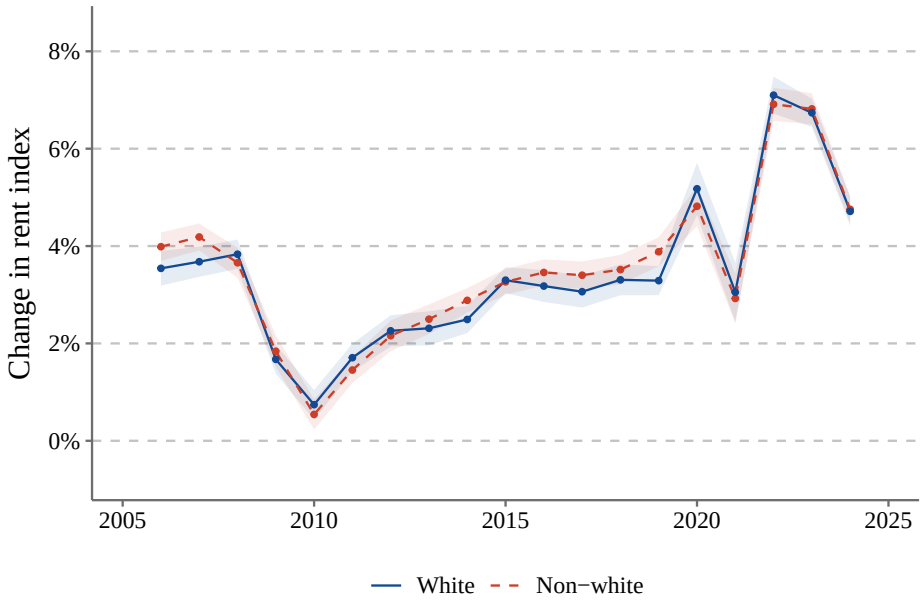


Figure A5: Annual rent inflation by household group in the ACS, 2006–2024 (continued)

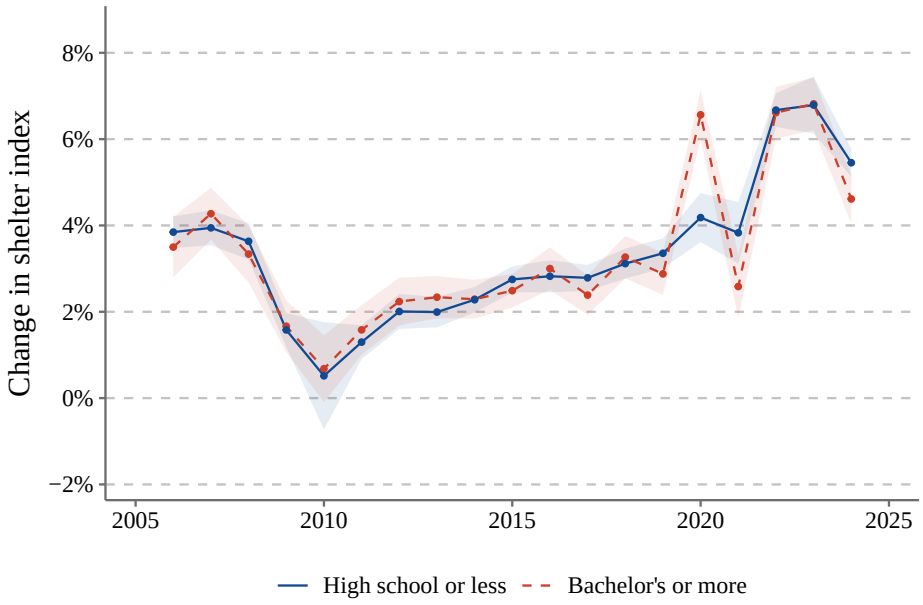
(C) Race



Notes: This figure presents annual rent inflation in the ACS from equation (2) by the education, age, and race of the householder. Shaded areas present 95 percent confidence intervals constructed using ACS replicate weights. Panels A and B omit the middle education group (some college) and the middle age group (45 to 64).

Figure A6: Annual shelter inflation by household group in the ACS, 2006–2024

(A) Education



(B) Age

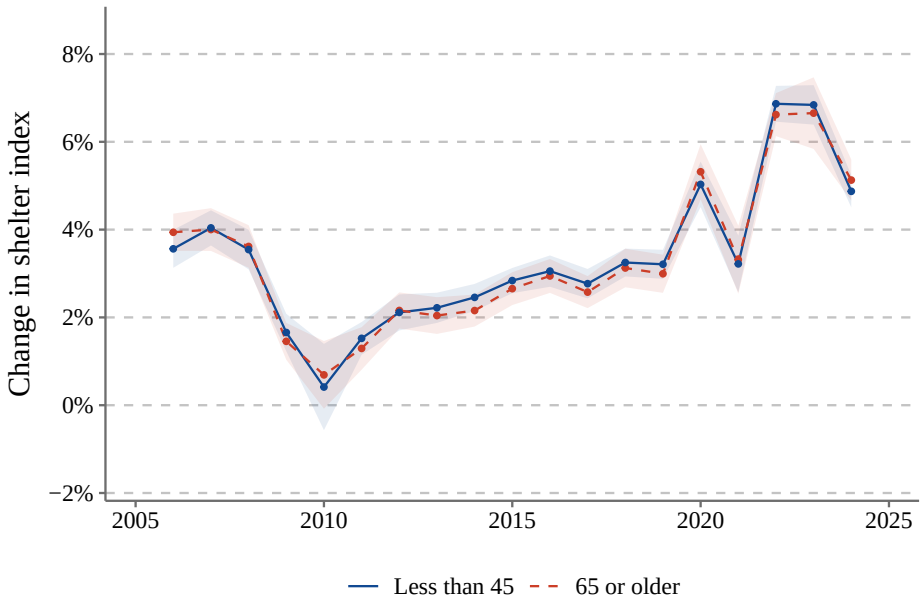
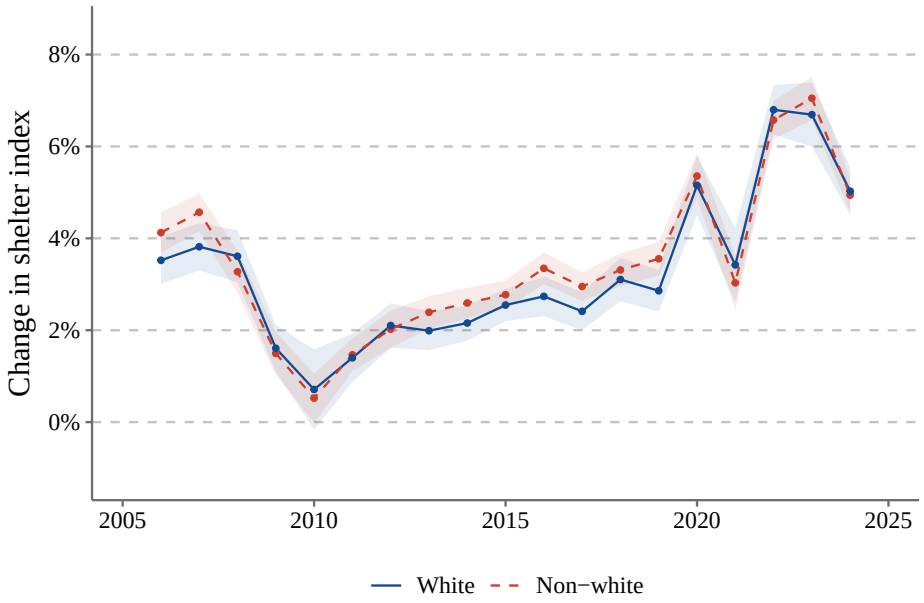


Figure A6: Annual shelter inflation by household group in the ACS, 2006–2024 (continued)

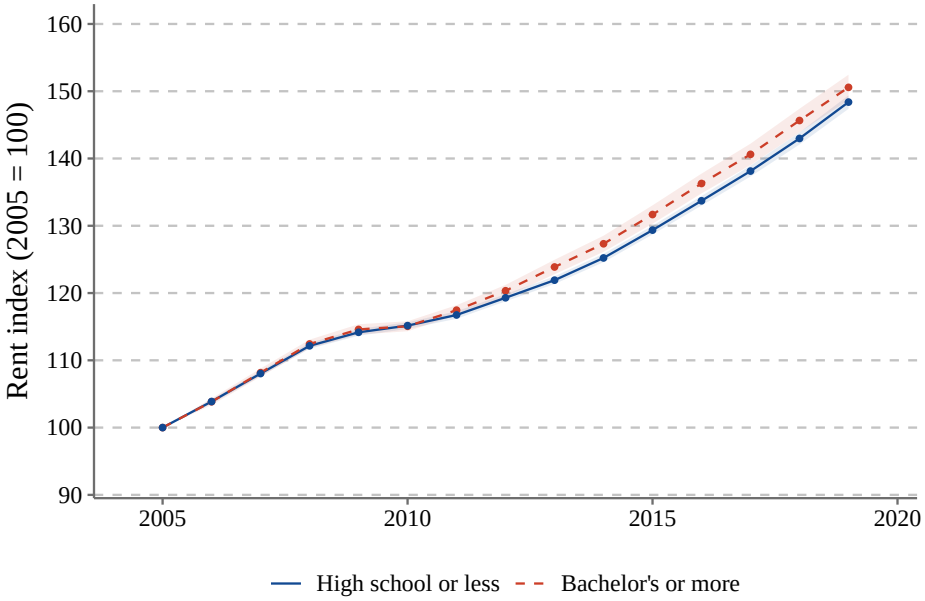
(C) Race



Notes: This figure presents annual shelter inflation in the ACS from equation (2) by the education, age, and race of the householder. Shaded areas present 95 percent confidence intervals constructed using ACS replicate weights. Panels A and B omit the middle education group (some college) and the middle age group (45 to 64).

Figure A7: Rent index by household group in the ACS, 2005–2019

(A) Education



(B) Age

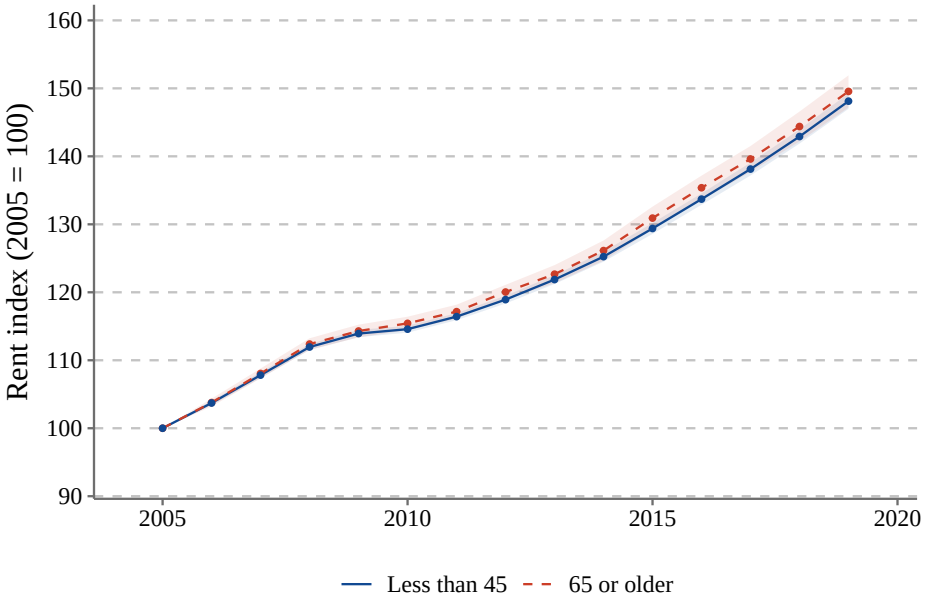
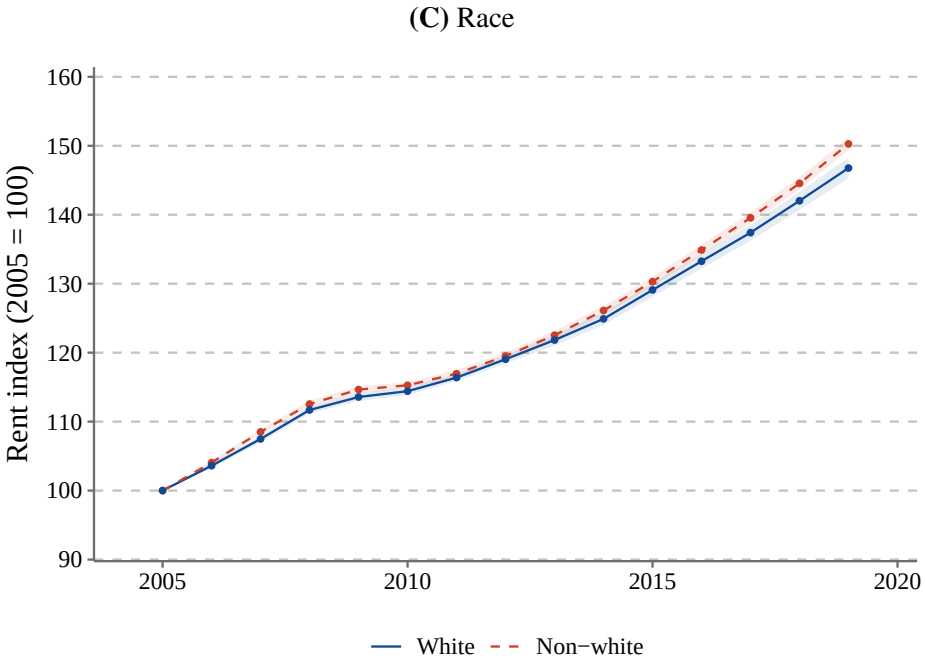


Figure A7: Rent index by household group in the ACS, 2005–2019 (continued)



Notes: This figure presents the ACS rent index by the education, age, and race of the householder, chained from the annual inflation rates in equation (2) and normalized to 100 in 2005. Shaded areas present 95 percent confidence intervals constructed using ACS replicate weights. Panels A and B omit the middle education group (some college) and the middle age group (45 to 64).

B Studies that use Census and ACS rents

Because repeat-rent panels are scarce, much of what economists know about how rents change over time comes from the rents reported in the decennial Census and the ACS. In this appendix, I review studies that use these rents to measure changes in rental prices, focusing on those published after 2010 in the *American Economic Review*, the *American Economic Journals*, *Econometrica*, the *Journal of Political Economy*, the *Quarterly Journal of Economics*, the *Review of Economic Studies*, and the *Review of Economics and Statistics*.[63] Table B1 groups these studies by the comparison of rents they make: (1) over time; and (2) before and after local shocks and policies. This review is not intended to be comprehensive.

Table B1: Studies that use Census or ACS rents

Study	Rent data	Level of aggregation
Panel A: Rents across time		
Gyourko, Mayer, and Sinai (2013)	1970–2000 Census	MSA, place
Moretti (2013)	1980–2000 Census	MSA
Diamond (2016)	1980–2000 Census	MSA
Hsieh and Moretti (2019)	1960–1970 Census; 2008–2009 ACS	MSA
Colas and Hutchinson (2021)	1980–2000 Census; 2005–2007 ACS	CBSA
Piyapromdee (2021)	1990–2000 Census; 2005–2007 ACS	MSA
Su (2022)	1990 Census; 2007–2011 ACS	Tract
Hornbeck and Moretti (2024)	1980–2000 Census; 2008–2012 ACS	MSA
Panel B: Rents and local shocks		
Davis (2011)	1990–2000 Census	Block
Boustan (2012)	1970–1980 Census	Block
Busso, Gregory, and Kline (2013)	1980–2000 Census	Tract
Bento, Freedman, and Lang (2015)	1990–2000 Census	Tract
Suárez Serrato and Zidar (2016)	1980–2000 Census; 2009 ACS	County group
Feler and Senses (2017)	1990–2000 Census; 2005–2009 ACS	Commuting zone
Allcott and Keniston (2018)	1940, 1970–2000 Census; 2005–2014 ACS	County
Bartik et al. (2019)	2000 Census; 2009–2013 ACS	County
Keiser and Shapiro (2019)	1970–2000 Census	Tract
Aaronson, Hartley, and Mazumder (2021)	1930–2000 Census; ACS	Tract, block group
Zabek (2024)	1980–2000 Census; 2006–2008 ACS	Commuting zone

Notes: This table presents the studies reviewed in Appendix B. Rent data lists the years of the decennial Census or the ACS from which each study draws rents. Level of aggregation is the geographic unit at which each study compares rents over time.

[63] I include studies that use Census or ACS rents as an outcome or as a price in a structural model. Some of these studies hedonically adjust rents for changes in the dwelling characteristics of the rental stock (e.g., Busso, Gregory, and Kline, 2013; Suárez Serrato and Zidar, 2016; Colas and Hutchinson, 2021).

C ACS index construction

C.1 Hedonic regressions

Equation (1) restricts the log rent of a variety to be additive in dwelling type, size, and age within each PUMA. As a comparison, I estimate a fully saturated specification with a fixed effect for each variety (i.e., each combination of PUMA, dwelling type, dwelling size, and dwelling age). The squared correlation between hedonic and actual rents is 0.60 for the saturated specification, compared to 0.56 for equation (1). Therefore, the saturated specification fits the data only slightly better. However, since the saturated specification identifies each variety's rent only from the renters in that variety, its hedonic rents, and in particular their annual changes, are noisy.

Because the fixed effects in equation (1) are only identified up to a normalization within each PUMA, I only predict rents for varieties whose hedonic rent does not depend on that normalization, given the combinations of type, size, and age observed among renters in the PUMA. These can include varieties with no renters (e.g., some owner-occupied varieties) when the rental varieties observed in the PUMA pin down the relevant combination of fixed effects. This issue is analogous to disconnected fixed effects in matched employer-employee data, where worker and firm effects are only identified within groups of workers and firms connected by worker mobility (Abowd, Creecy, and Kramarz, 2002).

C.2 Expenditure shares

The shelter index weights each variety by the hedonic rent of its households, thereby valuing owner-occupied dwellings at the rent of comparable rental units. As a validation exercise, I benchmark the shelter shares of total expenditure implied by these weights to shelter shares calculated using the CEX. For each year from 2005 to 2024, I divide the survey-weighted shelter expenditure of renters and owners in the CEX by their survey-weighted total expenditure. Shelter expenditure is rent paid for renters and owners' equivalent rent for owners.

Since the ACS does not record expenditure, I predict each household's total expenditure from its income. More specifically, for each year, I regress CEX total expenditure on a natural cubic spline of household income before taxes with knots at the quintiles of income.[64] I then predict the total expenditure of each ACS household from its household income, using the regression for the same year. Figure C1 shows that the fitted values closely track average total expenditure by income percentile. Total expenditure is greater than income for low-income households and less than income for high-income households, likely due to transfers and taxes. For each year and tenure, the ACS shelter share is the survey-weighted sum of annual hedonic rents divided by the

[64] In both surveys, I set negative household income to zero.

survey-weighted sum of predicted total expenditure.

Figure C2 presents the shelter shares in the ACS and the CEX. For renters, I find that the ACS share of rent in total expenditure almost exactly matches the CEX, averaging 26 percent in the ACS and 26 percent in the CEX from 2005 to 2024. In contrast, for owners, the ACS hedonic rent is a smaller share of expenditure than owners' equivalent rent in the CEX (21 versus 30 percent). One explanation is that owners in the CEX systematically overestimate the rent that their home would fetch. Consistent with this explanation, Diamond and Diamond (2024) find that rent-to-price ratios of owner-occupied homes are lower when rents are imputed using housing characteristics in the AHS than when owners estimate them in the CEX.

C.3 Dwelling characteristics by household group

To measure how the dwellings of different household groups differ, and how much of the difference reflects where each group lives, I estimate household-level regressions of each dwelling characteristic on an indicator for the comparison group interacted with year. For household i in year t, I estimate:

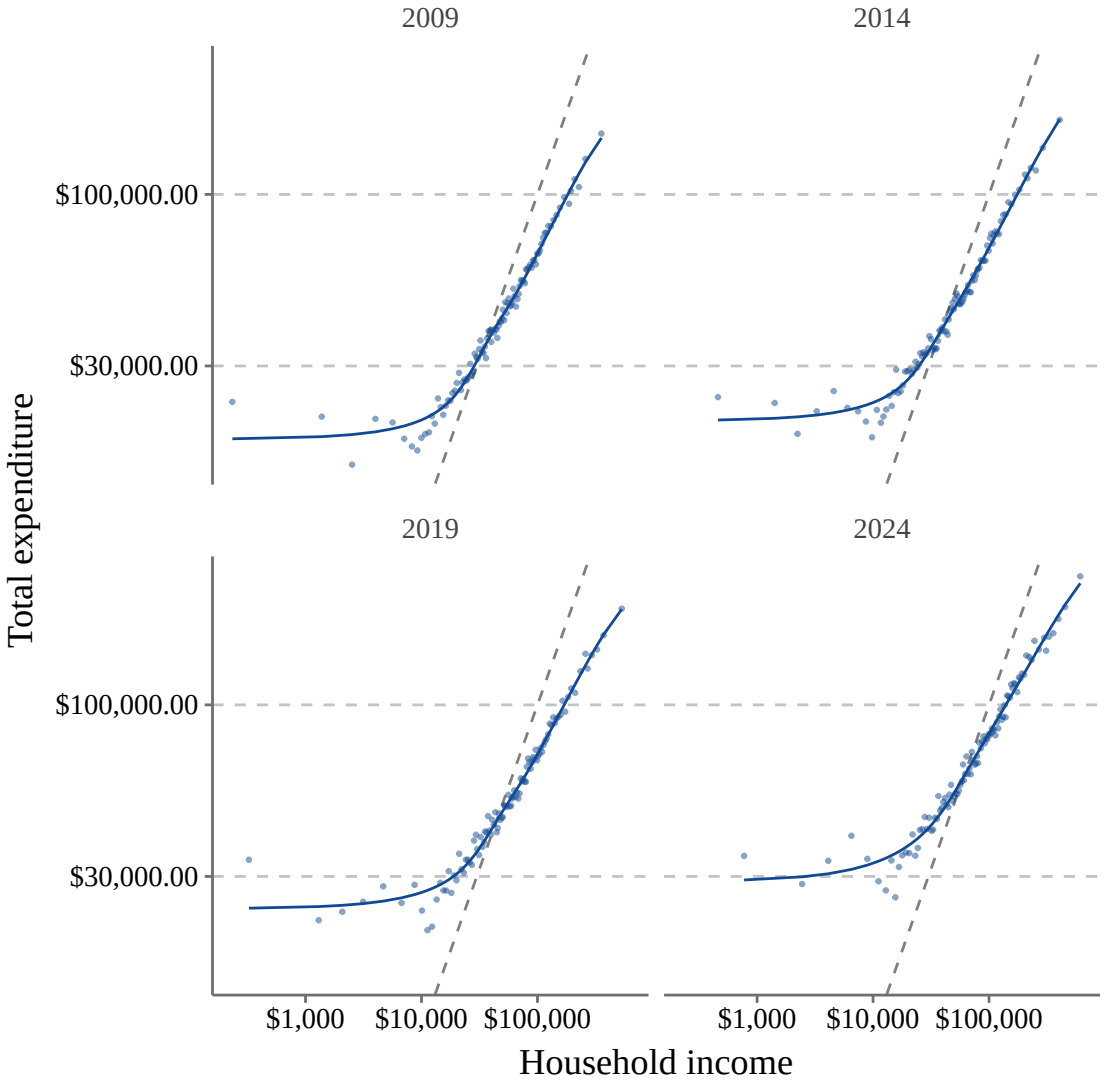
$$y_{it} = \sum_{\tau} \beta_{\tau} \cdot \mathbf{1}\{G_i = 1\} \cdot \mathbf{1}\{\tau = t\} + \mu_{c(i),t} + \varepsilon_{it}, \tag{13}$$

where y_{it} is the number of bedrooms, an indicator for a single-family home, or the year the structure was built; $G_i = 1$ indicates the comparison group; and $\mu_{c(i),t}$ is a fixed effect for the geography $c(i)$ of household i in year t. The coefficient β_{τ} is therefore the difference in the dwelling characteristic between the two groups in year τ, among households in the same geography.

I estimate equation (13) with four definitions of a geography: (1) the U.S., so that $\mu_{c(i),t}$ is a year fixed effect (None); (2) CBSA (CBSA); (3) PUMA (PUMA); and (4) PUMA interacted with the two dwelling characteristics other than the outcome (PUMA $\times$ Variety). The last definition compares households in the same PUMA with otherwise similar dwellings, and mirrors the housing varieties of Section 3.1. I weight each household by its survey weight and cluster standard errors by household.

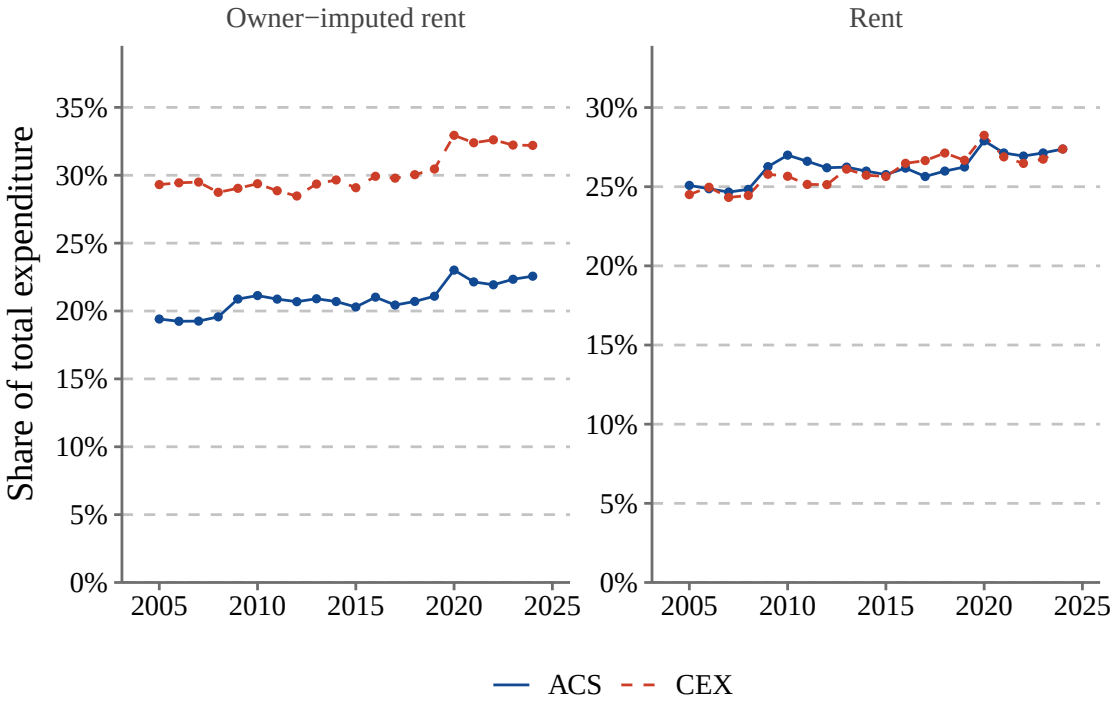
I estimate equation (13) on all households in the 2005–2024 ACS for three pairs of groups, dropping the middle education and age groups: householders 65 or older relative to householders under 45 (Figure C3); householders with at least a bachelor's degree relative to householders with at most a high school degree (Figure C4); and non-white relative to white householders (Figure C5).

Figure C1: Total expenditure and income in the CEX



Notes: This figure presents average annual total expenditure by average household income for each income percentile in the CEX Interview Survey in 2009, 2014, 2019, and 2024. Lines present fitted values from a regression of total expenditure on a natural cubic spline of income with knots at the quintiles of income, estimated separately by year. The dashed line is the 45-degree line.

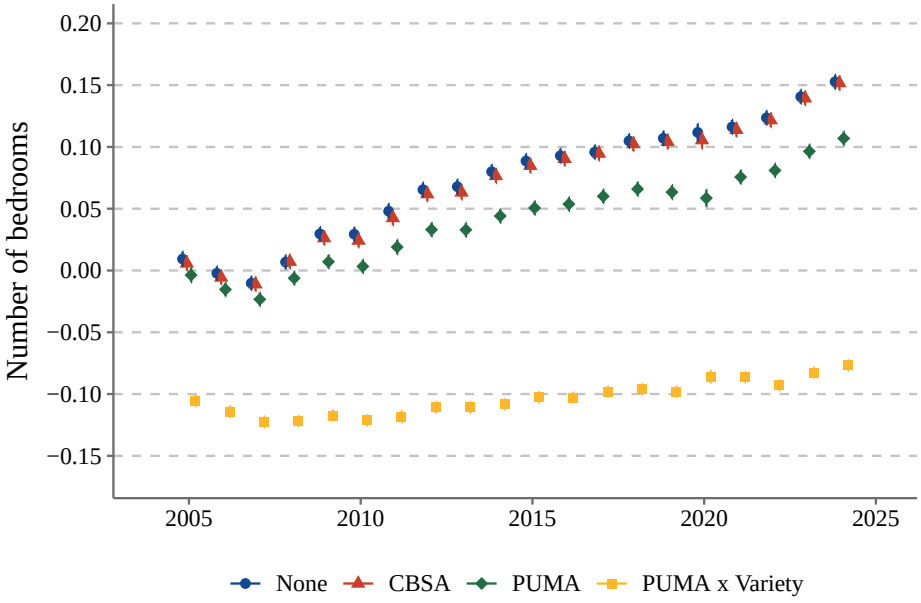
Figure C2: Shelter shares of total expenditure in the ACS and the CEX, 2005–2024



Notes: This figure presents shelter shares of total expenditure for renters and owners across two samples: (i) the ACS; and (ii) the CEX. The ACS series divide the sum of annual hedonic rents of renters (right panel) or owners (left panel) by the sum of their predicted total expenditure from Figure C1. The CEX series divide rent paid by renters, or owners' equivalent rent reported by owners, by total expenditure of the same households.

Figure C3: Dwelling characteristics of older relative to younger householders, 2005–2024

(A) Bedrooms



(B) Single-family

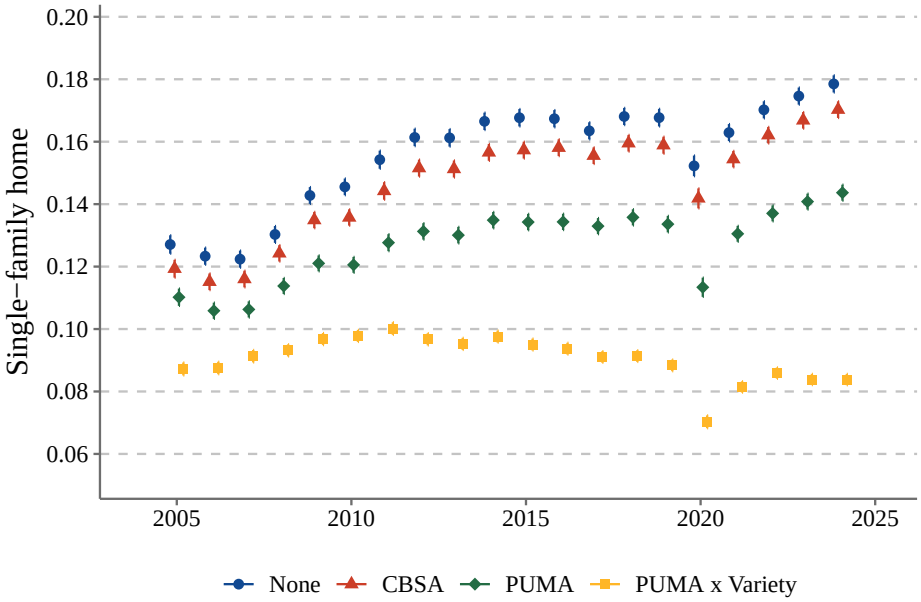
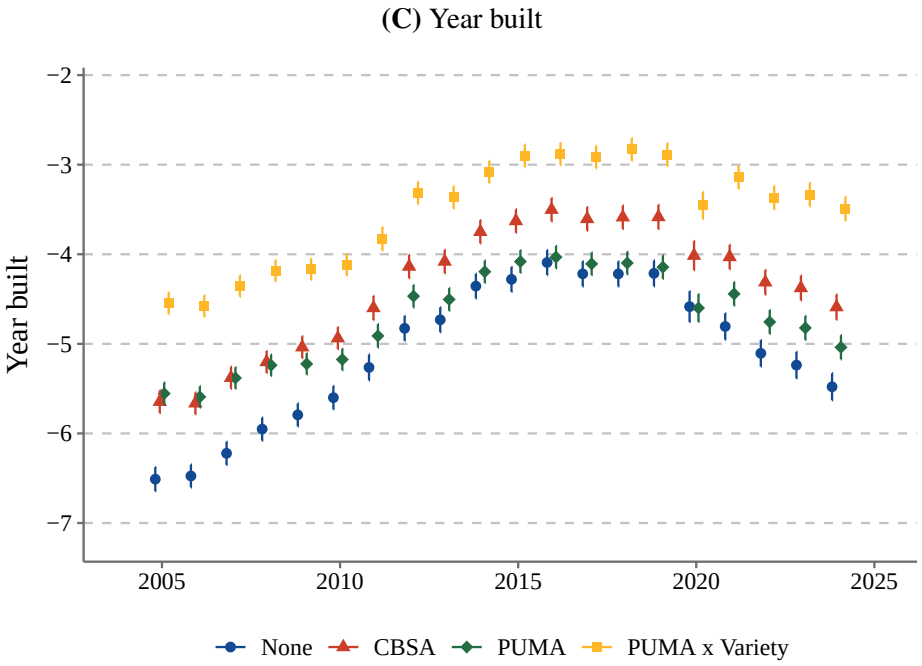


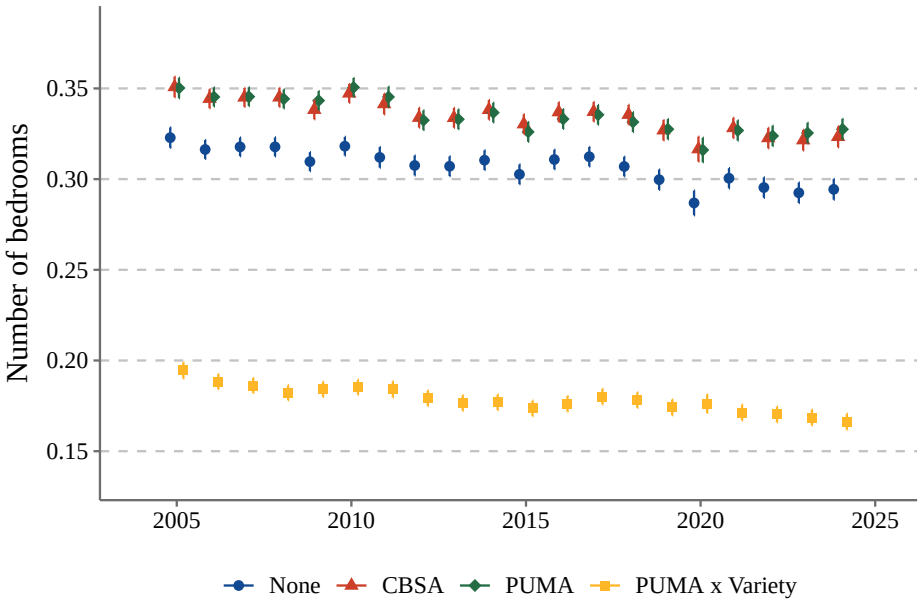
Figure C3: Dwelling characteristics of older relative to younger householders, 2005–2024 (continued)



Notes: This figure presents the coefficients β_τ from equation (13), where the comparison group is householders 65 or older and the reference group is householders under 45. Bars present 95 percent confidence intervals.

Figure C4: Dwelling characteristics of high- relative to low-education householders, 2005–2024

(A) Bedrooms



(B) Single-family

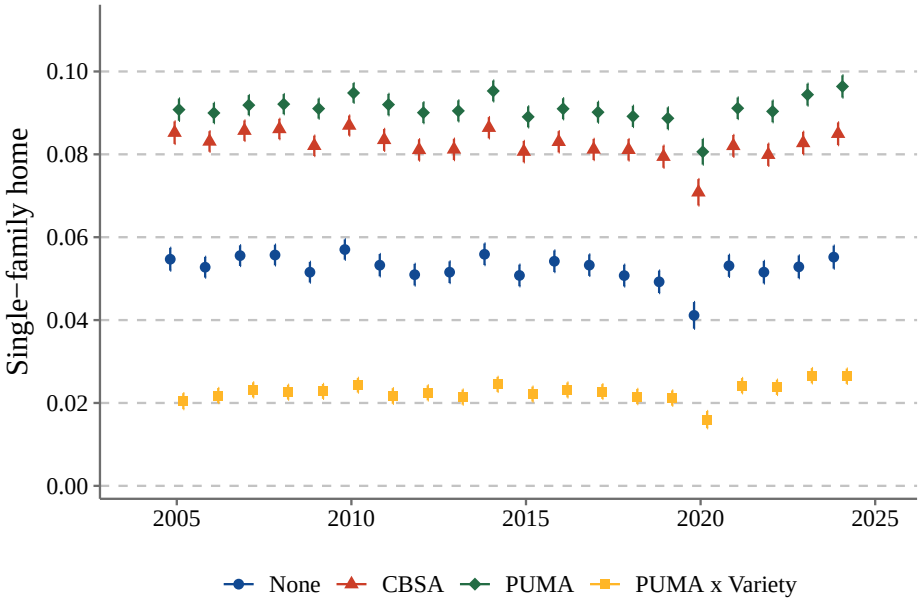
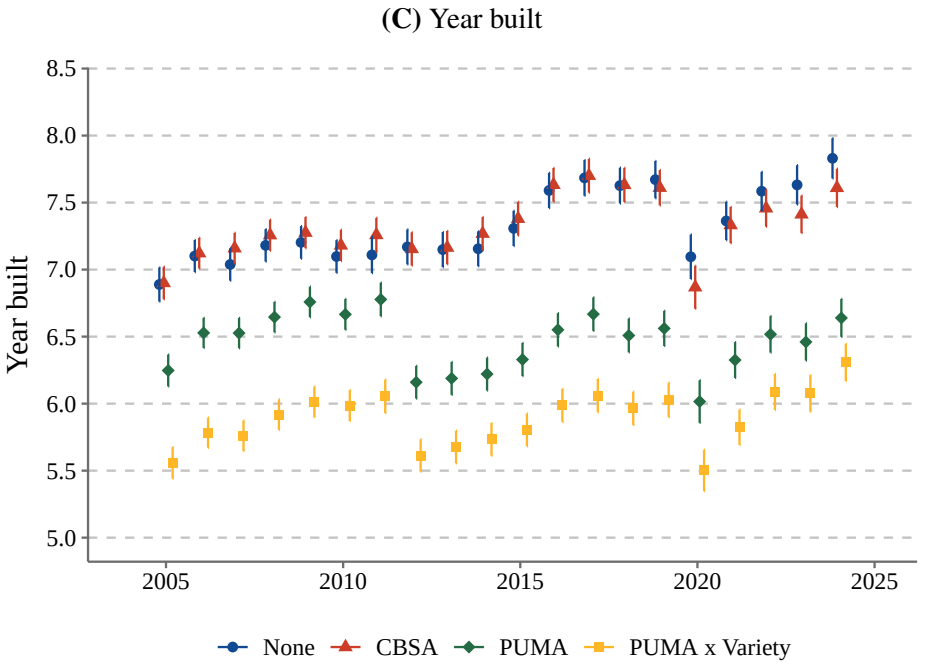


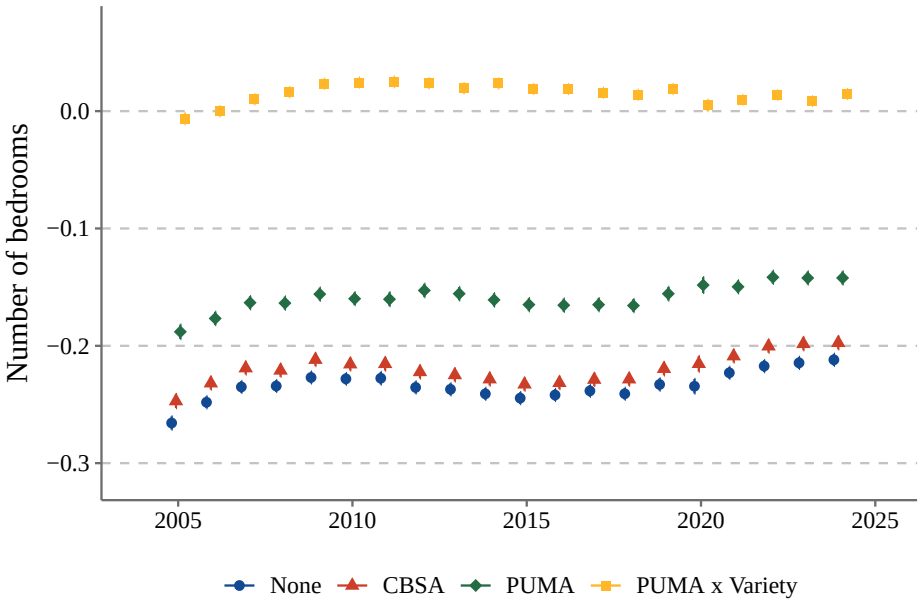
Figure C4: Dwelling characteristics of high- relative to low-education householders, 2005–2024 (continued)



Notes: This figure presents the coefficients β_τ from equation (13), where the comparison group is householders with at least a bachelor's degree and the reference group is householders with at most a high school degree. Bars present 95 percent confidence intervals.

Figure C5: Dwelling characteristics of non-white relative to white householders, 2005–2024

(A) Bedrooms



(B) Single-family

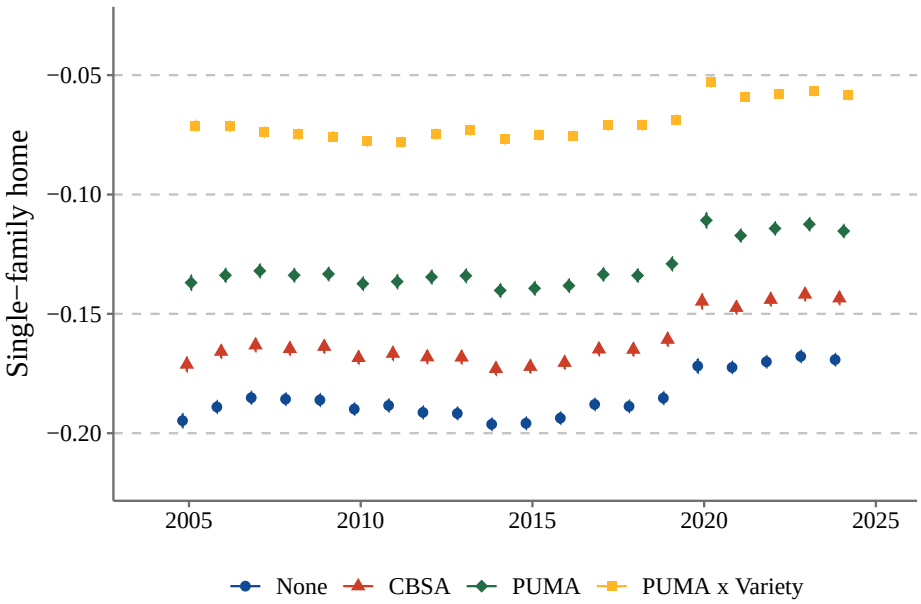
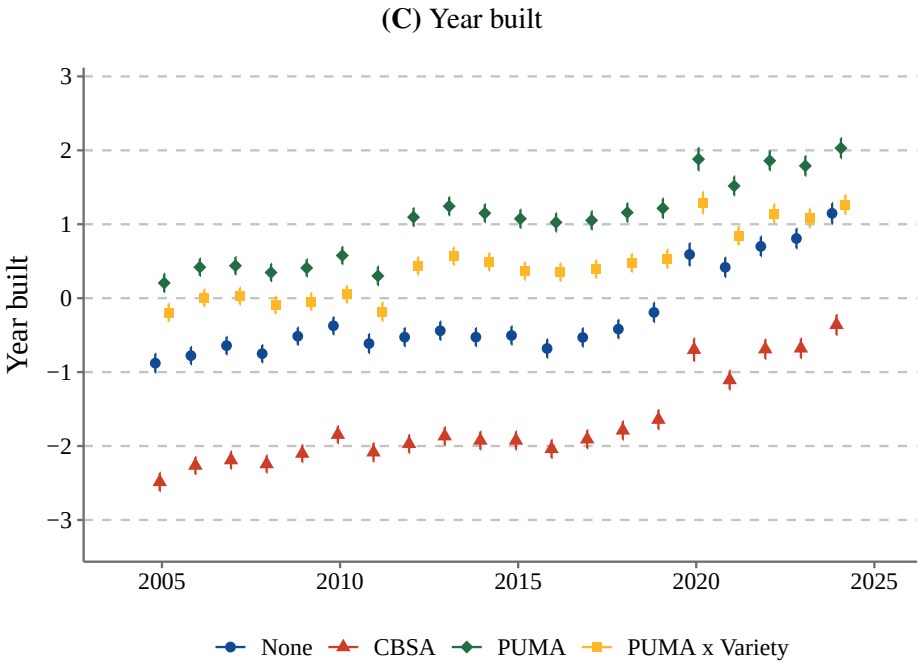


Figure C5: Dwelling characteristics of non-white relative to white householders, 2005–2024 (continued)



Notes: This figure presents the coefficients β_τ from equation (13), where the comparison group is non-white householders and the reference group is white householders. Bars present 95 percent confidence intervals.

D Repeat-rent indices from the AHS

Sample. I use the national AHS in odd-numbered years from 1975 to 2023, and limit my sample to units rented for cash with a valid contract rent. I define a rent change as the log change in contract rent for the same unit between consecutive observations, which are typically two years apart. I weight each rent change by the unit's survey weight at the start of the interval, and assign each rent change to the group of the householder at the start of the interval. I further drop rent changes whose base rent is top-coded.[65]

Estimator. To construct repeat-rent indices, I use the repeat-sales methodology of Case and Shiller (1987). Let R_{it} denote the monthly contract rent of unit i in year t, and let $s < t$ denote consecutive survey years in which unit i is observed. I estimate:

$$\log R_{it} - \log R_{is} = \sum_{g} \sum_{\tau} \psi_{\tau}^{g} \cdot \mathbf{1}\{G_{is} = g\} \cdot (\mathbf{1}\{\tau = t\} - \mathbf{1}\{\tau = s\}) + \varepsilon_{ist}, \tag{14}$$

where G_{is} is the group of the householder living in unit i in year s, and ψ_{τ}^{g} is the log rent index of group g in year τ, normalized to zero in the first year of each panel. Since the estimator uses only changes in the rent of the same unit, it holds dwelling characteristics fixed. Assigning each rent change to the householder at the start of the interval mirrors the base-period expenditure weights in equation (2). For example, a unit rented by a non-white householder in 2019 and by a white householder in 2021 contributes its 2019–2021 rent change to the non-white index. For the national index, I pool all householders into a single group.

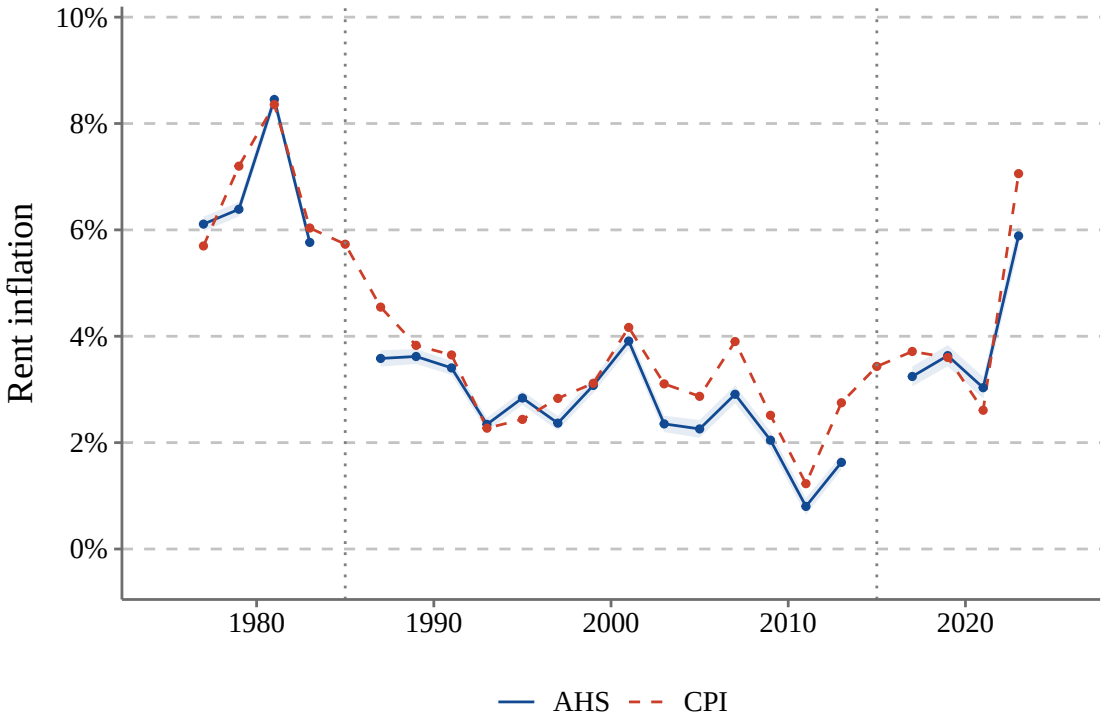
I estimate equation (14) separately for each of the three panels of the AHS (i.e., 1975–1983, 1985–2013, and 2015–2023). To limit the influence of outliers, I use a Huber robust regression rather than least squares, so that rent changes with large residuals receive less weight.[66] Annual rent inflation for group g between years s and t is then $(\psi_t^g - \psi_s^g)/(t-s)$. Its standard error is clustered by unit and accounts for the covariance of the two index levels.

National benchmark. As a validation exercise, Figure D1 benchmarks the national AHS index to the Rent of Primary Residence CPI over the same two-year intervals. From 1987 to 2023, the two series have a correlation of 0.92 and a mean absolute deviation of 0.51 percentage points per year. Figure D2 presents the group-specific indices over the full span of the survey.

[65] Top-coding affects only a small number of rent changes.

[66] I use the Huber M-estimator with tuning constant 1.345, estimated by iteratively reweighted least squares with a robust estimate of the residual scale. A rent change whose absolute standardized residual is below the tuning constant keeps its full survey weight, whereas a rent change with a larger absolute residual is downweighted in inverse proportion to its absolute residual.

Figure D1: Rent inflation in the AHS and the CPI, 1977–2023



Notes: This figure presents annualized rent inflation from the AHS repeat-rent index for cash renters, with 95 percent confidence intervals, and the annualized log change in the second-half average of the Rent of Primary Residence CPI over the same two-year intervals. Dotted lines mark the AHS sample redesigns in 1985 and 2015.

Figure D2: Rent inflation by household group in the AHS, 1977–2023

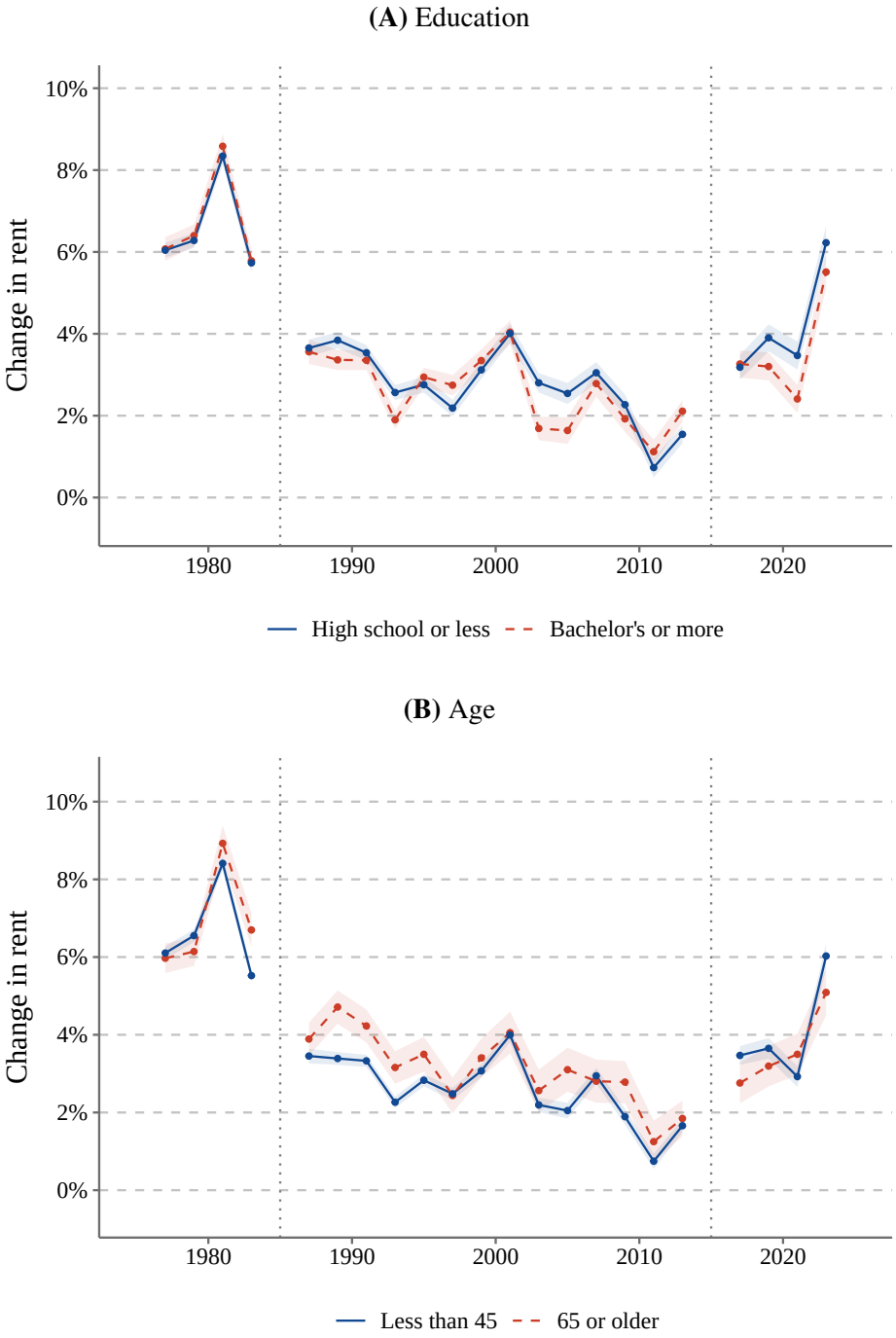
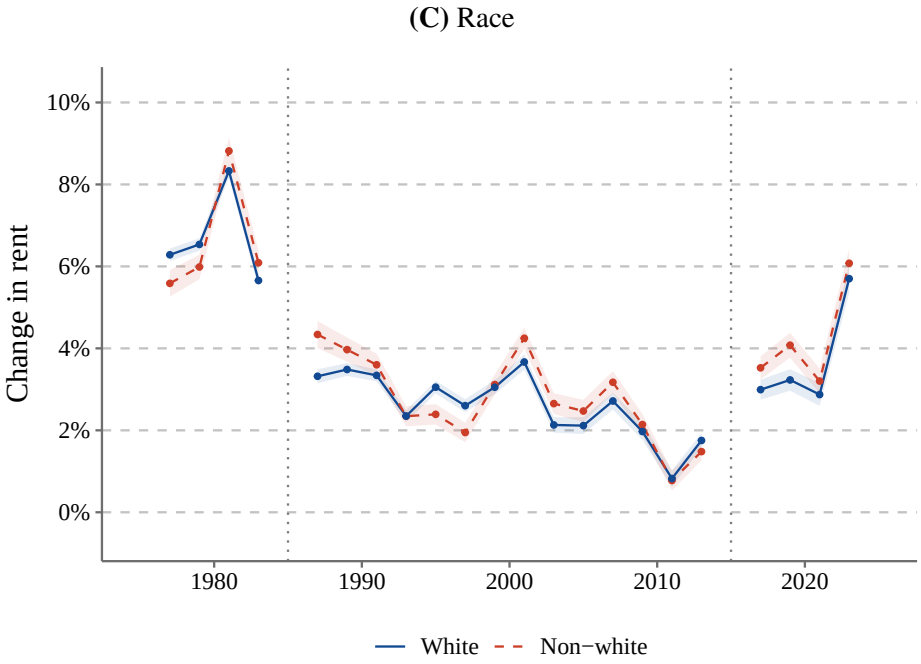


Figure D2: Rent inflation by household group in the AHS, 1977–2023 (continued)



Notes: This figure presents annualized rent inflation from the AHS repeat-rent index by the group of the householder at the start of each two-year interval, with 95 percent confidence intervals. Dotted lines mark the AHS sample redesigns in 1985 and 2015.

E Repeat-rent indices from the MLS

Sample. I use Cotality MLS rental listings from 2006 to 2024, and limit my sample to single-family homes, condominiums, and residential income properties. I measure monthly rent as the listed rent per square foot times the square footage of the unit. I identify each unit by its street address and ZIP code, and assign its address to the corresponding CBSA. Unlike the AHS, the MLS does not return to a unit at fixed intervals. Instead, a unit enters the sample in each year in which it is listed for rent, so the indices below measure the growth of new-tenant rents, rather than the rents of all renters that the ACS, the AHS, and the CPI measure.

Quality. Let R_{it} denote the listed rent of unit i in year t, and let $c(i)$ denote its CBSA. To classify units into quality groups, I first estimate:

$$\log R_{it} \;=\; \alpha_i \;+\; \delta_{c(i)t} \;+\; \varepsilon_{it},$$

where α_i is a unit fixed effect and δ_{ct} is a CBSA-by-year fixed effect. Since the CBSA-by-year fixed effect is common to all units in a CBSA, ranking the units in a CBSA by $\hat{\alpha}_i$ is the same as ranking them by their predicted rent in any year, so a unit does not need to be listed in 2010 to be classified. The CBSA-by-year fixed effects also ensure that the rank of a unit does not depend on when it is listed relative to the other units in its CBSA. I classify unit i as below-median, $B_i = 1$, if $\hat{\alpha}_i$ is below the median among units in its CBSA, and as above-median, $B_i = 0$, otherwise.[67]

Estimator. I then estimate:

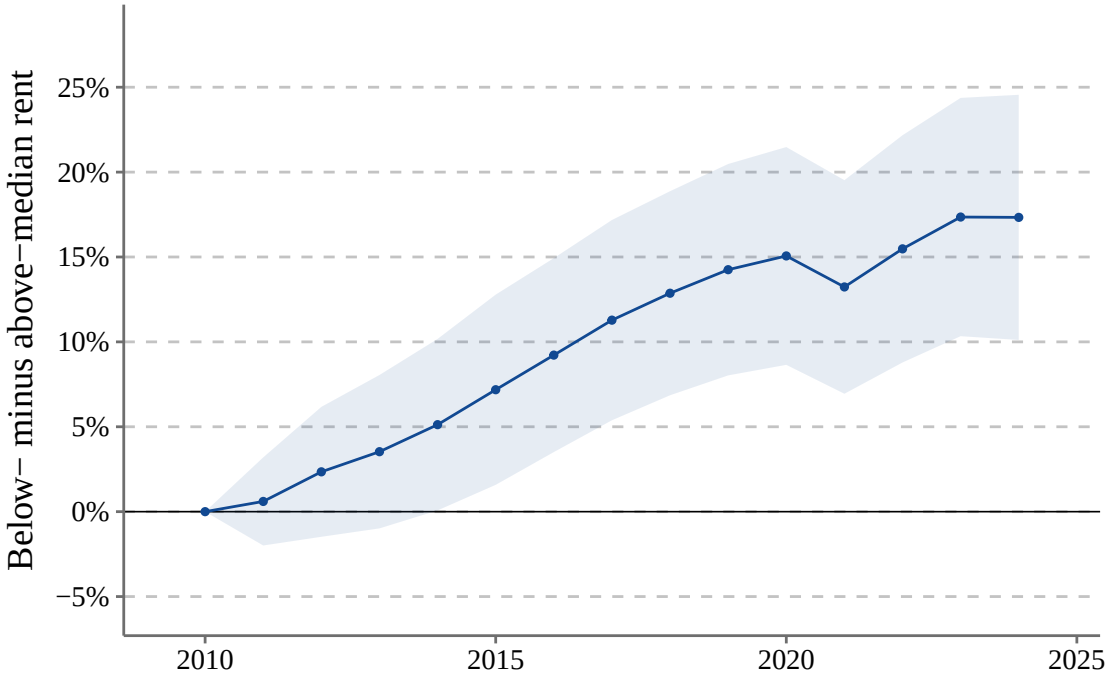
$$\log R_{it} \;=\; \alpha_i \;+\; \delta_{c(i)t} \;+\; \sum_{\tau \neq 2010} \psi_\tau \cdot B_i \cdot \mathbf{1}\{\tau = t\} \;+\; \varepsilon_{it}, \tag{15}$$

where ψ_τ is the log rent index of below-median relative to above-median units in year τ, normalized to zero in 2010. As in the AHS, the unit fixed effects hold the dwelling fixed, so the index compares the rent of the same unit across listings. The CBSA-by-year fixed effects absorb all differences in rent growth across CBSAs, so ψ_τ compares below- and above-median units listed in the same CBSA and year.[68] Since geographic coverage becomes comprehensive in 2010, I present the index from 2010 onward, but still use listings from 2006 to 2009 to estimate the unit fixed effects. I cluster standard errors by CBSA.

[67] I drop units that are listed only once, since they do not contribute to the repeat-rent index.

[68] That is, ψ_τ is a weighted average of the differences within each CBSA, which places more weight on CBSAs with more repeat listings.

Figure E1: Repeat-rent index of below- relative to above-median units in the MLS, 2010–2024



Notes: This figure presents the estimates of ψ_τ from equation (15), the log rent of below-median relative to above-median units in the same CBSA, normalized to zero in 2010. Shaded areas present 95 percent confidence intervals, with standard errors clustered by CBSA.

F Model calibration and solution

This appendix describes how I invert qualities in the model of Section 5, presents additional validations, and explains the equivalence between household growth and housing depreciation.

Sorting and housing supply. Figures F1 and F2 present the two patterns in the ACS that verify the features of the model in Section 5. Figure F1 presents the mean rent in 2009 and the change in real household income from 2009 to 2019 across CBSAs, against the share of householders with at least a bachelor's degree. Figure F2 presents the contribution of varieties with a new and an old dwelling age to the growth in the housing stock of each CBSA from 2009 to 2019. Section 5.3 discusses the implications of both figures.

Quality by education. Given 2009 budgets and rents, the nested logit in equation (3) can be inverted in closed form for each education group (Berry, 1994). Let $s^e_{v|c}$ denote the observed share of group e in metropolitan area c that lives in variety v, and s^e_c the observed share of the group that lives in metropolitan area c. Then:

$$q_{ev} \;=\; \sigma\left(\lambda \log s^e_{v|c} + \log s^e_c\right) \;-\; \log\left(X_{ec} - R^0_v\right), \tag{16}$$

which reproduces the observed shares of group e exactly, up to a constant that does not affect its choices. The small number of varieties with $s^e_{v|c} = 0$ get $q_{ev} = -\infty$.

Common quality. To measure how much of the sorting can be explained by income alone, I also compute a quality q_v that is common to all education groups. Starting from $q_v = 0$, I iterate on a contraction mapping similar to that of Berry (1994):

$$q_v \;\leftarrow\; q_v \;+\; \sigma\lambda \log\left(S^0_v / D_v\right), \tag{17}$$

where $D_v = \sum_e N_e P_v(e)$ is demand at 2009 budgets and rents with $q_{ev} = q_v$ for every group, until $\max_v |\log(S^0_v/D_v)| \approx 0$. Since this quality is common to all education groups, the inversion targets only the total number of households in each variety. Figure F3 compares the resulting shares of each education group with the observed shares, and Figure F4 compares the quality of each education group from equation (16) with the common quality. If differences in budgets fully explain differences in sorting across education groups, the two qualities are equal up to a constant. I find that, for every education group, the quality of the group moves nearly one-for-one with the common quality.

To further measure how much of the sorting non-homotheticity explains on its own, I compare

the composition of each variety by education predicted with the common quality to the actual composition in the data. Since the common quality targets only the total number of households in each variety, it does not target this composition. In particular, the model predicts that a share $N_e P_v(e)/D_v$ of the households in variety v belong to group e, where D_v is demand with $q_{ev} = q_v$ as above. Figure F5 compares the actual and predicted shares for each education group. The slope of the actual on the predicted share is 0.60 for householders with at most a high school degree and 0.90 for householders with at least a bachelor's degree. Therefore, much of the sorting by education groups across metropolitan areas and housing varieties is rationalized by general non-homothetic preferences for housing.

Household growth and housing depreciation. A common increase in the number of households is isomorphic to a proportional reduction in housing supply. To see this, suppose the number of households in every education group grows by a factor $G > 1$, holding budgets and qualities fixed across the two experiments. Combining equations (4) and (6), market clearing can be written as:

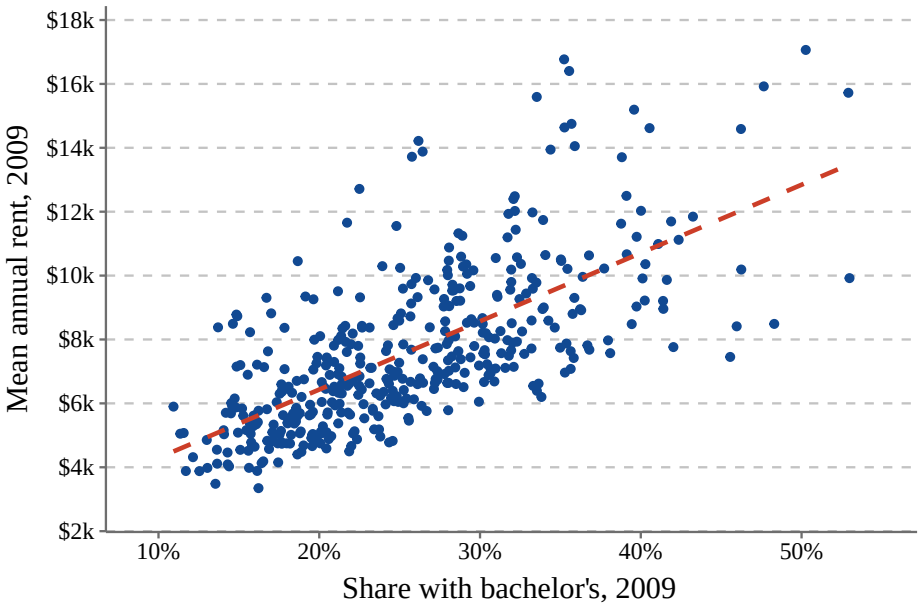
$$\sum_e G N_e P_v(e) = H_v \left(R_v/R_v^0\right)^{\gamma_v} \quad \Longleftrightarrow \quad \sum_e N_e P_v(e) = \frac{H_v}{G} \left(R_v/R_v^0\right)^{\gamma_v}.$$

The economy with more households therefore has the same equilibrium rents and choice probabilities as an economy with the original number of households and supply shifters H_v/G. Interpreting the latter as depreciation, the fraction of the stock lost at unchanged rents is $\delta = 1 - G^{-1}$. Thus household growth of $n = G - 1$ is equivalent to depreciation of $\delta = n/(1+n)$, or $\delta \approx n$ for small changes. Both reduce housing available per household and create pressure for rents to rise, with more elastic varieties accommodating more of the demand through a supply response. Aggregate housing quantities differ by the factor G, but housing quantities per household are identical.

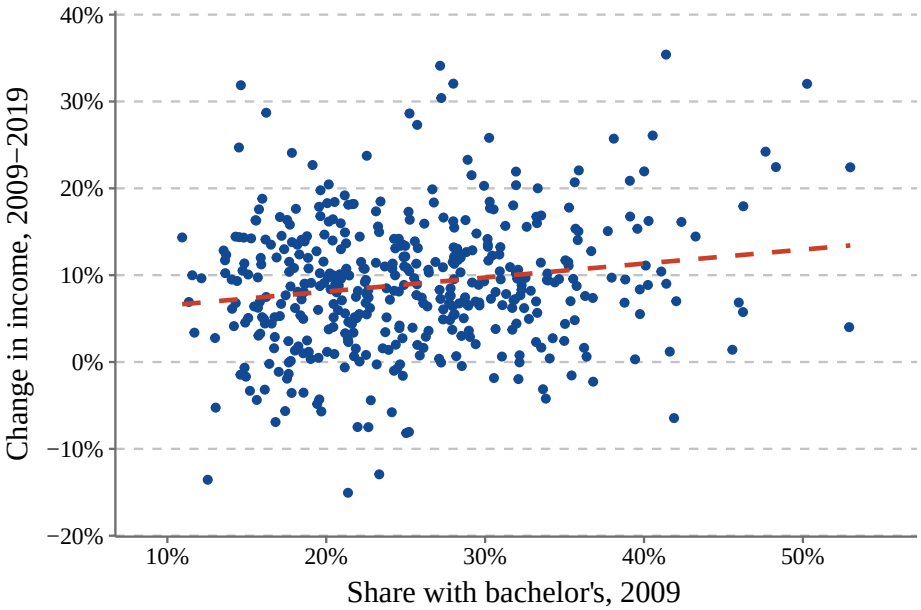
This equivalence treats depreciation as a proportional reduction in the supply shifter of every variety, leaving quality and supply elasticities unchanged. It does not describe deterioration in dwelling quality or changes in the composition of households. In the calibrated counterfactual, household growth differs across education groups. Writing $N'_e = G N_e(\mu'_e/\mu_e)$, where $G = (\sum_e N'_e)/(\sum_e N_e)$ and μ_e and μ'_e are the initial and final national education shares, separates the common growth factor from the composition change. Only the common factor can be absorbed into proportional depreciation; the changes in education shares remain separate demand shifts.

Figure F1: Rents and income by educational composition across CBSAs, 2009–2019

(A) Mean rent, 2009

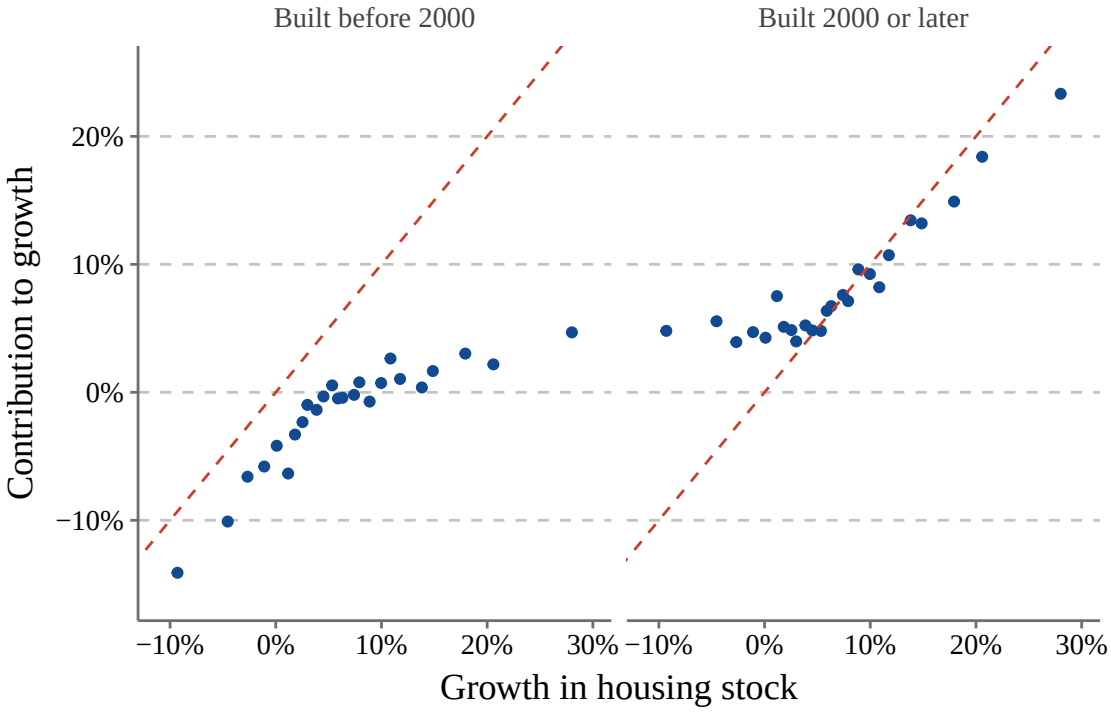


(B) Change in household income, 2009–2019



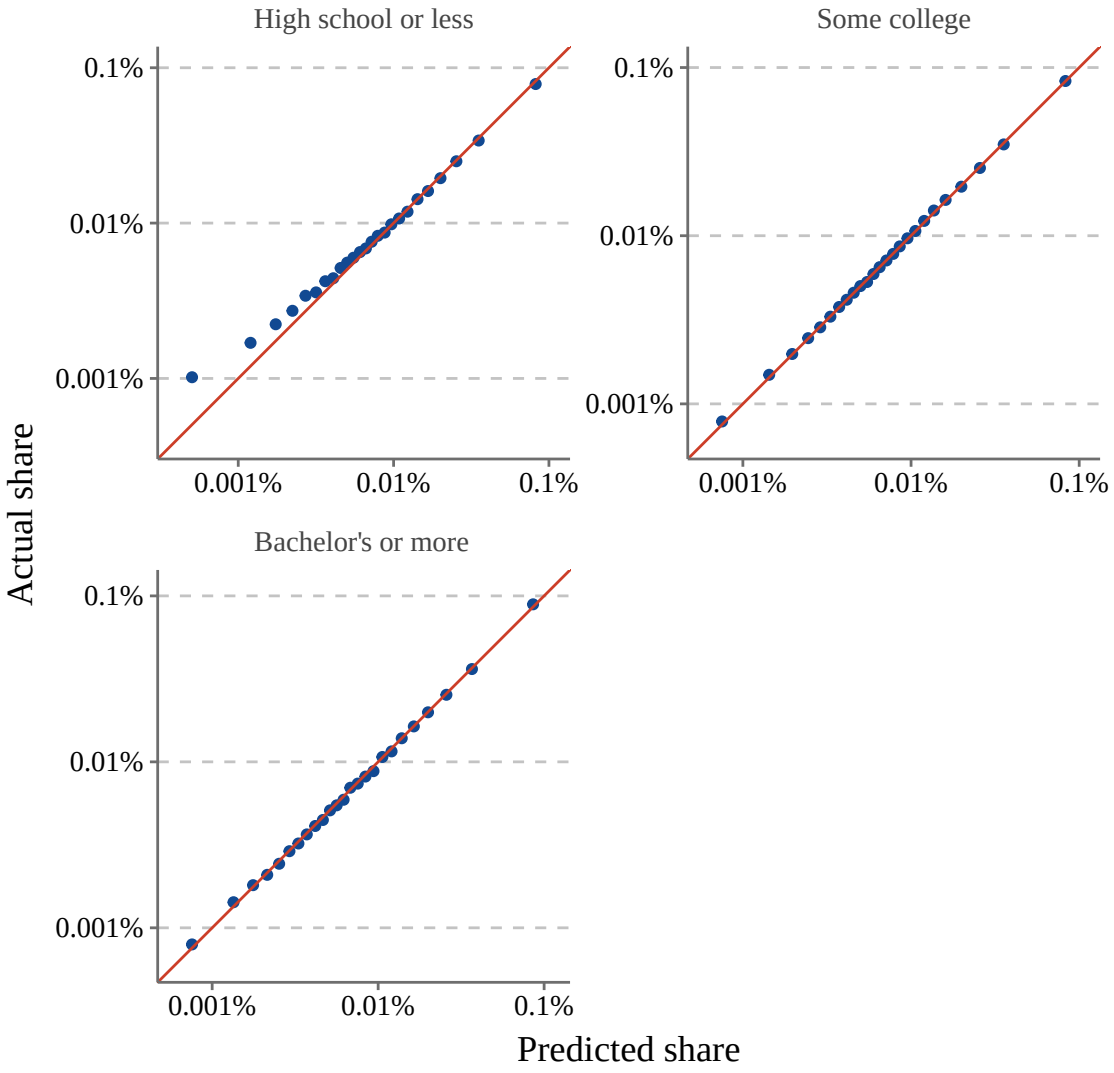
Notes: This figure presents scatter plots across CBSAs against the share of householders with at least a bachelor's degree in the 2009 ACS. Panel A presents the mean annual rent in each CBSA in 2009. Panel B presents the change in household income in each CBSA from 2009 to 2019, averaging across education groups with weights equal to their households in 2009. Income is deflated by the CPI for all items less shelter. The dashed line is the ordinary least squares fit.

Figure F2: Growth in the housing stock by dwelling age, 2009–2019



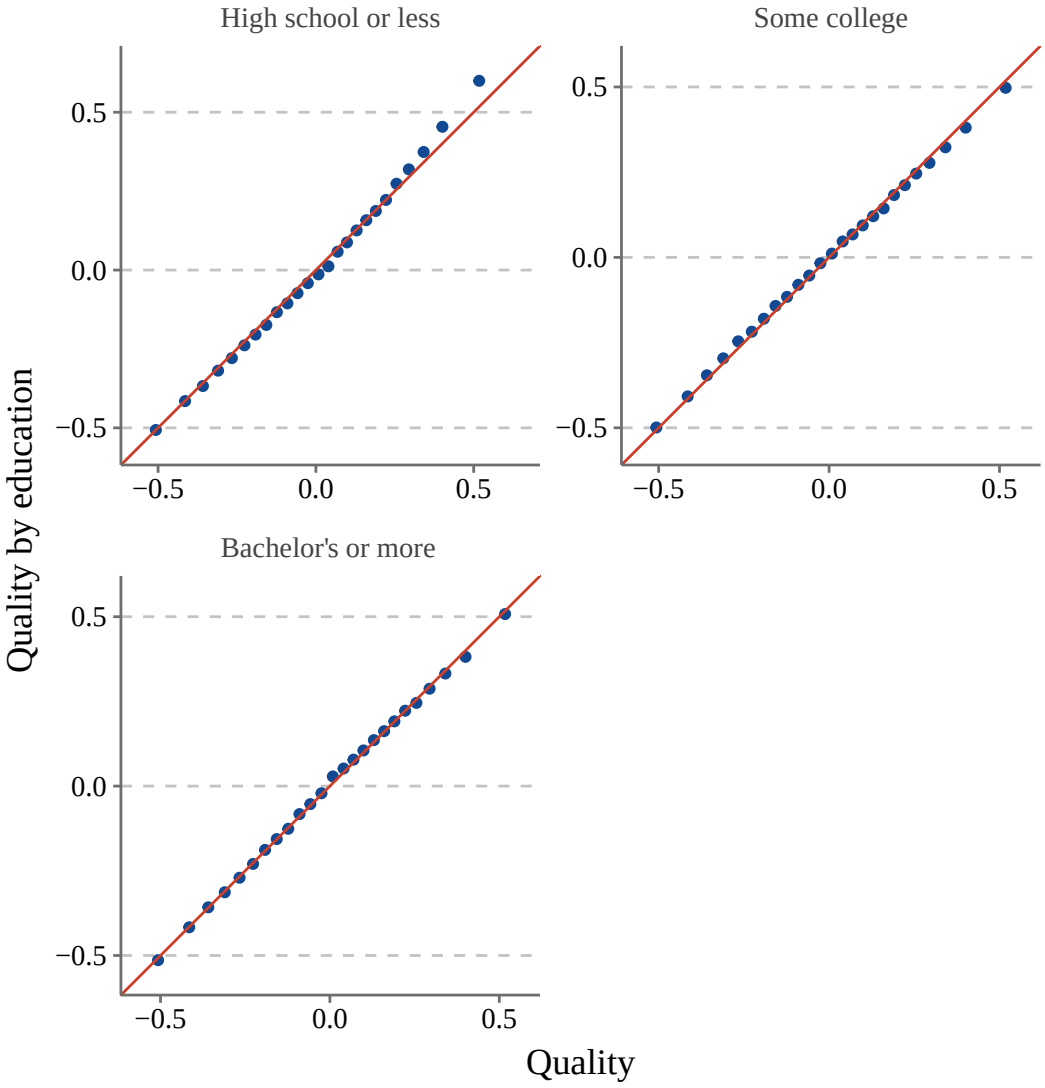
Notes: This figure presents binned scatter plots of the contribution of each dwelling age to the growth in the housing stock of each CBSA from 2009 to 2019, against the total growth in its housing stock. Dwellings built in 2000 or later correspond to varieties with a new dwelling age in 2009. The contribution of each dwelling age is the change in its stock divided by the total stock of the CBSA in 2009, so the two contributions sum to total growth. The dashed line is the 45-degree line.

Figure F3: Shares by education implied by income alone, 2009



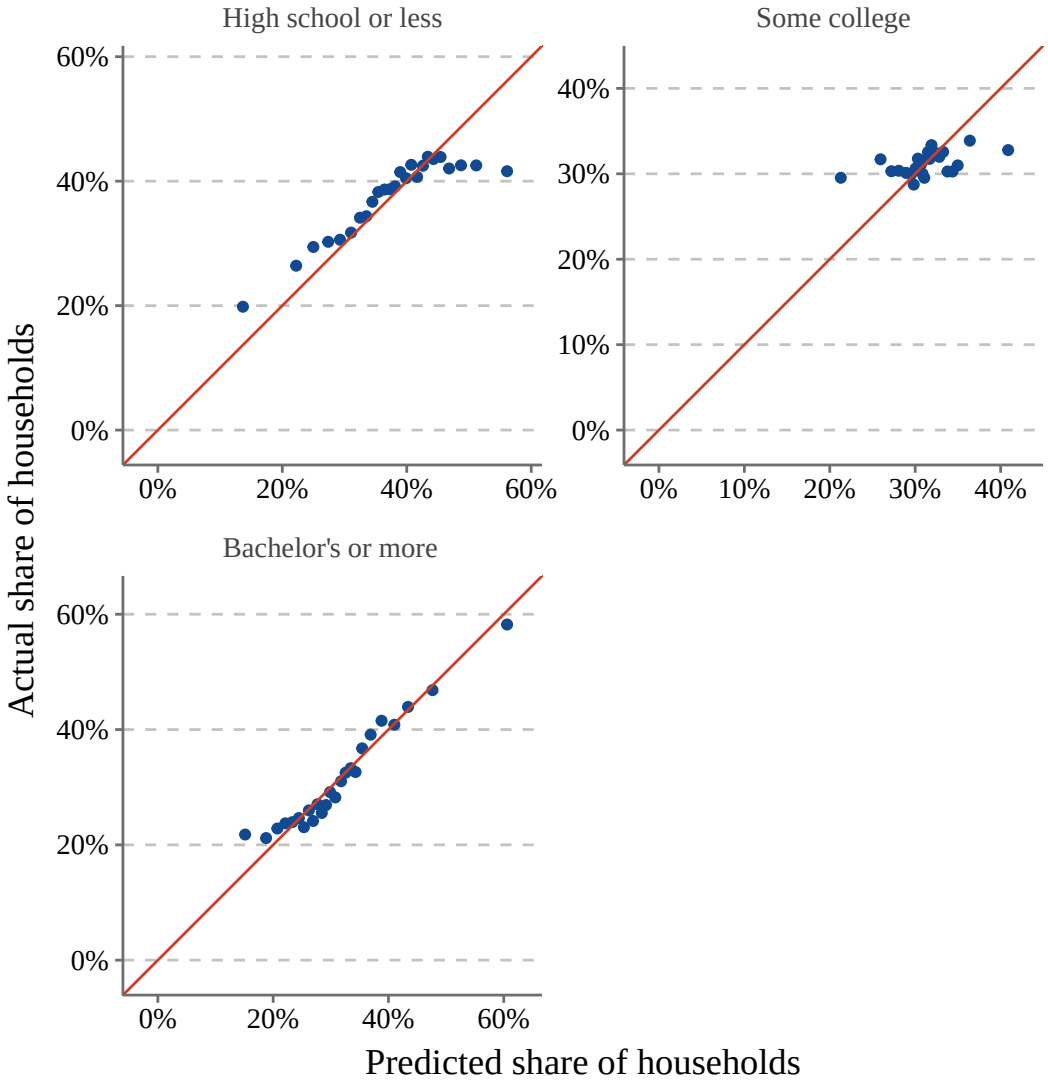
Notes: This figure presents binned scatter plots of the observed share of each education group that lives in each variety in the 2009 ACS against the share predicted by equation (3) with the quality q_v common to all education groups. The solid line is the 45-degree line.

Figure F4: Quality by education in the model, 2009



Notes: This figure presents binned scatter plots of the quality q_{ev} of each education group from equation (16), which rationalizes the observed choices of the group alone, against the quality q_v common to all groups from equation (17). The solid line is the 45-degree line.

Figure F5: Composition by education implied by income alone, 2009



Notes: This figure presents binned scatter plots of the observed share of the households in each variety that belong to each education group in the 2009 ACS, against the share predicted by equation (3) with the quality q_v common to all education groups. The solid line is the 45-degree line.

G Analytical results on sorting and shelter inflation

This appendix provides the derivations and proofs for Section 5.2. All results are first-order changes evaluated at an initial equilibrium, holding quality and supply shifters fixed. With common quality and $\lambda = 1$, equation (3) becomes:

$$P_{ecv} = \frac{\exp(q_{cv}/\sigma)(X_{ec} - R_{cv})^{1/\sigma}}{\sum_{d,w} \exp(q_{dw}/\sigma)(X_{ed} - R_{dw})^{1/\sigma}}. \tag{18}$$

As in Section 5.2, initial incomes are common across metropolitan areas, $X_{ec} = X_e$. I further assume $X_H > X_L > \max_{c,v} R_{cv}$, so every household can afford every housing variety. At the initial equilibrium, rents equal the base-year rents in equation (4), $R_{cv} = R^0_{cv}$, so the stock of each variety equals its supply shifter, $S^0_{cv} = H_{cv}$. Equilibrium requires:

$$S^0_{cv} \left(R_{cv}/R^0_{cv}\right)^{\gamma_{cv}} = \sum_e N_e P_{ecv},$$

so the initial stock of each variety equals its initial demand, and a rent change ρ_{cv} raises supply by $S^0_{cv}\gamma_{cv}\rho_{cv}$.

G.1 Decomposing shelter inflation

The initial expenditure shares defined in Section 5.2 are:

$$s_{ec} = \frac{\sum_v P_{ecv} R_{cv}}{\sum_{d,w} P_{edw} R_{dw}}, \qquad \theta_{ec} = \frac{P_{ecn} R_{cn}}{\sum_v P_{ecv} R_{cv}}.$$

Holding the initial choices of group e fixed, its local shelter inflation is the expenditure-weighted mean of local rent inflation, $\pi_{ec} = (1 - \theta_{ec})\rho_{co} + \theta_{ec}\rho_{cn}$, and its shelter inflation is $\pi_e = \sum_c s_{ec}\pi_{ec}$. Let ω_{ec} be the share of housing expenditure in metropolitan area c spent by education group e:

$$\omega_{ec} = \frac{N_e \sum_v P_{ecv} R_{cv}}{\sum_f N_f \sum_v P_{fcv} R_{cv}}.$$

Mean local rent inflation is then $\bar{\rho}_c = \omega_{Lc}\pi_{Lc} + \omega_{Hc}\pi_{Hc}$. Adding and subtracting $\sum_c s_{ec}\bar{\rho}_c$ in π_e gives equation (7).

Since $s_{eB} = 1 - s_{eA}$, the between-area gap is:

$$\pi^B_H - \pi^B_L = (s_{HA} - s_{LA})(\bar{\rho}_A - \bar{\rho}_B).$$

The local gap in equation (8) follows from subtracting π_{Lc} from π_{Hc}. Since $\omega_{Lc} + \omega_{Hc} = 1$, $\pi_{Hc} -$

$\bar{\rho}_c = \omega_{Lc}(\pi_{Hc} - \pi_{Lc})$ and $\pi_{Lc} - \bar{\rho}_c = -\omega_{Hc}(\pi_{Hc} - \pi_{Lc})$. Substituting into equation (7), the within-area gap is:

$$\pi_H^W - \pi_L^W = \sum_c (s_{Hc}\omega_{Lc} + s_{Lc}\omega_{Hc})(\pi_{Hc} - \pi_{Lc}).$$

That is, the within-area gap is a positively weighted sum of the local gaps.

G.2 Sorting

Proof of Proposition 1. Consider any two housing alternatives j and k, in the same or different metropolitan areas, with $R_j > R_k$. By equation (18), the denominator cancels from each group's choice odds:

$$\frac{P_{ej}}{P_{ek}} = \exp\left(\frac{q_j - q_k}{\sigma}\right)\left(\frac{X_e - R_j}{X_e - R_k}\right)^{1/\sigma}.$$

Common quality then cancels from the ratio of these odds across groups:

$$\frac{P_{Hj}/P_{Hk}}{P_{Lj}/P_{Lk}} = \left[\frac{(X_H - R_j)(X_L - R_k)}{(X_H - R_k)(X_L - R_j)}\right]^{1/\sigma} > 1. \tag{19}$$

The inequality holds because the numerator inside the brackets exceeds the denominator by $(X_H - X_L)(R_j - R_k) > 0$. Hence, $P_{Hj}P_{Lk} > P_{Lj}P_{Hk}$ for every such pair.

Within metropolitan area c, take j to be new housing and k old housing. Each group's expenditure odds are $\theta_{ec}/(1-\theta_{ec}) = P_{ecn}R_{cn}/(P_{eco}R_{co})$. Rents cancel from the ratio of these odds across groups, so equation (19) gives:

$$\frac{\theta_{Hc}/(1-\theta_{Hc})}{\theta_{Lc}/(1-\theta_{Lc})} = \frac{P_{Hcn}/P_{Hco}}{P_{Lcn}/P_{Lco}} > 1.$$

Since the odds increase in θ_{ec}, $\theta_{Hc} > \theta_{Lc}$.

Across metropolitan areas, since $s_{eB} = 1 - s_{eA}$, $s_{HA} > s_{LA}$ if and only if $s_{HA}s_{LB} > s_{LA}s_{HB}$. The denominators of the shares cancel, so this condition is:

$$\sum_{j\in A}\sum_{k\in B} R_j R_k P_{Hj} P_{Lk} > \sum_{j\in A}\sum_{k\in B} R_j R_k P_{Lj} P_{Hk}.$$

If every rent in A exceeds every rent in B, then $R_j > R_k$ for every pair with $j \in A$ and $k \in B$, so $P_{Hj}P_{Lk} > P_{Lj}P_{Hk}$ holds term by term and $s_{HA} > s_{LA}$.

The second result relies on equal initial incomes across metropolitan areas. When incomes differ across areas, rents alone do not determine sorting across them.

G.3 Rent inflation across metropolitan areas

As in Proposition 2, consider one housing variety per metropolitan area, and hold the national number of households in each education group, N_e, fixed. Let $p_e = P_{eA}$ be the probability that a household in group e chooses metropolitan area A, and define:

$$\ell_e = \frac{N_e p_e (1-p_e)}{\sigma}, \qquad F_c = S_c^0 \gamma_c, \qquad K_c = \sum_e \ell_e \frac{R_c}{X_e - R_c},$$
$$T = \sum_e \ell_e \left(\frac{dX_{eA}}{X_e - R_A} - \frac{dX_{eB}}{X_e - R_B} \right).$$

Here, F_c and K_c are the supply and demand responses of metropolitan area c to its rent, and T is the shift in demand toward metropolitan area A at fixed rents. Since $X_e > R_c$ and $p_e \in (0,1)$, $\ell_e > 0$ and $K_c > 0$.

Proof of Proposition 2. With two alternatives, p_e is a binary logit in the systematic utilities $V_{ec} = [\log(X_{ec} - R_c) + q_c]/\sigma$, so $dp_e = p_e(1-p_e)(dV_{eA} - dV_{eB})$, where:

$$dV_{ec} = \frac{dX_{ec} - R_c \rho_c}{\sigma (X_e - R_c)}.$$

The change in demand for metropolitan area A is therefore:

$$\sum_e N_e \, dp_e = T - K_A \rho_A + K_B \rho_B.$$

Since N_e is fixed, demand for B changes by the same amount with the opposite sign. Equating these changes with the supply responses $F_A \rho_A$ and $F_B \rho_B$ gives:

$$\begin{pmatrix} F_A + K_A & -K_B \\ -K_A & F_B + K_B \end{pmatrix} \begin{pmatrix} \rho_A \\ \rho_B \end{pmatrix} = \begin{pmatrix} T \\ -T \end{pmatrix}.$$

The determinant, $F_A F_B + F_A K_B + F_B K_A$, is positive since $\gamma_A, \gamma_B > 0$. Solving the system gives:

$$\rho_A = \frac{F_B T}{F_A F_B + F_A K_B + F_B K_A}, \qquad \rho_B = -\frac{F_A T}{F_A F_B + F_A K_B + F_B K_A}.$$

If income grows in A alone, then $dX_{eB} = 0$, and $dX_{eA} \geq 0$ for both groups with a strict increase for at least one. Hence, $T = \sum_e \ell_e \, dX_{eA}/(X_e - R_A) > 0$, so $\rho_A > 0 > \rho_B$.

G.4 Rent inflation within a metropolitan area

As in Proposition 3, consider old and new housing within one metropolitan area and suppress c. Let $N = \sum_e N_e$ be the initial local number of households, $\mu_e = N_e/N$ the initial local share of education group e, p_e the probability that a household in group e chooses new housing given its metropolitan area, and $\zeta = \sum_e \mu_e p_e$. Since the initial stock equals initial demand, ζ is the initial share of new housing in the local stock. Define the demand responses:

$$k = \sum_e \frac{\mu_e p_e (1-p_e)}{\sigma} \frac{R_n}{X_e - R_n}, \qquad l = \sum_e \frac{\mu_e p_e (1-p_e)}{\sigma} \frac{R_o}{X_e - R_o}.$$

Since $R/(X_e - R)$ increases in R and $R_n > R_o$, $k > l > 0$. Also define the scaled supply responses, a and b, and:

$$a = \zeta \gamma_n, \qquad b = (1-\zeta)\gamma_o, \qquad \bar{\gamma} = a + b, \qquad D = ab + al + bk.$$

Here, $\bar{\gamma}$ is the stock-weighted mean supply elasticity in Proposition 3. Since at least one supply elasticity is positive, $D > 0$.

Proof of Proposition 3. When incomes grow at the common rate g, $dX_e = gX_e$. Since $X_e/(X_e - R) = 1 + R/(X_e - R)$, differentiating p_e gives:

$$dp_e = \frac{p_e(1-p_e)}{\sigma} \left[g \left(\frac{R_n}{X_e - R_n} - \frac{R_o}{X_e - R_o} \right) - \frac{R_n}{X_e - R_n} \rho_n + \frac{R_o}{X_e - R_o} \rho_o \right].$$

At fixed rents, income growth therefore shifts demand toward new housing by $g(k-l)$ per initial household. With a common household growth rate n, the shares μ_e remain fixed, and demand for new housing, $\sum_e N_e p_e$, changes by $N[\zeta n + \sum_e \mu_e \, dp_e]$. Supply of new housing changes by $S_n^0 \gamma_n \rho_n = Na\rho_n$, and likewise for old housing. Equating changes in supply and demand, divided by N, gives:

$$\begin{pmatrix} a+k & -l \\ -k & b+l \end{pmatrix} \begin{pmatrix} \rho_n \\ \rho_o \end{pmatrix} = \begin{pmatrix} \zeta n + g(k-l) \\ (1-\zeta)n - g(k-l) \end{pmatrix}.$$

The determinant of the system is D, and solving it gives:

$$\rho_n = \frac{nl + b\zeta n + bg(k-l)}{D}, \qquad \rho_o = \frac{nk + a(1-\zeta)n - ag(k-l)}{D}.$$

Subtracting gives the rent gap:

$$\rho_o - \rho_n = \frac{n\zeta(1-\zeta)(\gamma_n - \gamma_o) + (n - \bar{\gamma} g)(k-l)}{D}. \tag{20}$$

If $n > \bar{\gamma} g \geq 0$ and $\gamma_n \geq \gamma_o$, the first term in the numerator is nonnegative and the second is positive, so $\rho_o > \rho_n$.

Equation (20) has two terms. The first is a supply channel: household growth raises demand for both varieties, and a more elastic supply of new housing absorbs more of it. The second is a demand channel: since demand for new housing responds more to its rent (i.e., $k > l$), household growth raises old rents more than new rents, whereas income growth shifts demand toward new housing. With no household growth and positive income growth, the first term vanishes and the second is negative for any admissible supply elasticities, so $\rho_n > \rho_o$.

To isolate the role of each supply elasticity, I rearrange equation (20). Old rents grow faster than new rents if and only if:

$$\zeta \gamma_n [(1-\zeta)n - g(k-l)] > (1-\zeta)\gamma_o [\zeta n + g(k-l)] - n(k-l).$$

The bracketed terms are the changes in demand for old and new housing at fixed rents. Holding all else fixed, a higher γ_n makes this inequality easier to satisfy if and only if demand for old housing rises at fixed rents (i.e., $(1-\zeta)n > g(k-l)$). Otherwise, a more elastic supply of new housing absorbs the households who upgrade and makes the inequality harder to satisfy.